QUANTUM MECHANICS 3

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Lecture 1. State vectors in Hilbert space

Here we learn that the state of a quantum particle can be represented as a vector in Hilbert space. We use Dirac notations for these vectors and introduce the inner (scalar) product in Hilbert space ("bra(c)ket") in terms of "ket"- and "bra"-vectors. Mathematical part of this lecture is a revision of material known from Maths2B.

1.1 State of a quantum particle

A classical state of a particle with f degrees of freedom ($f = 3$ for a particle without internal structure in a usual 3D space) is described by f coordinates $\{x_i\} \equiv (x_1, x_2, \ldots, x_f)$ and f (generalised) velocities $(\dot{x}_1, , \dot{x}_2, \ldots, \dot{x}_f)$ or, equivalently to the velocities, momenta $(p_1, p_2, \ldots, p_f)$. Therefore a state of such a classical particle is given by $2f$ numbers in total.

We know that in the quantum world simultaneous *exact* knowledge of particle's coordinates and momenta is impossible due to Heisenberg Uncertainty Principle. Moreover, although we can measure particle's coordinates with an arbitrary accuracy (losing information about momenta in the process), we can predict the results of measurements only *probabilistically* even when we know the state of the particle (e.g., its wavefunction) prior to the measurement. All available probabilistic information about coordinates and momenta of a quantum particle is contained in its wave function, $\Psi(\{x_i\})$, that fully characterises its *quantum state* at a given time, say $t{=}0$. The square of the wave function, $|\Psi(x_1, x_2, \ldots, x_f)|^2$, determines the statistical distribution of random coordinates: e.g., the probability that a particle moving in one dimension ($f = 1$) is found within the interval $[a, b]$ is given by

$$\mathrm{Prob}[a < x < b] = \int_a^b |\Psi(x)|^2 \, \mathrm{d}x \,. \tag{1.1}$$

Similarly, the momenta are also probabilistic quantities in QM. Although we don't know how to infer their distribution from the wave function,[1] some properties, like the average momentum, $\langle p \rangle$, or its variance, $\langle p^2 \rangle - \langle p \rangle^2$, are calculated for a given state using the operator $\hat{p} = -\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}x}$, so that

$$\langle p \rangle = \int_a^b \Psi^*(x) \hat{p} \Psi(x) \, \mathrm{d}x = \int_a^b \Psi^*(x) \left(-\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}x} \right) \Psi(x) \, \mathrm{d}x \tag{1.2}$$

$$\langle p^2 \rangle = \int_a^b \Psi^*(x) \hat{p}^2 \Psi(x) \, \mathrm{d}x = \int_a^b \Psi^*(x) \left(-\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}x} \right)^2 \Psi(x) \, \mathrm{d}x \,. \tag{1.3}$$

The time evolved wavefunction, $\Psi(x, t)$, obeys (as you know) the *linear* time-dependent Schrödinger equation, where the Hamiltonian $\hat{H}$ is the operator of full energy of the system:

$$\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}t} \Psi(x, t) = \hat{H} \Psi(x, t) \,. \tag{1.4}$$

The principal property of this equation is linearity: if $\Psi_1(x, t)$ and $\Psi_2(x, t)$ are its solutions, so it is their linear combination, $\alpha \Psi_1(x, t) + \beta \Psi_2(x, t)$ with any complex coefficients α and β. You know that many fundamental differential equations in physics are also linear, e.g., Maxwell's equations of

[1] we will learn how to do it very soon

electrodynamics, and solutions of the latter are electric and magnetic *vector* fields $\boldsymbol{E}$ and $\boldsymbol{B}$. In general, the linearity is the defining property of a vector space. Hence, it is tempting to say that solutions to the Schrödinger equations are vectors in a certain vector space.

This new definition of quantum states as *vectors* was introduced by Werner Heisenberg (1901-1976) and Paul Dirac (1902-1984). Familiar vectors in three-dimensional space are given by a set of three components, real (velocity, acceleration) or complex (electric and magnetic fields). Although vectors are geometric invariants, their components depend on the choice of coordinates (reference frame). Wavefunctions $\Psi(x)$ in quantum mechanics can be associated with state vectors which, in general, have *infinitely many complex components*: we shall see that values of $\Psi(x)$ at different points could be considered as components of the state vector in an infinitely dimensional vector space. Such a vector space that exists for any particular Hamiltonian $\hat{H}$ is called **Hilbert space**,[2] provided that the inner product of state vectors is defined.

1.2 Dirac notations

We know that vectors appear under different guises in maths and physics. The most simple but fundamental way to denote a vector is to draw an arrow. This gives a nice representation of physical quantities like displacements or velocities which have both magnitude (represented by the length of the arrow) and direction. It is also useful to give a name to such a vector, like $\vec{v}$ or $\boldsymbol{v}$.

To manipulate vectors, for example add and subtract them one can proceed in two ways. One is to use geometry by combining arrows as shown in Figure 1, or to use their components to perform calculations. In the latter case the sum of two vectors $\vec{a} = (a_1, a_2, a_3)$, $\vec{b} = (b_1, b_2, b_3)$ is $\vec{a} + \vec{b} = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$.

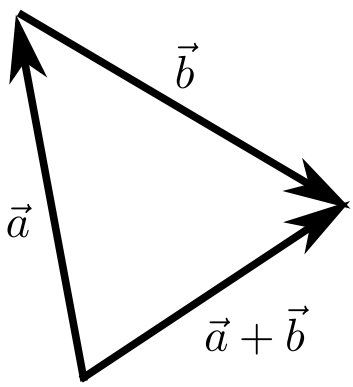


Figure 1: Vectors $\vec{a}$, $\vec{b}$ and their sum.

Both these two methods have their advantages and disadvantages. Manipulating arrows is very visual and intuitive, however it requires very good graphical skills and can be used only with given vectors. Using components allows perform general calculations when vectors are functions of time or other parameters. However, one should remember that, while a vector is geometric invariant, its components depend on the choice of coordinates (basis): the *same* vector will have *different* components if a different set of coordinates x, y, z (basis vectors) is chosen. The ability to use different bases is hugely advantageous for solving physical problems, including quantum mechanical ones as we shall see. (Note in passing that using different bases – frames – is one of the milestones of the Theory of Relativity, including Electrodynamics).

While many, if not all, physical calculations require choosing a particular basis at some stage,

[2]The mathematics of Hilbert space is a standard material and can be found in any decent book on linear algebra. A very good and concise exposition of it can be found in Chaper 1 of Shankar's book and I encourage everyone to use it as a reference for mathematical apparatus of Quantum Mechanics. It also provides a motivation to **Dirac notations** for vectors in Hilbert space. However, the minimal necessary information is given here.

the geometric method, which is completely independent of the choice of basis, is extremely useful for manipulations with vectors that can be added, subtracted, or scalar-multiplied; if needed, their components can be obtained at any stage by choosing an appropriate basis and projecting the vectors along the corresponding axes.

Such a basis-independent geometric approach to usual three-dimensional vectors has motivated Dirac notations for vectors in the Hilbert space. Of course it is impossible to draw arrows in infinite dimensional space (even for three dimensions it is difficult as paper has only two). However one can still introduce convenient notations, superseding those like $\vec{a}$ or $\boldsymbol{a}$ for usual three- (or two-) dimensional vectors in real spaces. The Dirac notations for a vector in Hilbert space became paramount:

$$\big|\Psi\big\rangle \tag{1.5}$$

pronounced "ket psi", uses (not necessarily, but often) Greek letters placed between vertical bar and the right angle brac**ket**. Despite using the same Greek letter Ψ as for the wave function in the beginning of this lecture the state vector $\big|\Psi\big\rangle$ and wave function $\Psi(x)$ are related but different objects: there is no reference to coordinate x in the former, while the latter is a representation of the state in the coordinate basis.

To summarise, we formulate the first QM postulate:

Postulate I. The state of the quantum particle is represented by vector $\big|\Psi\big\rangle$ in a Hilbert Space.

1.3 Properties of vectors in Hilbert space

Ket-vectors (or simply kets) in Hilbert space can be added together and multiplied by *complex* numbers. The result is also a vector. This constitutes the property of

Linearity: if $\big|\Psi\big\rangle$ and $\big|\Phi\big\rangle$ are vectors in a Hilbert space, then for any complex numbers $\alpha, \beta \in \mathbb{C}$ the linear combination $\alpha\big|\Psi\big\rangle + \beta\big|\Phi\big\rangle$ is also a vector in the same Hilbert space.

In particular $\big|\Psi\big\rangle - \big|\Psi\big\rangle = 0\big|\psi\big\rangle = \big|0\big\rangle$ is a unique zero ket-vector and often we will write simply $\big|\Psi\big\rangle - \big|\Psi\big\rangle = 0$.

Linear independence: Vectors $\big|\Psi_1\big\rangle$, $\big|\Psi_1\big\rangle$, ..., $\big|\Psi_j\big\rangle$, ... are linearly independent if the condition

$$\alpha_1\big|\Psi_1\big\rangle + \alpha_2\big|\Psi_2\big\rangle + \ldots + \alpha_k\big|\Psi_k\big\rangle + \ldots = \sum_{j=1}^{\infty} \alpha_j\big|\Psi_j\big\rangle = 0 \tag{1.6}$$

implies *all* the coefficients in this linear combination are zero, $\alpha_j = 0$.

A set $\big|\eta_n\big\rangle$, $n = 1, 2, \ldots$ of linearly independent vectors can be used as a *basis* so that any vector in the Hilbert space can be written as a linear combination,

$$\big|\Psi\big\rangle = \alpha_1\big|\eta_1\big\rangle + \alpha_2\big|\eta_2\big\rangle + \alpha_2\big|\eta_2\big\rangle + \ldots = \sum_{j=1}^{\infty} \alpha_j\big|\eta_j\big\rangle\,. \tag{1.7}$$

One can say that state $\big|\Psi\big\rangle$ is a quantum superposition of states $\big\{\big|\eta_j\big\rangle\big\}$.

In order to be able to represent *any* vector in this way, the basis must be *complete*.[3] The criterion of completeness of a basis will be given later.

The complex numbers α_1, α_2,... are called *components* of vector $\big|\Psi\big\rangle$ in the basis $\big\{\big|\eta_j\big\rangle\big\}$.

Show that the expansion (1.7) is unique in a given basis.

1.4 Scalar product in Hilbert Space. Bra and ket vectors.

Any ket-vector can be represented as a linear combination (1.7). The complex numbers α_n are components of $\big|\Psi\big\rangle$ in the basis $\big|\eta_n\big\rangle$. In standard three-dimensional space spanned by ***orthonormal*** (*i.e.*, all of unit length and orthogonal to each other) basis vectors $\hat{e}_1, \hat{e}_2, \hat{e}_3$ we find the components of a vector $\vec{a} = a_1\hat{e}_1 + a_2\hat{e}_2 + a_3\hat{e}_3$ by taking scalar products: $a_j = \hat{e}_j \cdot \vec{a}$. A similar procedure is implemented in Hilbert space, where we need to define the inner (a.k.a. scalar) product, of two vectors denoted by

$$\langle \Psi \mid \Phi \rangle$$

(pronounced "bra-ket psi phi").

Unlike the case of real vectors, *the order of arguments in the inner product does matter*, and the scalar $\langle \Psi \mid \Phi \rangle$ is, in general, a complex number. The first property (axiom) of the inner product is

Skew-symmetry: $\langle \Psi \mid \Phi \rangle = \langle \Phi \mid \Psi \rangle^*$.

This ensures that "square length" of a vector, $\langle \Psi \mid \Psi \rangle$, is a real number. For the QM applications we demand this number to be non-negative,[4] so that the following property holds

Positive semidefiniteness: $\langle \Psi \mid \Psi \rangle \geq 0$,

where zero is obtained only for zero vector. The third property is

Linearity in ket: $\big\langle \Psi \big| \Big(\alpha \big| \Phi_1 \big\rangle + \beta \big| \Phi_2 \big\rangle \Big) \equiv \langle \Psi \mid \alpha\Phi_1 + \beta\Phi_2 \rangle = \alpha \langle \Psi \mid \Phi_1 \rangle + \beta \langle \Psi \mid \Phi_2 \rangle$.

for any two complex numbers α, β.

The inner product allows us to define *orthogonality* of vectors: $\big|\Psi\big\rangle$ and $\big|\Phi\big\rangle$ are orthogonal if their inner product vanishes. We can also define the *norm* or length of the vector $\big|\Psi\big\rangle$ by taking square root of its inner product with itself, $\|\Psi\| \equiv \sqrt{\langle \Psi \mid \Psi \rangle}$.

The inner product makes the new object, $\big\langle \Psi \big|$ called "bra-vector" or simply "bra", to appear. One can think about it as of a different type of vectors, dual to ket-vectors. In the more formal mathematical language bras are linear functionals, *i.e.* objects which act on ket vectors to produce a number with the help of the inner product we defined. The convenience is in the fact that we can deal with $\langle \Phi \mid \psi \rangle$ as, indeed, with a product where both the left and the right part are separate objects.

[3] As an example of an incomplete basis consider the following situation in the regular three-dimensional space: take two vectors lying in the xy plane. They do not form a complete basis in three dimensions because no linear combination of these two vectors can represent a vector not parallel to this plane (*i.e.* having a non-zero z component).

[4] This is not a general property of the inner product in a vector space. E.g., in Special Relativity the "square length" of a vector is always real but could be negative or zero.

It is important to distinguish bras from kets. They are related by the fact that both $\big|\Psi\big\rangle$ and $\big\langle\Psi\big|$ describe the same quantum state; however, they live in different Hilbert spaces and cannot appear together in linear combinations, *i.e.* expressions like $\big|\Psi\big\rangle + \big\langle\Phi\big|$ make no sense!

It follows from skew-symmetry that

$$\begin{aligned}\langle\alpha\Phi_1 + \beta\Phi_2 \,|\, \Psi\rangle &= \langle\Psi \,|\, \alpha\Phi_1 + \beta\Phi_2\rangle^* \\ &= \alpha^*\langle\Psi \,|\, \Phi_1\rangle^* + \beta^*\langle\Psi \,|\, \Phi_2\rangle^* \\ &= \alpha^*\langle\Phi_1 \,|\, \Psi\rangle + \beta^*\langle\Phi_2 \,|\, \Psi\rangle\,.\end{aligned} \tag{1.8}$$

This property is called *antilinearity in bra*. It can be seen as a rule to produce a bra-vector:

$$\big\langle\alpha\Phi_1 + \beta\Phi_2\big| = \alpha^*\big\langle\Phi_1\big| + \beta^*\big\langle\Phi_2\big| \tag{1.9}$$

from bras $\big\langle\Phi_1\big|$ and $\big\langle\Phi_2\big|$. One may ask: "What are they?", namely given a ket-vector $\big|\Psi\big\rangle$ how to make the corresponding[5] bra vector $\big\langle\Psi\big|$?

The simplest way to answer this question is to use some basis, say $\big|n\big\rangle$, ($n = 1, 2, \ldots$) in the Hilbert space of ket-vectors. We will often omit letters for basis vectors. From now on we always assume that the basis is ***orthonormal***

$$\langle n \,|\, m\rangle = \delta_{nm} = \begin{cases} 1, & n = m \\ 0, & n \neq m \end{cases}\,. \tag{1.10}$$

Using this formula we can associate the bra-vector $\big\langle n\big|$ to each basis ket-vector $\big|n\big\rangle$. Now write a linear expansion of a ket-vector:

$$\big|\Psi\big\rangle = \alpha_1\big|1\big\rangle + \alpha_2\big|2\big\rangle + \ldots = \sum_{m=1}^{\infty}\alpha_m\big|m\big\rangle \tag{1.11}$$

and the corresponding bra-vectors follows from the antilinearity:

$$\big\langle\Psi\big| = \alpha_1^*\big\langle 1\big| + \alpha_2^*\big\langle 2\big| + \ldots = \sum_{m=1}^{\infty}\alpha_m^*\big\langle m\big|\,. \tag{1.12}$$

To extract the component α_n of $\big|\Psi\big\rangle$ in a given basis we take the inner product:

$$\langle n \,|\, \Psi\rangle = \sum_{m=1}^{\infty}\alpha_m\langle n \,|\, m\rangle = \sum_{m=1}^{\infty}\alpha_m\delta_{nm} = \alpha_n\,. \tag{1.13}$$

Given another vector expanded in the same basis,

$$\big|\Phi\big\rangle = \beta_1\big|1\big\rangle + \beta_2\big|2\big\rangle + \ldots = \sum_{m=1}^{\infty}\beta_m\big|m\big\rangle \tag{1.14}$$

we can calculate the inner product,

$$\langle\Psi \,|\, \Phi\rangle = \sum_{m=1}^{\infty}\alpha_m^*\big\langle m\big|\sum_{m'=1}^{\infty}\beta_{m'}\big|m'\big\rangle = \sum_{m=1,m'=1}^{\infty}\alpha_m^*\beta_{m'}\langle m \,|\, m'\rangle \tag{1.15}$$

[5] mathematicians use word "adjoint" or "conjugate"

$$= \sum_{m=1,m'=1}^{\infty} \alpha_m^* \beta_{m'} \delta_{mm'} = \sum_{m=1}^{\infty} \alpha_m^* \beta_m. \tag{1.16}$$

In particular, the vector norm is

$$\langle \Psi \mid \Psi \rangle = \sum_{m=1}^{\infty} \alpha_m^* \alpha_m = \sum_{m=1}^{\infty} |\alpha_m|^2 . \tag{1.17}$$

Let us stress that not only $\left|\Psi\right\rangle$ and $\left\langle\Psi\right|$ but also $\left|\alpha\Psi\right\rangle$ and $\left\langle\alpha\Psi\right|$ represent the same quantum state, albeit different vectors in the appropriate Hilbert space.

To narrow down the representation of quantum state, we introduce the normalised states,

$$\left|\widetilde{\Psi}\right\rangle \equiv \frac{\left|\Psi\right\rangle}{\sqrt{\langle\Psi|\,\Psi\rangle}}. \tag{1.18}$$

This still does not make the representation unique as $\left|\mathrm{e}^{i\phi}\widetilde{\Psi}\right\rangle$ is also normalised for any real number ϕ. Moreover, dealing with normalised states at any intermediate stage of calculations is not convenient since, *e.g.*, the sum of normalised states is not necessarily normalised.

A ket-vector can be defined in terms of its components in a given basis that can be arranged in a form of a matrix column:

$$\left|\Psi\right\rangle \leftrightarrow \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \end{pmatrix} \qquad \left|\Phi\right\rangle \leftrightarrow \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \end{pmatrix} . \tag{1.19a}$$

Here $\leftrightarrow$ means correspondence within a given basis.[6] The corresponding bra-vectors are then obtained by Hermitian conjugation, i.e. by transposing and taking complex conjugate:

$$\left\langle\Psi\right| \leftrightarrow \left(\alpha_1^*, \alpha_2^*, \ldots\right) \qquad \left\langle\Phi\right| \leftrightarrow \left(\beta_1^*, \beta_2^*, \ldots\right) . \tag{1.19b}$$

The inner product becomes the standard matrix multiplication:

$$\langle \Psi \mid \Phi \rangle = \left(\alpha_1^*, \alpha_2^*, \ldots\right) \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \end{pmatrix} = \alpha_1^* \beta_1 + \alpha_2^* \beta_2 + \ldots . \tag{1.20}$$

It is very useful to remember this correspondence between state vectors (bras and kets) and the corresponding row and column vectors, but we should always bear in mind that this correspondence relies on the choice of basis (as usual when one uses components to define a vector). So the matrix expressions in (1.19) give *representations* of the ket or bra state vectors in a given basis. If we have chosen a different basis, $\left|\tilde{n}\right\rangle$ we would end up with different components $\alpha_{\tilde{n}}$,$\beta_{\tilde{n}}$ for the vectors $\left|\Psi\right\rangle$, $\left|\Phi\right\rangle$. Of course the inner product is invariant under such a change, and the inner product in Dirac notations, $\langle \Psi \mid \Phi \rangle$, is truly basis-independent.

[6] We shall formalise this correspondence in Lecture 2.

Lecture 2. Operators in Hilbert space. 2nd and 3rd QM postulates.

Here we introduce operators in Hilbert space acting on ket- and bra-vectors and give both their matrix representation in a particular basis and the Dirac basis-independent representation. This allows us to define self-adjoint (Hermitian) operators representing observables, and unitary operators corresponding to allowed transformations.. As this is a revision of material given in Y2 Maths, I give it here for completeness but will scant over this part in the class, focusing on the QM postulates of measurements.

2.1 Linear operators

Once we have established the description of quantum mechanical states as vectors in Hilbert space we move to the next stage and want to manipulate them in order to be able to describe quantum dynamics. For this we need *operators*, *i.e.* rules to transform vectors into one another. We know examples of operators in regular three-dimensional Euclidean space like rotations. In Hilbert space operators transform state vectors like this:

$$\big|\Psi\big\rangle \xrightarrow{\hat{O}} \big|\Phi\big\rangle\,. \tag{2.1}$$

We will write $\big|\Phi\big\rangle = \hat{O}\big|\Psi\big\rangle = \big|\hat{O}\Psi\big\rangle$ which means that operator $\hat{O}$ acts on the state-vector $\big|\Psi\big\rangle$ and produces Φ. Operators are denoted by small hats, but sometimes these are omitted if there is no confusion.

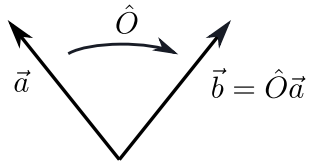


Figure 2: Rotation.

We will only deal with *linear* operators which take linear superpositions into linear superpositions of transformed states:

$$\hat{O}\big|\alpha\Psi_1 + \beta\Psi_2\big\rangle = \alpha\hat{O}\big|\Psi_1\big\rangle + \beta\hat{O}\big|\Psi_2\big\rangle = \alpha\big|\Phi_1\big\rangle + \beta\big|\Phi_2\big\rangle\,, \tag{2.2}$$

for two complex numbers α, β and $\hat{O}\big|\Psi_{1,2}\big\rangle = \big|\Phi_{1,2}\big\rangle$. Operators can also act on bra-vectors and we will return to this point later.

The simplest, but very important example of a linear operator is the identity operator $\hat{I}$, which leaves the state-vector unchanged: $\hat{I}\big|\Psi\big\rangle = \big|\Psi\big\rangle$. We have seen that any orthonormalised complete basis $\{\eta_n\}$ satisfies $\sum_n \big|\eta_n\big\rangle\big\langle\eta_n\big| = \hat{I}$.

Operators can be combined into products:

$$\hat{O}_2\hat{O}_1\big|\Psi\big\rangle = \hat{O}_3\big|\Psi\big\rangle = \hat{O}_2\big|\hat{O}_1\Psi\big\rangle\,, \tag{2.3}$$

so that $\hat{O}_3 = \hat{O}_2\hat{O}_1$, *i.e.* operator $\hat{O}_3$ is the combined result of acting first with $\hat{O}_1$ and then with $\hat{O}_2$. {The order of the operators' action is *from right to left* like in Semitic languages, *i.e.* in the product $\hat{A}\hat{B}$ the operator $\hat{B}$ acts first. The order is important: in general, $\hat{O}_1\hat{O}_2 \neq \hat{O}_2\hat{O}_1$. A good example is two consecutive rotations in three dimensions around different axes.

The difference $\hat{O}_1\hat{O}_2 - \hat{O}_2\hat{O}_1 \equiv [\hat{O}_1, \hat{O}_2]$ is called the **commutator** of $\hat{O}_1$ and $\hat{O}_2$.

One defines the **inverse** operator $\hat{O}^{-1}$ such that $\hat{O}^{-1}\hat{O} = \hat{O}\hat{O}^{-1} = \hat{I}$. This operator reverts the action of $\hat{O}$. The **inverse of a product**

$$\left(\hat{O}_2\hat{O}_1\right)^{-1} = \hat{O}_1^{-1}\hat{O}_2^{-1} \tag{2.4}$$

since $\hat{I} = \left(\hat{O}_2\hat{O}_1\right)^{-1}\left(\hat{O}_2\hat{O}_1\right) = \hat{O}_1^{-1}\hat{O}_2^{-1}\hat{O}_2\hat{O}_1 = \hat{O}_1^{-1}\hat{O}_1$. In simple words one has to undo the action of the last operator that acted on the state.

2.2 Matrix and Dirac representation of operators

To specify an operator we must define how it transforms *any* vector in a Hilbert space. One way of doing this is to choose a complete ***orthonormal*** basis $\{|n\rangle\}$ and define the action of an operator $\hat{B}$ on each basis vector:

$$\hat{B}|n\rangle = |\widetilde{n}\rangle = \sum_j \beta_{jn}|j\rangle\,. \tag{2.5}$$

As any vector can be spanned in basis $\{|n\rangle\}$, we have expanded $|\widetilde{n}\rangle$ above. The expansion coefficients β_{jn} depend on two indices, the first is the summation index and the second indicates on which basis vector we have acted by $\hat{B}$. Now we multiply both sides of (2.5) by the bra-vector $\langle m|$ and use the orthonormality of basis $\{|n\rangle\}$:

$$\langle m\,|\,\hat{B}\,|\,n\rangle = \langle m|\sum_{j=1}^{\infty}\beta_{jn}|j\rangle = \sum_{j=1}^{\infty}\beta_{jn}\underbrace{\langle m\,|\,j\rangle}_{=\delta_{mj}} = \beta_{mn}, \tag{2.6}$$

as the Krönecker symbol δ_{mj} kills all the terms in the sum but that with $j = m$. We call complex numbers $\beta_{mn} = \langle m|\hat{B}|n\rangle$ **matrix elements** of the operator $\hat{B}$ in the basis $|n\rangle$. Obviously, the set $\{\beta_{mn}\}$ can be considered as a matrix, possibly infinitely-dimensional. Moreover, the operator actions can be represented by matrix multiplication.

To see this, let us take the representation of an arbitrary state $|\Psi\rangle$ in the same basis $\{|n\rangle\}$:

$$|\Psi\rangle = \sum_{n=1}^{\infty}\alpha_n|n\rangle \text{ with components } \alpha_n = \langle n\,|\,\Psi\rangle. \tag{2.7}$$

The transformed state $|\Phi\rangle = \hat{B}|\Psi\rangle$ is obtained by acting with the operator on each basis vector in the sum. Using (2.5), we find

$$|\Phi\rangle = \hat{B}\sum_{n=1}^{\infty}\alpha_n|n\rangle = \sum_{n=1}^{\infty}\alpha_n\hat{B}|n\rangle = \sum_{n=1}^{\infty}\alpha_n\sum_{j=1}^{\infty}\beta_{jn}|j\rangle = \sum_{j=1}^{\infty}\left(\sum_{n=1}^{\infty}\beta_{jn}\alpha_n\right)|j\rangle \equiv \sum_j \gamma_j|j\rangle, \tag{2.8}$$

where $\{\gamma_j = \langle j\,|\,\Phi\rangle\}$ is the column-matrix representation of $|\Phi\rangle$. Hence, indeed, we see the equivalence of the operator action to the matrix multiplication in the chosen basis:

$$|\Phi\rangle = \hat{B}|\Psi\rangle \quad\Leftrightarrow\quad \gamma_j = \sum_{n=1}^{\infty}\beta_{jn}\alpha_n \quad\Leftrightarrow\quad \begin{pmatrix}\gamma_1\\ \gamma_2\\ \vdots\end{pmatrix} = \begin{pmatrix}\beta_{11} & \beta_{12} & \dots\\ \beta_{21} & \beta_{22} & \dots\\ \vdots & \vdots & \ddots\end{pmatrix}\begin{pmatrix}\alpha_1\\ \alpha_2\\ \vdots\end{pmatrix}. \tag{2.9}$$

This we have established the correspondence between operators in Hilbert space and matrices. This correspondence relies on a particular choice of basis, albeit it holds in any complete orthonormal basis. Now we shall use Dirac notations for specifying operator actions in a basis-independent way.

Let us return to (2.7) and substitute the expression for α_n into the expansion:

$$\big|\Psi\big\rangle = \sum_{n=1}^{\infty} \alpha_n \big|n\big\rangle = \sum_{n=1}^{\infty} \langle n \,|\, \Psi\rangle \big|n\big\rangle = \sum_{n=1}^{\infty} \big|n\big\rangle\langle n \,|\, \Psi\rangle = \left(\sum_{n=1}^{\infty} \big|n\big\rangle\big\langle n\big|\right)\big|\Psi\big\rangle\,. \tag{2.10}$$

by moving the ket-vectors to the front. In this form it becomes clear that the right hand side is a result of an operator $\sum_i \big|n\big\rangle\big\langle n\big|$ acting on $\big|\Psi\big\rangle$. Since this operator acting on $\big|\Psi\big\rangle$ produces $\big|\Psi\big\rangle$ again, it is nothing but the identity operator:

$$\hat{I} = \sum_n \big|n\big\rangle\big\langle n\big|\,. \tag{2.11}$$

This complicated way to write the operator which does nothing is called the **resolution of unity**[1] and will become one of the most useful tricks in doing calculations with Dirac notations. I deliberately left out the limits in the summation sign: the resolution of unity works as long as basis $\big\{\big|n\big\rangle\big\}$, whether it comprises a finite or infinite number of basis vectors, is complete in a Hilbert space. Actually, this is nothing more than the criterion of completeness:

An orthonormal set, $\big\{\big|n\big\rangle\big\}$, is complete in a Hilbert space iff $\sum_n \big|n\big\rangle\big\langle n\big| = \hat{I}$

Let us use (2.11) to represent an operator $\hat{B}$ in a basis-independent Dirac form:

$$\hat{B} = \hat{I}\hat{B}\hat{I} = \left(\sum_i \big|i\big\rangle\big\langle i\big|\right)\hat{B}\left(\sum_j \big|j\big\rangle\big\langle j\big|\right) = \sum_{ij} \big|i\big\rangle\beta_{ij}\big\langle j\big|. \tag{2.12}$$

On the face of it, this expression still depends on the basis, but this is not the case: if the basis changed the matrix elements β_{ij} would change as well while the operator remains the same. We will come to this point later when we will discuss basis transformations.

Naturally, the Dirac representation of (2.12) is consistent with the matrix representation of (2.6). Indeed,

$$\big\langle m \,\big|\, \hat{B} \,\big|\, n\big\rangle = \big\langle m \,\big|\left(\sum_{ij} \big|i\big\rangle\beta_{ij}\big\langle j\big|\right)\big|\, n\big\rangle = \sum_{ij} \underbrace{\langle m \,|\, i\rangle}_{=\delta_{mi}} \beta_{ij} \underbrace{\langle j \,|\, n\rangle}_{=\delta_{jn}} = \beta_{mn}.$$

Remember: in Dirac notations a bra-ket $\langle\Phi_1 \,|\, \Phi_2\rangle$ is a number (scalar product), while a ket-bra $\big|\Phi_2\big\rangle\big\langle\Phi_1\big|$ *is an operator*: if it acts on a ket-state $\big|\Psi\big\rangle$ from the left, then the bra $\big\langle\Phi_1\big|$ with the the ket $\big|\Psi\big\rangle$ will produce a number $\alpha = \langle\Phi_1 \,|\, \Psi\rangle$ and the result will be this number times $\big|\Phi_2\big\rangle$:

$$\big|\Psi\big\rangle \longrightarrow \big|\Phi_2\big\rangle\langle\Phi_1 \,|\, \Psi\rangle = \alpha\big|\Phi_2\big\rangle\,. \tag{2.13}$$

[1] a.k.a. the **resolution of identity**

Of course, a sum of such terms is also an operator. Another way to see this is to use the correspondence of kets with column vectors, $\big|\Phi_1\big\rangle \mapsto \{\alpha_n\}^{\mathrm{T}}$ and bras with row vectors, $\big\langle\Phi_2\big| \mapsto \{\beta_n^*\}$ in the same basis. The standard matrix multiplication of column vector (on the left) and row vector (on the right) is a matrix representation of the operator $\big|\Phi_2\big\rangle\big\langle\Phi_1\big|$:

$$\big|\Phi_2\big\rangle\big\langle\Phi_1\big| \mapsto \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \end{pmatrix} \big(\beta_1^*, \beta_2^*, \ldots\big) = \begin{pmatrix} \alpha_1\beta_1^* & \alpha_1\beta_2^* & \ldots \\ \alpha_2\beta_1^* & \alpha_2\beta_2^* & \ldots \\ \vdots & \vdots & \ddots \end{pmatrix} . \tag{2.14}$$

A special case is when we use the ***eigenset*** of operator $\hat{B}$ as a basis, provided that it is complete. The eigenset is a set of eigenvectors, $\big\{\big|\beta_n\big\rangle\big\}$ of $\hat{B}$, that satisfy the eigenvalue equation (EVE):

$$\hat{B}\big|\beta_n\big\rangle = \beta_n\big|\beta_n\big\rangle \tag{2.15}$$

(we use here a more specific notation, $\big|\beta_n\big\rangle$ instead of generic $\big|n\big\rangle$). Completeness means, as usual, that $\sum_n \big|\beta_n\big\rangle\big\langle\beta_n\big| = \hat{I}$. Then we have

$$\hat{B} = \hat{B}\hat{I} = \sum_n \hat{B}\big|\beta_n\big\rangle\big\langle\beta_n\big| = \sum_n \beta_n\big|\beta_n\big\rangle\big\langle\beta_n\big|. \tag{2.16}$$

In particular, from the most famous EVE in physics, the time-independent Schrödinger equation, follows the Dirac representation of the Hamiltonian in its own eigenvalue basis, which is the eigenenergy basis:

$$\hat{H}\big|\varepsilon_n\big\rangle = \varepsilon_n\big|\varepsilon_n\big\rangle \qquad \Rightarrow \qquad \hat{H} = \sum_n \varepsilon_n\big|\varepsilon_n\big\rangle\big\langle\varepsilon_n\big|. \tag{2.17}$$

2.3 Adjoint and self-adjoint (Hermitian) operators

In the previous lecture we introduced quantum mechanical operators acting on vectors in Hilbert space. We have only considered their action on ket-vectors resulting in another ket-vector:

$$\big|\Phi\big\rangle = \hat{O}\big|\Psi\big\rangle = \big|\hat{O}\Psi\big\rangle\,. \tag{2.18}$$

Operators can also act on bra-vectors transforming them into another bra-vector:

$$\big\langle\tilde{\Phi}\big| = \big\langle\Psi\big|\hat{O}\,. \tag{2.19}$$

I have deliberately used tilde in $\big\langle\tilde{\Phi}\big|$ to distinguish the resulting bra-vector from $\big\langle\Phi\big|$ corresponding to the ket-vector $\big|\Phi\big\rangle$. For a general operator $\hat{O}$, bra-vectors $\big\langle\Phi\big|$ and $\big\langle\tilde{\Phi}\big|$ are different. To demonstrate this, consider operator $\hat{O} = \big|X\big\rangle\big\langle Y\big|$ acting on two different vectors $\big|X\big\rangle$ and $\big|Y\big\rangle$. We have

$$\big|\Phi\big\rangle = \big|X\big\rangle\langle Y \mid \Psi\rangle\,, \qquad \big\langle\tilde{\Phi}\big| = \langle\Psi \mid X\rangle\big\langle Y\big|\,. \tag{2.20}$$

The scalar products are just numbers so that $\big|\tilde{\Phi}\big\rangle$ is proportional to $\big|Y\big\rangle$ which is not equal to $\big|\Phi\big\rangle$, proportional to $\big|X\big\rangle$.

If we want to get the bra-vector $\langle\Phi|$ corresponding to $|\Phi\rangle = |\hat{O}\Psi\rangle$ we must act on the bra-vector $\langle\Psi|$ with the *adjoint operator*

$$\langle\Phi| = \langle\Psi|\hat{O}^\dagger = \langle\hat{O}\Psi|\,. \tag{2.21}$$

The "dagger" symbol in $\hat{O}^\dagger$ means Hermitian conjugation of the operator. In most cases, adjoint and Hermitian-conjugate operator means the same.[2]

To make the reason why $\hat{O}^\dagger$ is Hermitian conjugate of $\hat{O}$, let us calculate the matrix elements of $\hat{O}^\dagger$ in a basis $\{|i\rangle\}$, using the skew-symmetry of scalar product, $\langle\Psi\,|\,\Phi\rangle = \langle\Phi\,|\,\Psi\rangle^*$:

$$\hat{O}^\dagger_{ij} \equiv \langle i|\hat{O}^\dagger|j\rangle = \langle\hat{O}i\,|\,j\rangle = \langle j\,|\,\hat{O}i\rangle^* = \langle j|\hat{O}|i\rangle^* = O^*_{ji}\,, \tag{2.22}$$

This relation shows that the matrix corresponding to the adjoint operator $\hat{O}^\dagger$ is the Hermitian conjugate of the matrix corresponding to $\hat{O}$, *i.e.* is obtained by transposing O_{ij} and taking the complex conjugate of its elements:

$$\hat{O} \longleftrightarrow \begin{bmatrix} O_{11} & O_{12} & \dots \\ O_{21} & O_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}, \qquad \hat{O}^\dagger \longleftrightarrow \begin{bmatrix} O^*_{11} & O^*_{21} & \dots \\ O^*_{12} & O^*_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}. \tag{2.23}$$

The adjoint of a product of two operators is the product of their adjoints in reversed order:

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger\,. \tag{2.24}$$

Let us prove it. Take a ket-vector $|\Psi\rangle$ and act on it with the product $\hat{A}\hat{B}$. The obtained ket-vector $|\hat{A}\hat{B}\Psi\rangle = \hat{A}\hat{B}|\Psi\rangle$ is converted to the corresponding bra-vector and we write

$$\langle\hat{A}\hat{B}\Psi| = \langle\Psi|(\hat{A}\hat{B})^\dagger\,, \tag{2.25}$$

treating $\hat{A}\hat{B}$ as one operator. Now we do this step by step. First we extract $\hat{A}^\dagger$,

$$\langle\hat{A}\hat{B}\Psi| = \langle\hat{B}\Psi|\hat{A}^\dagger\,, \tag{2.26}$$

and use the fact that $\langle\hat{B}\Psi| = \langle\Psi|\hat{B}^\dagger$ so that

$$\langle\hat{A}\hat{B}\Psi| = \langle\Psi|\hat{B}^\dagger\hat{A}^\dagger \tag{2.27}$$

Since Ψ is arbitrary this proves our statement.

The operators which coincide with their adjoints are called **self-adjoint** or **Hermitian**: $\hat{H} = \hat{H}^\dagger$.

They play a major role in quantum mechanics as their eigenvalues are real[3]. Indeed, their *expectation* value (average) in a state $|\Psi\rangle$ is

$$\langle\hat{O}\rangle \equiv \langle\Psi|\hat{O}|\Psi\rangle = \langle\Psi\,|\,\hat{O}\Psi\rangle = \langle\hat{O}\Psi\,|\,\Psi\rangle^* = \langle\Psi|\hat{O}^\dagger|\Psi\rangle^* = \langle\Psi|\hat{O}|\Psi\rangle^*\,, \tag{2.28}$$

[2]We will not consider situations when they are not exactly the same.

[3]Similarly we define **anti-Hermitian** operators, $\hat{A}^\dagger = -\hat{A}$, whose eigenvalues happen to be pure imaginary numbers

which is real. From this follows that their *eigenvalues* λ are all real:

$$\hat{O}\big|\lambda_i\big\rangle = \lambda_i\big|\lambda_i\big\rangle \quad \Rightarrow \quad \big\langle\lambda_i\big|\hat{O}\big|\lambda_i\big\rangle = \lambda_i\langle\lambda_i \mid \lambda_i\rangle\,. \tag{2.29}$$

The left-hand side is real as well as the scalar product in the right-hand side. Therefore, all λ_i are real. Hence, they are suitable to represent physically measurable quantities – *observables*. Of course, there are non-Hermitian operators that have only real eigenvalues. However, if we demand that the solutions of the EVE for operator $\hat{O}$ with real eigenvalues are orthonormalised, then its Dirac representation is

$$\hat{O} = \sum_n \lambda_n\big|\lambda_n\big\rangle\big\langle\lambda_n\big| \quad \Rightarrow \quad \hat{O}^\dagger = \sum_n \lambda_n^*\big|\lambda_n\big\rangle\big\langle\lambda_n\big| = \sum_n \lambda_n\big|\lambda_n\big\rangle\big\langle\lambda_n\big| \quad \Rightarrow \quad \hat{O}^\dagger = \hat{O}^\dagger$$

There exist non-Hermitian operators with real eigenvalues. Indeed, making a similarity transformation of $\hat{H}_\mathrm{d} = \mathrm{diag}\,(\lambda_1 \cdots \lambda_n)$, we find $\hat{H} = \hat{P}\hat{H}_\mathrm{d}\hat{P}^{-1}$ to be, in general, non-Hermitian. To see why the Hermitian operators play a special role in QM, we need also to see why the unitary operators are equally important.

2.4 Unitary operators

Another type of operators related to their adjoints are **unitary** operators defined by

$$\hat{U}^\dagger = \hat{U}^{-1} \qquad \text{or} \qquad \hat{U}^\dagger\hat{U} = \hat{U}\hat{U}^\dagger = \hat{I}\,. \tag{2.30}$$

They play a role similar to that of complex numbers of unit modulus, $u = e^{\mathrm{i}\theta}$. These operators preserve scalar product of vectors they act on: let $\big|\Phi\big\rangle_1 = \hat{U}\big|\Psi\big\rangle_1$ and $\big|\Phi\big\rangle_2 = \hat{U}\big|\Psi\big\rangle_2$, then

$$\langle\Phi_1 \mid \Phi_2\rangle = \big\langle\Psi_1\big|\hat{U}^\dagger\hat{U}\big|\Psi_2\big\rangle = \big\langle\Psi_1\big|\hat{I}\big|\Psi_2\big\rangle = \langle\Psi_1 \mid \Psi_2\rangle\,. \tag{2.31}$$

You might remember from Maths that any unitary operator, $\hat{U}$, can be represented as an exponent of some Hermitian operator, $\hat{h}$:

$$\hat{U} \equiv \mathrm{e}^{i\hat{h}} \quad \Rightarrow \quad \hat{U}^\dagger = \mathrm{e}^{-i\hat{h}^\dagger} = \mathrm{e}^{-i\hat{h}} = \hat{U}^{-1}. \tag{2.32}$$

All basis transformations in QM must be unitary to preserve normalisation. If we unitary transform $\hat{H}_\mathrm{d} = \mathrm{diag}\,(\lambda_1 \cdots \lambda_n)$ with real λ_n, we find $\hat{H} = \hat{U}\hat{H}_\mathrm{d}\hat{U}^\dagger$ to be Hermitian:

$$\hat{H}^\dagger = \Big(\hat{U}\hat{H}_\mathrm{d}\hat{U}^\dagger\Big)^\dagger = \hat{U}\hat{H}_\mathrm{d}^*\hat{U}^\dagger = \hat{H}$$

since $\hat{H}_\mathrm{d}^\dagger = \hat{H}_\mathrm{d}^* = \hat{H}_\mathrm{d}$ as all the eigenvalue are real.

The requirement of all the eigenvalues being real and all the basis transformation being orthogonal leads to the formulation of the next two QM postulates.

2.5 Quantum Measurements and the 2$^{\rm nd}$ and 3$^{\rm rd}$ QM Postulates

QM Postulate II. Any observable is described by a corresponding Hermitian operator; measurable values of this observable are eigenvalues of this operator.

This is the second of the four postulates[4] that lie in the foundation of QM. Naturally, a postulate is not provable but something taken for granted.

QM got its name since measurements of many (albeit not all) microscopic observables return well defined ***quantised*** values. The discreteness of atomic optical spectra, in particular of hydrogen, could not be explained classically and prompted a revolutionary new explanation by Niels Bohr, which is now known as semiclassical. This was great but not sufficient, and entirely new quantum mechanics emerged ten years later in seminal works of Erwin Schrödinger – wave mechanics, and Werner Heisenberg – matrix mechanics. Both these approaches explained known "quantum" experiments equally well but seemed mutually contradictory competing theories until Heisenberg and Dirac proved their full equivalence.

While everybody (qualified to judge) is happy with the 2$^{\rm nd}$ postulate, the next one was and remains to be most controversial part of the QM.

QM Postulate III. A measurement of observable $\hat{A}$ in a QM system in state $\big|\Psi\big\rangle$ yields one of its eigenvalues, A_n, with the probability $P_n = |\langle A_n \,|\, \Psi\rangle|^2$. As a result of the measurement, the state will instantaneously collapse from $\big|\Psi\big\rangle$ into $\big|A_n\big\rangle$.

Here an observable $\hat{A}$ (which could be energy, momentum, position, angular momentum, *etc*) obeys the EVE $\hat{A}\big|A_n\big\rangle = A_n\big|A_n\big\rangle$, while state $\big|\Psi\big\rangle$ is assumed to be normalised and the eigenset $\big\{\big|A_n\big\rangle\big\}$ is orthonormal, $\langle A_n \,|\, A_m\rangle = \delta_{nm}$, and complete, $\sum_n \big|A_n\big\rangle\big\langle A_n\big| = \hat{I}$. This means that any $\big|\Psi\big\rangle$ can be represented as a ***quantum superposition*** of the eigenstates $\big\{\big|A_n\big\rangle\big\}$

$$\big|\Psi\big\rangle = \hat{I}\big|\Psi\big\rangle = \Big(\sum_n \big|A_n\big\rangle\big\langle A_n\big|\Big)\big|\Psi\big\rangle = \sum_n \langle A_n \,|\, \Psi\rangle\big|A_n\big\rangle \tag{2.33}$$

The formulation given here is known as the Copenhagen interpretation of QM (wholeheartedly accepted by the ***qualified*** majority[5]). The collapse of state $\big|\Psi\big\rangle$ means that the measurement of $\hat{A}$ that gives an eigenvalue A_n ***projects*** $\big|\Psi\big\rangle$ into the corresponding eigenstate $\big|A_n\big\rangle$ and implies that any subsequent measurement of $\hat{A}$ performed *immediately after the first one* will result in the same particular value A_n as in the first measurement.[6] Equation (2.33) ensures $\sum_n P_n = \sum_n |\langle A_n \,|\, \Psi\rangle|^2 = 1$, and the probabilities defined in the 3$^{\rm rd}$ Postulate give the *expectation* (average) value of the observable in the standard QM way:

$$\langle A\rangle = \sum_n A_n P_n = \sum_n A_n |\langle A_n \,|\, \Psi\rangle|^2 = \sum_n \langle \Psi \,|\, A_n\rangle A_n \langle A_n \,|\, \Psi\rangle = \big\langle\Psi\big|\hat{A}\big|\Psi\big\rangle. \tag{2.34}$$

In the next lecture, we discuss why this postulate is so controversial and look in more detail on the simplest example of quantum measurements following the 3$^{\rm rd}$ postulate.

[4] In some books, postulates are grouped as 5 or 6 statements, but their contents is the same.

[5] The minority that never accepted this interpretation included Einstein and Schrödinger himself.

[6] The "immediately after" clause is important as the collapsed state may change as a result of evolution which we will discuss soon.

Lecture 3. Two-level systems. Quantum measurements.

In this lecture, we consider consequences of the 2$^{\rm nd}$ and 3$^{\rm rd}$ postulates and start discussing inevitable controversies around the probabilistic nature of Quantum Mechanics.

A full physical interpretation of Quantum Mechanics emerged as a result of intensive discussions between Bohr and Heisenberg and is now known as the ***Copenhagen interpretation***, accepted by the overwhelming majority of physicists.[1] The focus of this interpretation is the intrinsically probabilistic character of QM measurements. When more than one outcome of a measurement may occur, it is impossible to predict it with certainty. According to the QMP3, only the probability of the outcome could be given (equal to $|\langle\Psi\,|\,A_n\rangle|^2$ for measuring the eigenvalue A_n of $\hat{A}$ in state $\big|\Psi\big\rangle$) – that is the fraction of its appearance after repeating the measurement multiple times.

A random result is unpleasant from classical viewpoint (*God does not play dice* according to Einstein) and controversy related to such a statement was made quite apparent by Schrödinger, offering the eponymous cat as a counter-example. The *Schrödinger's cat* appeared as a reaction to the equally famous objection to QM (or to its Copenhagen interpretation) in the paper by Einstein, Podolsky, and Rosen which became known as the *EPR "paradox"*. Before going to consider these objections and their resolution, let us consider in more detail how the QM postulates work using the simplest possible QM system.

3.1 Two-level system.

The two-level system (TLS), which is now much better known under the name *qubit*, is generated by an operator, say $\hat{\sigma}$, that has only two non-degenerate eigenvalues, equal ± 1 in proper units (or spin up-down for a real spin projection equal $\pm\frac{1}{2}\hbar$ or any "pseudo-spin" which is a possible way of mapping a TLS onto). The corresponding Hilbert space is two-dimensional, and the EVE for $\hat{\sigma}$ and its Dirac representation can be written as

$$\begin{aligned} &\hat{\sigma}\big|\sigma\big\rangle = \sigma\big|\sigma\big\rangle, \quad \text{with } \sigma = \pm 1 \quad \Longleftrightarrow \quad \hat{\sigma}\big|\pm\big\rangle = \pm\big|\pm\big\rangle \\ \Rightarrow \quad &\hat{\sigma} = \sum_{\sigma=\pm 1} \big|\sigma\big\rangle\sigma\big\langle\sigma\big| = \big|+\big\rangle\big\langle+\big| - \big|-\big\rangle\big\langle-\big| \equiv \big|\uparrow\big\rangle\big\langle\uparrow\big| - \big|\downarrow\big\rangle\big\langle\downarrow\big| \equiv \big|1\big\rangle\big\langle 1\big| - \big|0\big\rangle\big\langle 0\big|. \end{aligned} \tag{3.1}$$

Here we use the flexibility of Dirac's notations: two eigenstates could be denoted in an arbitrary manner, so that we have chosen $\big|1\big\rangle \equiv \big|\uparrow\big\rangle \equiv \big|+\big\rangle$ and $\big|0\big\rangle \equiv \big|-\big\rangle \equiv \big|\downarrow\big\rangle$.

A spin $\frac{1}{2}$ particle in a magnetic field (traditionally, along the z-axis) is the simplest realisation of TLS , about which you have learned a bit in QM2. This justifies notations: $\big|+\big\rangle \equiv \big|\uparrow\big\rangle$ or $\big|-\big\rangle \equiv \big|\downarrow\big\rangle$ are the states with the spin projection up or down, *i.e.* in the positive or negative direction along the

[1] There is no democracy in science, and a truthfulness – or otherwise – of a physical theory is not decided by the majority vote. Objections of a *qualified* minority (rather than people in the street) could be very important and relevant. The whole modern building of QM has emerged notwithstanding (but rather due to) the objections of the minority that never accepted the Copenhagen interpretation. This minority included, amongst others, Einstein, Schrödinger, Louis de Broglie – all are the Nobel Prize winners with tremendous contributions to QM.

z-axis. The dimensionality of spin = that of the angular momentum is $[p \times r]$ which is the same as that of $\hbar$, and the spin-projection operator $\hat{S}_z$ is related to $\hat{\sigma}$ by $\hat{S}_z = \dfrac{\hbar}{2}\hat{\sigma}$.

The matrix representation of the operator $\hat{\sigma}$ in the 2-dimensional Hilbert space is given by

$$\langle \sigma \,|\, \hat{\sigma} \,|\, \sigma' \rangle = \begin{pmatrix} \langle\uparrow| \left(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|\right) |\uparrow\rangle & \langle\uparrow| \left(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|\right) |\downarrow\rangle \\ \langle\downarrow| \left(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|\right) |\uparrow\rangle & \langle\downarrow| \left(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|\right) |\downarrow\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{3.2}$$

where you should recognise (from QM2 and/or Maths2B) Pauli's σ_z matrix. The simplification of the matrix element are based on the orthonormality, *i.e.* $\langle\uparrow|\downarrow\rangle = \langle\downarrow|\uparrow\rangle = 0$ and $\langle\uparrow|\uparrow\rangle = \langle\downarrow|\downarrow\rangle = 1$. Its eigenstates, $|\sigma\rangle = |\pm\rangle$, in matrix representation are

$$|+\rangle \equiv |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad |-\rangle \equiv |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.3}$$

Any state of a TLS (qubit, spin $\frac{1}{2}$, *etc*) is a quantum superposition of its eigenstates, $|\Psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$. The requirement for $|\Psi\rangle$ to be normalised gives

$$1 = \langle\Psi\,|\,\Psi\rangle = |\alpha|^2 \underbrace{\langle\uparrow|\uparrow\rangle}_{=1} + |\beta|^2 \underbrace{\langle\downarrow|\downarrow\rangle}_{=1} + \alpha^*\beta \underbrace{\langle\uparrow|\downarrow\rangle}_{=0} + \alpha\beta^* \underbrace{\langle\downarrow|\uparrow\rangle}_{=0} = |\alpha|^2 + |\beta|^2 ,$$

from which follows that α and β could be chosen as sine and cosine of some angle with an arbitrary phase factor. We shall learn (much later in the course) that for spin $\frac{1}{2}$ this angle must be chosen as $\theta/2$ when projections measured along a straight line at angle θ to the z-axis,[2] so that the quantum superposition $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$ could be written as

$$|\Psi\rangle = \mathrm{e}^{i\phi}\cos(\theta/2)|\uparrow\rangle + \mathrm{e}^{i\varphi}\sin(\theta/2)|\downarrow\rangle , \tag{3.4}$$

where phase angles ϕ and φ are arbitrary. According to the 3rd QMP, a measurement can only return

$$|\uparrow\rangle \text{ with the probability } \cos^2(\theta/2) \qquad \text{or} \qquad |\downarrow\rangle, \text{ with the probability } \sin^2(\theta/2) \tag{3.5}$$

while the particle is in the superposition (3.4), *i.e.* simultaneously in states $|\uparrow\rangle$ and $|\downarrow\rangle$ prior to the measurement. Naturally, there is nothing special about the z-axis, and the same probabilities are for the measurement at a given angle θ to any direction.

Finally, the Hamiltonian of any TLS can be written in its own (energy) basis as

$$\hat{H} = \sum_{n=0,1} \varepsilon_n |n\rangle\langle n| = \varepsilon_0|0\rangle\langle 0| + \varepsilon_1|1\rangle\langle 1|.$$

The standard EVEs immediately follow from this notation. E.g., the EVE for ε_0:

$$\hat{H}|0\rangle = \varepsilon_0|0\rangle\langle 0|\,|0\rangle + \varepsilon_1|1\rangle\langle 1|\,|0\rangle \equiv \varepsilon_0\,|0\rangle \underbrace{\langle 0|\,0\rangle}_{=1} + \varepsilon_1\,|0\rangle \underbrace{\langle 1\,|\,0\rangle}_{=0} = \varepsilon_0\,|0\rangle \ .$$

[2] If we measure an orbital angular momentum, the angle in superposition (3.4) would be θ: the reason for an additional $\frac{1}{2}$ is that spin is not transformed as a vector; one need a rotation through 4π rather than 2π to return to the same spin state, as we shall see later in the course.

Example 3.1. *The TLS is described by the Hamiltonian* $\hat{H} = -\varepsilon\big|0\big\rangle\big\langle 0\big| + \varepsilon\big|1\big\rangle\big\langle 1\big|$. *It is acted upon by an operator* $\hat{A}$ *which has eigenvalues* $A_{1,2} = \pm 1$ *with eigenvectors* $\big|A_{1,2}\big\rangle$ *related to those of* $\hat{H}$ *by*

$$\big|A_1\big\rangle = \tfrac{1}{2}\big|0\big\rangle + \tfrac{\sqrt{3}}{2}\big|1\big\rangle \text{ and } \big|A_2\big\rangle = \tfrac{\sqrt{3}}{2}\big|0\big\rangle - \tfrac{1}{2}\big|1\big\rangle \ .$$

(i) If the system was prepared in the groundstate, what is the probability to measure A_1*?*
(ii) What is the energy of the system after it was measured in state $\big|A_1\big\rangle$*?*
(iii) Write $\hat{A}$ *in the energy basis.*

According to the QMP3,

$$P_1 = |\langle A_1 \,|\, 0\rangle|^2 = \Big|\Big(\tfrac{1}{2}\langle 0| + \tfrac{\sqrt{3}}{2}\langle 1|\Big)|0\rangle\Big|^2 = \Big|\tfrac{1}{2}\underbrace{\langle 0 \,|\, 0\rangle}_{=1} + \tfrac{\sqrt{3}}{2}\underbrace{\langle 1 \,|\, 0\rangle}_{=0}\Big|^2 = \tfrac{1}{4}\,.$$

The energy in state $\big|\Psi\big\rangle$ *is* $\big\langle\Psi \,\big|\, \hat{H} \,\big|\, \Psi\big\rangle$. *In state* $\big|A_1\big\rangle$ *we find*

$$\begin{aligned}\big\langle\hat{H}\big\rangle &= \big\langle A_1 \,\big|\, \hat{H} \,\big|\, A_1\big\rangle = \Big(\tfrac{1}{2}\langle 0| + \tfrac{\sqrt{3}}{2}\langle 1|\Big)\hat{H}\Big(\tfrac{1}{2}|0\rangle + \tfrac{\sqrt{3}}{2}|1\rangle\Big) \\ &= \Big(\tfrac{1}{2}\langle 0| + \tfrac{\sqrt{3}}{2}\langle 1|\Big)\Big(-\tfrac{\varepsilon}{2}|0\rangle + \tfrac{\sqrt{3}\varepsilon}{2}|1\rangle\Big) = -\tfrac{\varepsilon}{4}\langle 0|\,0\rangle + \tfrac{3\varepsilon}{4}\langle 1|\,1\rangle = \tfrac{1}{2}\varepsilon\,.\end{aligned}$$

Expanding $\hat{A}$ *in the energy basis gives (using* $\hat{A}\big|A_n\big\rangle = A_n\big|A_n\big\rangle$*):*

$$\begin{aligned}\hat{A} &= A_1\big|A_1\big\rangle\big\langle A_1\big| + A_2\big|A_2\big\rangle\big\langle A_2\big| = \big|A_1\big\rangle\big\langle A_1\big| - \big|A_2\big\rangle\big\langle A_2\big| \\ &= \Big(\tfrac{1}{2}|0\rangle + \tfrac{\sqrt{3}}{2}|1\rangle\Big)\Big(\tfrac{1}{2}\langle 0| + \tfrac{\sqrt{3}}{2}\langle 1|\Big) - \Big(\tfrac{\sqrt{3}}{2}|0\rangle - \tfrac{1}{2}|1\rangle\Big)\Big(\tfrac{\sqrt{3}}{2}\langle 0| - \tfrac{1}{2}\langle 1|\Big) \\ &= -\tfrac{1}{2}|0\rangle\langle 0| + \tfrac{\sqrt{3}}{2}|0\rangle\langle 0|1 + \tfrac{\sqrt{3}}{2}|1\rangle\langle 1|0 + \tfrac{1}{2}|1\rangle\langle 1|\,.\end{aligned}$$

In matrix form,

$$\mathsf{A} = \frac{1}{2}\begin{pmatrix} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix} \ .$$

3.2 Controversies around the 3rd postulate

The measurement theory was codified by John von Neumann who clarified that the result of a *quantum* measurement can only be read by a *classical* device, *i.e.* a state $\big|\Psi\big\rangle$ collapses after the measurement as a result of the interaction of a quantum system with such a device. Hence, QM is incomplete by construction as a measurement (collapse) cannot be described by a QM equation.

This incompleteness was one of the main reasons for mighty criticism of QM by many leading physicists focused on unacceptability of the 3$^{\rm rd}$ postulate. The 3$^{\rm rd}$ postulate contains two statements, each being impossible in classical physics. First, the result of a quantum measurement is random: experimentalists know the probabilities of possible outcomes, but cannot predict which particular one will turn up after any particular measurement. So it goes against the following well-known statement: "*Insanity is doing the same thing, over and over again, but expecting different results*", which is often *misattributed* to Einstein,[3] who actually said "*God does not play dice with the Universe*".

[3]One could think that a classical throwing a coin also leads to an unpredictable result but this is only due to the fact

The second controversial statement – an instantaneous collapse of the wavefunction – apart from making QM incomplete seems, on the face of it, to contradict the Relativity Theory, as had been noticed in an extremely important, famous paper by Einstein, Podolsky and Rosen. Their statement subsequently became known as the EPR paradox, which we consider in the following lecture.

Some most spectacular objections to the 3$^{\rm rd}$ QM Postulate have been inspired by (even if not all directly connected to) the EPR paper, focusing on the *conjecture* of the ***hidden variables*** which would make QM complete. In this approach, QM is incomplete by our lack of the proper description, not by nature. The idea is that the measurement outcomes which only appear random but “in reality" are governed by these hidden variables. We postpone considerations of this conjecture (and its spectacular experimental failure) till the following lecture.

Here I start with briefly analyzing the famous ***Schrödinger's cat*** – an objection against the probabilistic nature based on the re-interpretation of the TLS results.

3.3 Schrödinger's cat.

Schrödinger, absolutely unhappy with the probabilistic interpretation of QM, has suggested the following *gedanken* experiment: consider a radioactive atom as a TLS. Over the time it might decay but the timing is unpredictable.[4] Hence, prior to the measurement the atom is in a quantum superposition of (3.4), where $|\uparrow\rangle$ is the undecayed atom and $|\downarrow\rangle$ is the decayed one. Further assume that the decay triggers an electronic device that opens an ampule with poison, and put this assembly into a non-transparent, isolated cage with a cat inside.[5] According to the 3$^{\rm rd}$ QMP, we do not know the state before the measurement (opening the door of the cage), so that not only the atom but the cat itself is in a quantum superposition. Hence we can now identify our states as

$$|\downarrow\rangle - \text{dead}, \qquad\qquad |\uparrow\rangle - \text{alive}. \tag{3.6}$$

We can define the vitality operator (related to $\hat{\sigma}$ by a shift of eigenvalues)

$$\hat{V}|\uparrow\rangle = 1|\uparrow\rangle \text{ alive} \qquad\qquad \hat{V}|\downarrow\rangle = 0|\downarrow\rangle \text{ dead} \tag{3.7}$$

Before making a classical measurement of the vitality (opening the door), we do not know whether the atom decayed or not. The measurement of $\hat{V}$ would resolve the issue and give the eigenvalue 1 (alive!) with probability $\cos^2(\theta/2)$ and 0 (dead) with probability $\sin^2(\theta?2)$. The average (over many experiments) vitality of the cat is (obviously) $\cos^2(\theta/2)$, while before any particular measurement the cat is in a quantum superposition state of being dead and alive simultaneously.

Obviously, such a conclusion is very unpleasant as we believe (with a good reason to be spelt

that, in general, it is impossible to exactly reproduce initial conditions, in contrast to some experiments in the microworld where the reproducibility of the initial conditions is not a problem but not the result of a measurement is unpredictable.

[4]This is by itself classically unpleasant: even when everything is known about the current state of, say, an uranium atom, QM states that it is in principle impossible to predict when it decays. But QM predicts with an arbitrary accuracy the average half-decay time, i.e. the time over which half of the atoms in any given lump of uranium decay.

[5]RSPA would prosecute those who try to make such an experiment in reality.

out in subsequent lectures) that QM only describes microscopic objects.[6] This remains the subject of (mostly) philosophical disputes even now.

However, for the majority of researchers the following Bohr's answer to Schrödinger was persuasive: the measurement is an interaction of a quantum object with a classical measuring device, whether this device is observed by experimentalists or not. From this viewpoint, triggering the classical electronic device that opens the ampule is already a measurement, even if not witnessed by a human being. Hence, the poor cat is certainly dead if the device was triggered and alive if it was not, which happens (or not) independently of opening the door.

Nevertheless, the Schrödinger cat became proverbial, known in popular literature, and used by physicists for describing most entangled states (see the next lecture) as *cat states*.

Exercise: write down a state for a Schrödinger's cat who is four times more likely to be alive than dead and calculate its vitality.

[6] There are exceptions like superfluidity or superconductivity in macroscopic systems which are quantum phenomena that cannot be explained classically; however, the Schrödinger cat does not belong to this category.

Lecture 4. Entanglement and (the absence of) hidden variables

Here we briefly consider the famous EPR paradox, introduce the concept of quantum entanglement and discuss the Bell's inequality as the criterion of existence (or otherwise) of hidden variables that would allow one to get rid of quantum randomness.

The Schrödinger cat, discussed in the previous lecture, has emerged as Schrödinger's reaction to the famous paper by Einstein, Podolsky and Rosen published in 1935, which questioned the completeness of QM. The conclusion of the paper became known as the EPR paradox, and we shall consider it here in a simplified (but equivalent) form suggested by Bohm.

4.1 EPR paradox

The Schrödinger cat was a macroscopic extension of a TLS exemplified by spin $\frac{1}{2}$. Now we consider the system that contains two identical spin $\frac{1}{2}$ particles, created by the decay of a 0 spin particle at rest in the laboratory frame. The momentum conservation means that these two particles are moving in the opposite directions at the same speed, while the angular momentum conservation means that they have the opposite spin projections.

From QM2, you know that the spin projection can be measured using the Stern-Gerlach (SG) magnet oriented along any axis. Let us assume that Alice and Bob measure these projections for particles A and B, with Bob being slightly farther away from the point of decay so that Alice makes her measurement slightly earlier. Let us further assume that they are separated by a spacelike interval so that Alice cannot communicate the result of her measurement to Bob before his measurement, even if she sends a signal at the speed of light.

Suppose that Alice and Bob have oriented their SG magnets along the z axis. Their measurements can return, according to the 3$^{\rm rd}$ QMP, only the eigenvalues of the spin-projection operator, *i.e.* $\pm\frac{\hbar}{2}$. Then if Alice measured $s_z = +\frac{\hbar}{2}$, Bob ***must*** measure $s_z = -\frac{\hbar}{2}$. Thus their measurements are ***correlated***, in spite of the fact that they could not exchange any information.

This is the essence of the EPR paradox: the particle B somehow 'knows' in what state was the particle A, although any information exchange is forbidden by the theory of relativity. So there is a correlation between A and B, regardless of how far apart are the particles at the time of the measurements. Such a correlation is now known as ***quantum entanglement***.

On the face of it, this implies action at a distance or transmission of information at a superluminal speed, unless any spin component of each particle has *a definite value* from the moment of decay, and the measurements simply reveal these ***predetermined*** values – obviously, in contradiction to the 3$^{\rm rd}$ QM Postulate. For Einstein and collaborators the existence of such objective properties independently from subsequent measurements seemed to be much more reasonable than throwing QM dice, not even speaking about the information exchange at a superluminal speed. Any theory that obey these requirements – measurements at remote points are not correlated as the real-world particles must have objective properties unrelated to measurements – would be called a ***realistic local theory.*** Obviously, the QM which is based, in particular, on the 3$^{\rm rd}$ postulate is neither local, as a state of the

particle depends on a state of the other particle after the measurement, nor 'realistic' as it remains undetermined prior to the measurement.

Bohr, however, has noticed that entanglement was in no contradiction with relativity: although the wavefunction instantaneously collapses on the first (Alice's) measurement and the result of the second (Bob's) measurement is thus predetermined, there is no way to let Bob know prior to his measurement about Alice's result (or even whether Alice has already done the measurement). While accepted by many, this was not persuasive for Einstein and Schrödinger, not mentioning others.

This disagreement led to a very animated debate (lasted for several years) between great and famous physicists[1] that held opposite views on the nature or very existence of entanglement. These views finessed in the debate can be summarised as follows:

1. QM is incomplete due to lack of our knowledge: Bob's and Alice's results are not random as both particles' states are fully determined by some local (inherent to each particle) hidden variables; hence no information is transferred as A and B measured predetermined states

2. QM is nonlocal (and incomplete by nature): The probabilistic description is as good as it gets: results of Bob's and Alice's measurements are not predetermined but mutually *entangled* despite the absence of information transfer between them

For almost thirty years since the start of this famous discussion (often quoted as Einstein vs Bohr, although quite a few almost equally famous physicists participated), it had no definite resolution, and gradually became a subject of philosophical rather than physical discussions, since most physicists were quite satisfied with the fact that QM works, whether there are hidden variables or not. But this has changed after publication in 1964 of work by John Stewart Bell.

4.2 Bell's inequality

After the splash caused by the EPR paper, the interest in foundational matters gradually faded: although a few models of hidden variables were suggested, none was experimentally verifiable (or falsifiable). The vague idea of eventually finding the 'correct' ones was certainly out of the focus but still viable. Then Bell has discovered that no particular model of hidden variables was needed: if they exist, then the measurements results must obey a certain inequality.

Namely, Bell has noticed in 1964 that a mere assumption (inherent to any realistic local theory) that the values of spin projections in any direction were predetermined[2] would lead to a certain verifiable (or falsifiable) analytic inequality *without specifying hidden variables*. The importance of this has been immediately realised by the physics community, and subsequently different Bell-like inequalities have emerged, some of them easier for the experiment than the original one. However, the proof of the original inequality is rather straightforward and we go along with it.

[1] Some quotes: Einstein to Schrödinger – "The talmudic philosopher (Bohr) doesn't give a hoot for 'reality' . . . ", Ehrenfest to Einstein – "Einstein, I am ashamed of you. . . "

[2] You have learned in QM2 that projections of angular momentum or spin in different directions cannot be simultaneously measured as the appropriate operators do not commute. The *hidden variables* conjecture does not assume that any two components can be measured together, but only that any of them is predetermined.

Note first that Alice's and Bob's SG magnets could be arbitrarily aligned with an angle θ between their orientations, say $\boldsymbol{n}_A$ and $\boldsymbol{n}_B$. We have seen that measuring the spin projection of a particle prepared in state $\left|\uparrow\right\rangle$ along the z-axis in the field directed at an angle θ would give $\left|\uparrow\right\rangle$ along the new direction with the probability $\cos^2(\theta/2)$. Now it was Alice's measurement that prepared the spin projection for Bob (in the opposite to her direction): if her measurement produced $s_{\boldsymbol{n}_A} = \frac{\hbar}{2}$, Bob's measurement would give $s_{\boldsymbol{n}_B} = \frac{\hbar}{2}$ with the probability $\cos^2(\theta/2)$. Hence, their measurements would be correlated unless $\theta = \pi/2$ (then Bob's measurements would give $\uparrow$ or $\downarrow$ with probability $\frac{1}{2}$).

Assume that Alice and Bob have agreed, prior to the experiment, to orient their SG magnets in ***any one*** of three ***coplanar*** directions, $\boldsymbol{n}_1$, $\boldsymbol{n}_2$ and $\boldsymbol{n}_3$. If local hidden variables exist, then we can denote particle A state as $\left|\sigma_1, \sigma_2, \sigma_3\right\rangle$, where all the three components are predetermined with each σ_i being either up or down[3], *e.g.*, $\left|\uparrow, \downarrow, \uparrow\right\rangle$. Similarly predetermined is the state of particle B, $\left|\overline{\sigma}_1, \overline{\sigma}_2, \overline{\sigma}_3\right\rangle$, where $\left|\overline{\uparrow}\right\rangle = \left|\downarrow\right\rangle$ to allow for fact that the total spin projection must be zero in any direction. Let denote the joined state of two particles as

$$\left|\sigma_1, \sigma_2, \sigma_3;\ \overline{\sigma}_1, \overline{\sigma}_2, \overline{\sigma}_3\right\rangle \equiv \left|\sigma_1, \sigma_2, \sigma_3\right\rangle . \tag{4.1}$$

Here we do not need explicitly write spin directions in any outcome of Bob's measurements as they are always opposite to Alice's. Let $f(\sigma_1, \sigma_2, \sigma_3)$ be the fraction of pairs in the above state for any particular values of σ_i for particle A (and corresponding $\overline{\sigma}_i$ for particle B, which we do not explicitly write). It is vital to stress that this correspondence is *predetermined* and measurements are totally local, since the outcome of Alice's measurement does not depend on the orientation of Bob's SG magnets.

Now we compute $F_{\uparrow\uparrow}(\boldsymbol{n}_i; \boldsymbol{n}_j)$, the fraction of outcomes where Alice's measurement along $\boldsymbol{n}_i$ and Bob's measurement along $\boldsymbol{n}_j$ give the up-projection of spin. Clearly, $i \neq j$ here as the projections along the same axis must be opposite. The spin projection conservation also fixes outcomes along these two directions for both A and B: Alice's expected spin projection for $\boldsymbol{n}_j$ is always $\downarrow$ as it is opposite to Bob's. Hence to get F one needs to sum over two possible outcomes in the remaining direction. Then, choosing three (out of the six possible) pairs of directions, we find

$$\begin{aligned}
F_{\uparrow\uparrow}(\boldsymbol{n}_1; \boldsymbol{n}_2) &= \sum_{\sigma_3} f(\uparrow, \downarrow, \sigma_3) = f(\uparrow, \downarrow, \uparrow) + f(\uparrow, \downarrow, \downarrow)\,; \\
F_{\uparrow\uparrow}(\boldsymbol{n}_1; \boldsymbol{n}_3) &= \sum_{\sigma_2} f(\uparrow, \sigma_2\, \downarrow) = f(\uparrow, \uparrow, \downarrow) + f(\uparrow, \downarrow, \downarrow)\,; \\
F_{\uparrow\uparrow}(\boldsymbol{n}_3; \boldsymbol{n}_2) &= \sum_{\sigma_1} f(\sigma_1, \downarrow, \uparrow) = f(\uparrow, \downarrow, \uparrow) + f(\downarrow, \downarrow, \uparrow)\,.
\end{aligned} \tag{4.2}$$

From this we obtain

$$\boxed{F_{\uparrow\uparrow}(\boldsymbol{n}_1; \boldsymbol{n}_3) + F_{\uparrow\uparrow}(\boldsymbol{n}_3; \boldsymbol{n}_2) - F_{\uparrow\uparrow}(\boldsymbol{n}_1; \boldsymbol{n}_2)} = f(\uparrow, \uparrow, \downarrow) + f(\downarrow, \downarrow, \uparrow) \geqslant 0 \tag{4.3}$$

since all the fractions f are non-negative. This is one of the famous Bell's inequalities.

[3]The 2nd QM postulate claim that only eigenvalues of operators can be measured is not disputed in any formulation.

Let me stress again: for any *realistic local theory* this inequality (and a plethora of similar ones) must be obeyed, whatever is the nature of local hidden variables. From this viewpoint, it is (almost) immediately clear that QM in the Copenhagen interpretation is not one of such theories. To see this, let us calculate the probabilities $P_{\uparrow\uparrow}(\boldsymbol{n}_i;\boldsymbol{n}_j) = P_{\uparrow\uparrow}(\theta_{ij})$, which are the QM counterparts of frequencies $F_{\uparrow\uparrow}(\boldsymbol{n}_i,\,\boldsymbol{n}_j)$ of (4.2). Here θ_{ij} is the angle between Alice's and Bob's SG magnets. It is straightforwardly based on the results of Section 3.1. Alice has the probability $1/2$ to get spin projection up along $\boldsymbol{n}_i$. Then for Bob's measurement particle B would have spin projection down along $\boldsymbol{n}_i$ with probability 1, and according to (3.5) he will measure $\left|\downarrow\right\rangle$ (*i.e.* the same projection in which particle has been prepared) with the probability $\sin^2(\theta_{ij}/2)$. Hence the QM counterpart of Bell's inequality (4.3) is

$$P_{\uparrow\uparrow}(\boldsymbol{n}_1;\boldsymbol{n}_3) + P_{\uparrow\uparrow}(\boldsymbol{n}_3;\boldsymbol{n}_2) - P_{\uparrow\uparrow}(\boldsymbol{n}_1;\boldsymbol{n}_2) = \sin^2(\theta_{13}/2) + \sin^2(\theta_{23}/2) - \sin^2(\theta_{12}/2) \geqslant 0 \quad (4.4)$$

Now it is trivial to find geometry in which this inequality clearly fails. Choosing, e.g., that $\boldsymbol{n}_3$ bisects the angle between $\boldsymbol{n}_1$ and $\boldsymbol{n}_2$, *i.e.* $\theta_{13} = \theta_{23} = \frac{1}{2}\theta_{12}$ reduces Bell's inequality to

$$2\sin^2(\theta_{13}/2) - \sin^2(\theta_{13}) = 1 - \cos(\theta_{13}) - \sin^2(\theta_{13}) \geqslant 0 \quad \Longleftrightarrow \quad \cos^2(\theta_{13}) \geqslant \cos(\theta_{13}).$$

This inequality breaks down when $\cos(\theta_{13}) \geqslant 0$, *i.e.* for $0 \leqslant \theta_{13} \leqslant \pi/2$.

Hence, the conclusion is that a (relatively) simple experiment could decisively choose between locality and incompleteness on the one hand and QM entanglement on the other.

4.3 Experimental verdict: no hidden variables

Richard Feynman: "It seems to be almost ridiculous that you can squeeze the difficulty of QM to a numerical question that one thing is bigger than another. But there you are – it is bigger…"

The inequality (4.3) was not easy (but still possible) to verify experimentally, but many other Bell-type inequalities have been developed. The first experiment was carried out in 1972 by John Clauser[4] and Stuart Freedman.[5] It confirmed the violation of one of Bell's inequalities (in form suggested by Clauser and Shimony and modified by Freedman), albeit not with a decisive accuracy.

A real breakthrough was due to experiments led by Alain Aspect[4] in 1982 (in collaboration with other three physicists), based on his laying out a strategy for such an experiment in his PhD in 1978. They were based on the most popular CHSH[6] form of Bell's inequalities. The experiment was done with photons that can have only two polarisations, so that they are in this aspect TLS systems. Instead of SG magnets, linear polarisers have been used, each with two possible orientations $\boldsymbol{n}$ and $\boldsymbol{n}'$. The role of spin up and down is played by horizontal (H), $\perp$ to $\boldsymbol{n}$, or vertical (V), $\parallel$ to $\boldsymbol{n}$, polarisation. The CHSH inequality is

$$|S| \leqslant 2 \text{ where } S = E(\boldsymbol{n}_A,\boldsymbol{n}_B) - E(\boldsymbol{n}_A,\boldsymbol{n}'_B) + E(\boldsymbol{n}'_A,\boldsymbol{n}_B) + E(\boldsymbol{n}'_A,\boldsymbol{n}'_B), \quad (4.5)$$

[4] 2022 Nobel Prize in physics awarded to A Aspect, JF Clauser and A Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science", 04 October 2022

[5] Passed away in 2012 – to get the Nobel Prize one should be clever, lucky and live long enough. Sadly, John Bell has passed away in 1990 but not before he had seen numerous experimental falsifications of his theorem. Now, *John Stewart Bell Prize* is awarded biannually for major advances in foundations of quantum mechanics.

[6] J Clauser, M Horne, A Shimony and R Holt

and E is the correlation between Alice's and Bob's measurements of polarisations,

$$E(A, B) = P_{VV} + P_{HH} - P_{VH} - P_{HV}$$

with, e.g., P_{VV} is the probability (or fraction) of measuring two vertical polarisation along the corresponding axes.

In contrast to this particular form of Bell's inequality, QM allows $|S|$ to reach $2\sqrt{2}$. The first experiments in early 80's produced $|S|$ around 2.8 with fidelity exceeding 5σ. The current value of $|S|$ is 2.8284 with fidelity exceeding 99σ, *i.e.* the probability that this result is wrong is less than $e^{-99} \approx 10^{-42}$. The argument of ***entanglement*** vs ***locality*** has been won by entanglement with knock out.

We will come back to learning more about entanglement (but, sadly, not about its applications – this would require a separate set of lectures) closer to the end of this course.

Lecture 5. Position and Momentum Operators

We shall return to entangled states later, after learning here how to deal with continuum. We introduce the position and momentum operators and formally show that the wavefunction $\Psi(x)$ are components of Dirac state $\big|\Psi\big\rangle$ in the continuous position basis. Then we find the representation of the momentum operator in this basis.

5.1 Continuous basis

Position and momentum are very important in quantum mechanics as they provide a way to write down Hamiltonians by simply replacing classical variables with the corresponding operators. To treat this properly, we should finally consider an infinite-dimensional Hilbert space after having dealt with the TLS - two discrete degrees of freedom correspond to two levels only and thus generate two-dimensional Hilbert space (not to be confused with two-dimensional real space where QM particles with two continuous degrees of freedom generate infinitely-dimensional Hilbert spaces).

We start with a particle in one dimension (the 3D generalisation will be given at the end of this lecture). In quantum mechanics the position x of the particle becomes an observable and is represented by the *position operator* $\hat{x}$. Similarly to any Hermitian operator the position operator defines its eigenbasis (set of eigenvectors):

$$\hat{x}\big|x\big\rangle = x\big|x\big\rangle\,. \tag{5.1}$$

This notation is a bit heavy, but it is pretty much standard in modern literature. In the equation above $\hat{x}$ is an operator, *any* real number x is its eigenvalue and $\big|x\big\rangle$ is the corresponding eigenvector. In the previous lectures we have assumed eigenvalues of an operator form a discrete, countable set, allowing us to use notations like $\big\{\big|n\big\rangle\big\}$ (or $\big\{\big|\beta_n\big\rangle\big\}$, if it were necessary to be more specific). On the contrary, the eigenvalues of the position operator are all real numbers (a QM particle can be anywhere) forming a *continuous set*. This leads to some important modifications. A summation over discrete eigenvalues goes over to an integration, so the **completeness** and **orthonormality** conditions in the case of the position eigenbasis becomes

$$\hat{I} = \int_{-\infty}^{\infty} \mathrm{d}x\,\big|x\big\rangle\big\langle x\big| \quad \text{and} \quad \langle x\,|\,x'\rangle = \delta(x-x')\,. \tag{5.2}$$

Here $\delta(x)$ is the Dirac delta function defined by the relation

$$\int_{-\infty}^{\infty} \mathrm{d}x' f(x')\delta(x-x') = f(x)\,. \tag{5.3}$$

Similar to the diagonal matrix representation of $\hat{B}$ in its eigenbasis, $\big\langle n\,\big|\,\hat{B}\,\big|\,m\big\rangle = \beta_n\delta_{nm}$, we have

$$\big\langle x'\,\big|\,\hat{x}\,\big|\,x\big\rangle = x\langle x'\,|\,x\rangle = x\delta(x-x') \quad \Rightarrow \quad \big\langle x'\,\big|\,\hat{V}(x)\,\big|\,x\big\rangle = V(x)\delta(x-x'). \tag{5.4}$$

where $\hat{V}(x) \equiv V(\hat{x})$ is any operator that depends on x only, *e.g.*, a potential energy operator, and $V(x)$ is simply the corresponding function of x, assumed to be smooth, so that the last equality above could be verified by Taylor's expansion.

Now we can represent a state $\big|\Psi\big\rangle$ of a particle in 1D using the orthonormal and complete position eigenbasis $\big\{\big|x\big\rangle\big\}$ similarly to representation with a discrete basis $\big\{\big|n\big\rangle\big\}$, using the resolution of unity:

$$\big|\Psi\big\rangle = \hat{I}\big|\Psi\big\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \big|x\big\rangle\langle x \,|\, \Psi\rangle \equiv \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi(x)\big|x\big\rangle \tag{5.5}$$

where we have identified the component $\langle x \,|\, \Psi\rangle$ of state vector Ψ in the position basis with the familiar **wavefunction** $\Psi(x)$. The skew symmetry of the inner product gives us its complex conjugate counterpart $\langle \Psi \,|\, x\rangle = \langle x \,|\, \Psi\rangle^* = \Psi^*(x)$. This correspondence produces the **inner product** of two state vectors in terms of the integral of wavefunctions as expected:

$$\langle \Psi \,|\, \Phi\rangle = \big\langle \Psi\big|\hat{I}\big|\Phi\big\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \langle \Psi \,|\, x\rangle\langle x \,|\, \Phi\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi^*(x)\Phi(x)\,. \tag{5.6}$$

Inserting $\hat{I}$ twice into the Dirac expression for the average of an x dependent observable $\hat{V}(x)$, we find (similar to the discrete case) the expected correspondence with wave-function notations:

$$\begin{aligned}\big\langle \Psi \,\big|\, \hat{V}(x) \,\big|\, \Psi\big\rangle &= \big\langle \Psi\big| \int_{-\infty}^{\infty} \mathrm{d}x'\big|x'\big\rangle\big\langle x'\big|\hat{V}(x) \int_{-\infty}^{\infty} \mathrm{d}x\, \big|x\big\rangle\big\langle x\big|\big|\Psi\big\rangle \\ &= \int_{-\infty}^{\infty} \mathrm{d}x' \int_{-\infty}^{\infty} \mathrm{d}x\, \langle \Psi \,|\, x'\rangle \underbrace{\big\langle x' \,\big|\, \hat{V}(x) \,\big|\, x\big\rangle}_{=V(x)\delta(x-x')} \langle x \,|\, \Psi\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi^*(x)V(x)\Psi(x)\,,\end{aligned} \tag{5.7}$$

again, as expected.

Summarising:

Wavefunction $\Psi(x) \equiv \langle x \,|\, \Psi\rangle$ ***is the component of*** $\big|\Psi\big\rangle$ ***in the position eigenbasis*** $\big\{\big|x\big\rangle\big\}$***, while*** $\Psi^*(x) \equiv \langle \Psi \,|\, x\rangle$***;*** x ***is a 'continuous index' similar to*** n ***in discrete basis*** $\big\{\big|n\big\rangle\big\}$***;***

The bra(c)ket: $\langle \Psi \,|\, \Psi\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi^*(x)\Psi(x)$, $\big\langle \Psi \,\big|\, \hat{V}(x) \,\big|\, \Psi\big\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi(x)V(x)\Psi(x)$.

5.2 Position measurement

According to the 3rd QMP, measuring an observable $\hat{A}$ in a QM system in state $\big|\Psi\big\rangle$ yields one of its eigenvalues, A_n, with the probability $P_n = |\langle A_n \,|\, \Psi\rangle|^2$. The probabilities can be added: the probability that a measurement gives *either* A_n *or* A_m is given by the sum of the corresponding probabilities $P_{n\,\mathrm{or}\,m} = P_n + P_m$.

For the position operator, the probability to measure one of its eigenvalues is meaningless – the probability to get any particular value in continuum is zero. Hence we define the probability that position x is in certain range, $a < x < b$, *i.e.* take any value in the interval $[a, b]$. Generalising the sum of probabilities for discrete eigenvalues to continuum replaces, as always, the sum by the integral:

$$\mathrm{Prob}\,[a \leqslant x \leqslant b] = \int_a^b \mathrm{d}x |\langle x \,|\, \Psi\rangle|^2 = \int_a^b \mathrm{d}x |\Psi(x)|^2\,. \tag{5.8}$$

This is the probability to find the particle in the interval $[a, b]$ as was stated in QM2. The function $|\langle x \,|\, \Psi\rangle|^2 = |\Psi(x)|^2$ is the ***probability density*** for the position of the particle.

All the definitions above are generalised to a 3d space by changing $x \to \boldsymbol{r} = (x, y, z)$. The wave-function depends on three independent coordinates, $\Psi(x) \to \Psi(\boldsymbol{r}) = \langle \boldsymbol{r} \,|\, \Psi\rangle$, and the integration is carried out over volume element $\mathrm{d}\boldsymbol{r} \equiv \mathrm{d}^3x$ which in Cartesian coordinates is $\mathrm{d}\boldsymbol{r} \equiv \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$.

5.3 Momentum operator

Position operator is diagonal in its own eigenbasis, Eq. (5.4). What are matrix elements of momentum operator $\hat{p}$? Recall that in QM2 we used

$$\hat{p}\Psi(x) = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\Psi(x)\,. \tag{5.9}$$

With the new knowledge that $\Psi(x) = \langle x \,|\, \Psi\rangle$ we rewrite the left hand side as

$$\hat{p}\Psi(x) = \big\langle x\big|\hat{p}\big|\Psi\big\rangle\,, \tag{5.10}$$

i.e. the operator $\hat{p}$ acts on $\big|\Psi\big\rangle$, modifies it, and the x-component of the modified state $\big|\hat{p}\Psi\big\rangle$ is calculated. Let us insert the resolution of unity in the position basis:

$$\hat{p}\Psi(x) = \big\langle x\big|\hat{p}\big|\Psi\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}x'\big\langle x\big|\hat{p}\big|x'\big\rangle\langle x' \,|\, \Psi\rangle = \int_{-\infty}^{\infty}\mathrm{d}x'\big\langle x\big|\hat{p}\big|x'\big\rangle\Psi(x')\,. \tag{5.11}$$

Comparing this with

$$-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\Psi = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\int_{-\infty}^{\infty}\mathrm{d}x'\delta(x-x')\Psi(x') = \int_{-\infty}^{\infty}\mathrm{d}x'\left(-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\delta(x-x')\right)\Psi(x'), \tag{5.12}$$

we identify the matrix element of the momentum operator in the position basis[1] $\big\{\big|x\big\rangle\big\}$:

$$\big\langle x\big|\hat{p}\big|x'\big\rangle = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\delta(x-x') \tag{5.13}$$

Unlike the position operator $\hat{x}$ diagonal in its eigenbasis, the momentum operator $\hat{p}$ is not: the difference between $\delta(x)$ and $\delta'(x)$ in the expressions for matrix elements is crucial!

Let us find the components of $\big|p\big\rangle$ in the position representation. We start with the EVE for $\hat{p}$ and *project* it into this representation by taking the inner product with $\big\langle x\big|$:

$$\hat{p}\big|p\big\rangle = p\big|p\big\rangle \quad\Rightarrow\quad \big\langle x\big|\hat{p}\big|p\big\rangle = p\langle x \,|\, p\rangle \equiv p\Psi_p(x)\,. \tag{5.14}$$

Here we identified the x-component of $\big|p\big\rangle$ with the wavefunction $\Psi_p(x)$, as Eq. (5.14) is a specific case of more general Eq. (5.10) with $\big|\Psi\big\rangle \to \big|p\big\rangle$. Using the resolution of unity and Eq. (5.13) we find

$$\big\langle x\big|\hat{p}\big|p\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}x'\big\langle x\big|\hat{p}\big|x'\big\rangle\langle x' \,|\, p\rangle = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\int_{-\infty}^{\infty}\mathrm{d}x'\delta(x-x')\Psi_p(x) = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\Psi_p\,. \tag{5.15}$$

Hence we write the EVE (5.14) as a familiar differential equation whose solution is the plain wave:

$$-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\Psi_p(x) = p\Psi_p(x) \quad\Rightarrow\quad \Psi_p(x) \equiv \langle x \,|\, p\rangle = A\mathrm{e}^{ipx/\hbar}\,. \tag{5.16}$$

The factor A should, in principle, be calculated from normalisation,

$$\langle p \,|\, p\rangle = \int \mathrm{d}x\langle p \,|\, x\rangle\langle x \,|\, p\rangle = \int \mathrm{d}x|\Psi_p(x)|^2 = |A|^2\int_{-\infty}^{\infty}\mathrm{d}x|\mathrm{e}^{ipx/\hbar}|^2. \tag{5.17}$$

[1] The derivative of Dirac delta function, $\delta'(x)$, is defined via its action on an arbitrary sufficiently smooth function $f(x)$ as $\int_{-\infty}^{\infty}\mathrm{d}x\,\delta'(x)f(x) = -\int_{-\infty}^{\infty}\mathrm{d}x\,\delta(x)f'(x) = -f'(0)$.

However, as $|\mathrm{e}^{i\alpha}| = 1$, this integral diverges: we already know from QM2 that the states belonging to continuous spectrum are not normalisable. Instead we specify coefficient A above by demanding the inner product $\langle p \,|\, p' \rangle = \delta(p - p')$, similar to that for position states,[2] Eq. (5.2)

$$\langle p' \,|\, p \rangle = |A|^2 \int_{-\infty}^{\infty} \mathrm{d}x \, \langle p' \,|\, x \rangle \langle x \,|\, p \rangle = |A|^2 \int_{-\infty}^{\infty} \mathrm{d}x \, \mathrm{e}^{i(p-p')x/\hbar} = 2\pi\hbar |A|^2 \delta(p - p') \,. \tag{5.18}$$

Hence we choose $A = 1/\sqrt{2\pi\hbar}$ which determines the wavefunction of momentum eigenstate[3]:

$$\psi_p(x) = \langle x \,|\, p \rangle = \frac{1}{\sqrt{2\pi\hbar}} \mathrm{e}^{ipx/\hbar} \,. \tag{5.19}$$

The generalisation to 3D is straightforward: $p \to \boldsymbol{p}$, while $px \to \boldsymbol{p} \cdot \boldsymbol{r}$ so that the momentum operator, its matrix elements and its eigenfunctions in the position representation are

$$\hat{\boldsymbol{p}} \Psi(\boldsymbol{r}) = -i\hbar \nabla \Psi(\boldsymbol{r}) \,; \quad \big\langle \boldsymbol{r} \big| \hat{\boldsymbol{p}} \big| \boldsymbol{r}' \big\rangle = -i\hbar \nabla \delta(\boldsymbol{r} - \boldsymbol{r}') \,; \quad \Psi_{\boldsymbol{p}}(\boldsymbol{r}) = \langle \boldsymbol{r} \,|\, \boldsymbol{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} \mathrm{e}^{i\boldsymbol{p}\cdot\boldsymbol{r}} \tag{5.20}$$

5.4 Momentum representation

The EVEs for x and p, Eqs. (5.4) and (5.14), are of the same structure. Hence, as we get any $\big|\Psi\big\rangle$ in the position representation by taking the inner product with $\big\langle x \big|$, we find $\big|\Psi\big\rangle$, and in particular $\big|x\big\rangle$, in the momentum representation by taking the inner product with $\big\langle p \big|$:

$$\langle p \,|\, \psi \rangle \equiv \Psi(p) \quad \Rightarrow \quad \psi_x(p) \equiv \langle p \,|\, x \rangle = \langle x \,|\, p \rangle^* = \psi_p^*(x) = \frac{1}{\sqrt{2\pi\hbar}} \mathrm{e}^{-ipx/\hbar} \,. \tag{5.21}$$

Note that $\Psi(p) = \langle p \,|\, \Psi \rangle$ plays the same role in the momentum representation as the wave function $\Psi(x) = \langle x \,|\, \Psi \rangle$ in the position representation. In particular, $|\Psi(p)|^2$ is the probability density for momentum, *i.e.* can be used to calculate the probability

$$\mathrm{Prob}\,[p_a \leqslant p \leqslant p_b] = \int_{p_a}^{p_b} \mathrm{d}p \, |\Psi(p)|^2 \tag{5.22}$$

to find a quantum particle in state $\big|\Psi\big\rangle$ in the momentum interval between p_a and p_b.

The wavefunctions in momentum and position representations are related by the Fourier transform, as we find by using the resolution of unity in both representations[4]:

$$\Psi(p) = \langle p \,|\, \Psi \rangle = \int_{-\infty}^{\infty} \mathrm{d}x \, \langle p \,|\, x \rangle \langle x \,|\, \Psi \rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \mathrm{d}x \, \mathrm{e}^{-ipx/\hbar} \Psi(x) \tag{5.23}$$

$$\Psi(x) = \langle x \,|\, \Psi \rangle = \int_{-\infty}^{\infty} \mathrm{d}p \, \langle x \,|\, p \rangle \langle p \,|\, \Psi \rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \mathrm{d}p \, \mathrm{e}^{ipx/\hbar} \Psi(p) \,. \tag{5.24}$$

[2]Reminder from Maths2: $\int_{-\infty}^{\infty} \mathrm{d}x \, \mathrm{e}^{ikx} = 2\pi\delta(k)$; then, $\delta(ak) = (1/|a|)\delta(k)$.

[3]Such a choice of A is not always convenient, since the WF $\psi_p(x)$ is still *not normalised*. Hence, it cannot be directly substituted when counted probabilities, since in the 3rd QMP all states assumed to be normalised. For example, the probability of a free particle to be in any particular interval $[a, b]$ in the *infinite* system is always 0 but if one substituted $\Psi_p(x)$ of Eq. (5.19) into Eq. (5.8), the answer would be finite, and of a wrong dimensionality. The way out is to consider a system of finite large length, when the momentum becomes discrete – a QM particle in a rectangular box with the normalisable WF.

[4]Here the choice of A in Eq. (5.10) and hence in Eq. (5.21) imposed equal coefficients in the direct and inverse FT.

Find dimensions of $\Psi(x)$ and $\Psi(p)$ and explain why they differ from those of $\psi_p(x)$ and $\psi_x(p)$.

In physical terms one notices that a state with well-defined position x is a linear superposition of infinitely many states with well defined momentum and vice versa. This is a consequence of the celebrated Heisenberg uncertainty principle, to which we shall return soon.

Note also that the completeness of the momentum eigenbasis (the resolution of unity),

$$\int_{-\infty}^{\infty} \mathrm{d}p \big|p\big\rangle\big\langle p\big| = \hat{I} \tag{5.25}$$

used in Eq. (5.24), follows from that in the position eigenbasis, Eq. (5.2) which can be demonstrated by taking the position matrix elements of the l.h.s. and r.h.s. of Eq. (5.25):

$$\text{l.h.s.: } \big\langle x'\big| \left(\int_{-\infty}^{\infty} \mathrm{d}p \big|p\big\rangle\big\langle p\big| \right) \big|x\big\rangle = \int_{-\infty}^{\infty} \mathrm{d}p \langle x' \,|\, p\rangle \langle p \,|\, x\rangle = \int_{-\infty}^{\infty} \frac{\mathrm{d}p}{2\pi\hbar} \mathrm{e}^{ip(x-x')/\hbar} = \delta(x'-x)$$

$$\text{r.h.s.: } \big\langle x'\big|\hat{I}\big|x\big\rangle = \langle x' \,|\, x\rangle = \delta(x'-x)\,.$$

Lecture 6. Heisenberg uncertainty principle

Here, after introducing spatial translations, we re-iterate that two observables can be measured simultaneously only if the appropriate operators commute, and go from that to the HUP.

6.1 Momentum operator and spatial translations

Momentum is related to translations. To see this we consider the unitary operator[1]

$$\hat{T}(a) = \mathrm{e}^{-ia\hat{p}/\hbar} = 1 - \frac{i}{\hbar}a\hat{p} - \frac{1}{2\hbar^2}a^2\hat{p}^2 + \dots\,, \tag{6.1}$$

where a is some length. Acting by $\hat{T}$ on position eigenket $\big|x\big\rangle$, we find:

$$\hat{T}(a)\big|x\big\rangle = \hat{I}\hat{T}(a)\big|x\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}p\big|p\big\rangle\big\langle p\big|\hat{T}(a)\big|x\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}p\big|p\big\rangle\big\langle p\big|\mathrm{e}^{-ia\hat{p}/\hbar}\big|x\big\rangle\,, \tag{6.2}$$

where the p-resolution of unity was used again.

Since $\hat{p}$ is Hermitian, $\hat{p} = \hat{p}^\dagger$, it can act on the bra as well as on the ket, giving the EVE $\big\langle p\big|\hat{p} = p\big\langle p\big|$. As we know from maths, $\big\langle p\big|f(\hat{p}) = f(p)\big\langle p\big|$ for any smooth function f. Hence we get

$$\begin{aligned}
&\big\langle p\big|\mathrm{e}^{-ia\hat{p}/\hbar}\big|x\big\rangle = \mathrm{e}^{-ipa/\hbar}\langle p\,|\,x\rangle = \frac{1}{\sqrt{2\pi\hbar}}\mathrm{e}^{-ip(x+a)/\hbar} = \langle p\,|\,x+a\rangle \\
&\Rightarrow\quad \hat{T}(a)\big|x\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}p\big|p\big\rangle\langle p\,|\,x+a\rangle = \big|x+a\big\rangle
\end{aligned} \tag{6.3}$$

Therefore, the action of $\hat{T}(a)$ on $\big|x\big\rangle$ is the translation $\big|x\big\rangle \to \big|x+a\big\rangle$.

6.2 Noncommuting variables & generalised Heisenberg uncertainty principle

The state $\big|x+a\big\rangle$ is an eigenstate of $\hat{x}$ corresponding to the eigenvalue $x+a$. Hence

$$\hat{x}\hat{T}(a)\big|x\big\rangle = \hat{x}\big|x+a\big\rangle = (x+a)\big|x+a\big\rangle\,. \tag{6.4}$$

Had we applied the position operator *before* acting with $\hat{T}(a)$ the result would be different:

$$\hat{T}(a)\hat{x}\big|x\big\rangle = \hat{T}(a)x\big|x\big\rangle = x\hat{T}(a)\big|x\big\rangle = x\big|x+a\big\rangle\,. \tag{6.5}$$

Subtracting Eq. (6.5) from Eq. (6.4), we find the commutator

$$\big[\hat{x},\hat{T}(a)\big]\big|x\big\rangle = a\big|x+a\big\rangle = a\hat{T}(a)\big|x\big\rangle \quad\Rightarrow\quad \big[\hat{x},\hat{T}(a)\big] = a\hat{T}(a)\,. \tag{6.6}$$

When displacement $a \to 0$, it follows from expansion (6.1) that the r.h.s. of Eq. (6.6), $a\hat{T}(a) \simeq a$, while the l.h.s. is given by

$$\big[\hat{x},\hat{T}(a)\big] \simeq \big[\hat{x}, 1 - i\frac{a\hat{p}}{\hbar}\big] = -\frac{ia}{\hbar}\big[\hat{x},\hat{p}\big] \text{ so that } \frac{a}{i\hbar}\big[\hat{x},\hat{p}\big] = a \quad\Rightarrow\quad [\hat{x},\hat{p}] = i\hbar\,. \tag{6.7}$$

[1] Recall that $\mathrm{e}^{i\hat{h}}$ is a unitary operator for any Hermitian $\hat{h}$, Eq. (2.32).

Hence we have recovered the familiar commutation relations between position and momentum, from which (as you know from QM2) follows the Heisenberg uncertainty principle (HUP),

$$\Delta x \Delta p \geqslant \hbar/2. \tag{6.8}$$

We shall now extend the HUP to any pair of non-commuting Hermitian operators. First, we define formally the uncertainty, ΔA, in measuring an observable $\hat{A}$ in a normalised state $\big|\Psi\big\rangle$ as the square root of the *standard deviation* or *variance*:

$$(\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 = \big\langle \Psi \big| \hat{A}^2 \big| \Psi \big\rangle - \big\langle \Psi \big| \hat{A} \big| \Psi \big\rangle^2 . \tag{6.9}$$

The variance $\langle \hat{A}^2 \rangle$ and the expectation value $\langle \hat{A} \rangle$ characterise the statistical distribution of results of repeated measurements of $\hat{A}$: the smaller ΔA, the more precisely one can know $\hat{A}$.

Let me stress here that the statistical averaging over many measurements, like $\langle \hat{A} \rangle$, and the QM expectation value in a given state, $\big\langle \Psi \,\big|\, \hat{A} \,\big|\, \Psi \big\rangle$, are directly connected. Indeed, using the resolution of unity in the eigenbasis $\big\{ \big|n\big\rangle \big\}$ of $\hat{A}$ defined by the EVE, $\hat{A}\big|n\big\rangle = a_n \big|n\big\rangle$, we find

$$\big\langle \Psi \big| \hat{A} \big| \Psi \big\rangle = \sum_n a_n |\langle n \,|\, \Psi \rangle|^2 = \sum_n a_n P_n, \text{ and } \big\langle \Psi \big| \hat{A}^2 \big| \Psi \big\rangle = \sum_n a_n^2 |\langle n \,|\, \Psi \rangle|^2 = \sum_n a_n^2 P_n \tag{6.10}$$

so that

$$(\Delta A)^2 = \sum_n a_n^2 P_n - \sum_{m,n} a_m P_m a_n P_n = \langle a_n^2 \rangle - \langle a_n \rangle^2 = \langle a_n - \langle a_n \rangle \rangle^2 \tag{6.11}$$

is indeed the standard deviation of a random variable taking values a_n with probabilities P_n.

The HUP, Eq. (6.8), is generalised to any pair of Hermitian operators $\hat{A}$ and $\hat{B}$ as[2]

$$\Delta A \Delta B \geqslant \tfrac{1}{2} \Big| \langle [\hat{A}, \hat{B}] \rangle \Big| = \tfrac{1}{2} \Big| \big\langle \Psi \big| [\hat{A}, \hat{B}] \big| \Psi \big\rangle \Big| \to \tfrac{1}{2} |c|, \tag{6.12}$$

the latter being valid if the commutator $\big[\hat{A}, \hat{B}\big]$ is a c-number. As this is true for $\big[\hat{x}, \hat{p}\big]$, in this case Eq. (6.12) reduces to Eq. (6.8). For two general noncommuting operators, the r.h.s. remains proportional to the expectation value of their commutator.

If the commutator of two observables vanishes, $[\hat{A}, \hat{B}] = 0$, they are said to be *compatible*. As it was shown in Maths2B, such operators can be diagonalised simultaneously, *i.e.* they share the same eigenbasis, say $\big\{ \big|n\big\rangle \big\}$, with different eigenvalues:

$$\hat{A}\big|n\big\rangle = a_n \big|n\big\rangle; \qquad \hat{B}\big|n\big\rangle = b_n \big|n\big\rangle. \tag{6.13}$$

In this case, if a measurement of $\hat{A}$ resulted in a_m, then the system collapses to state $\big|m\big\rangle$ so that an immediate measurement of $\hat{B}$ will give result b_m *with probability 1*. This is the meaning of the statement that compatible observables can be measured simultaneously.

[2] One of many possible variants of proving this is given at the end of this lecture.

On the contrary, if commutator $[\hat{A},\hat{B}]\neq 0$, the observables are *incompatible* and the common eigenbasis cannot be found. In this case, when measuring $\hat{A}$ returns a_n, $\big|\Psi\big\rangle$ collapses to the eigenstate $\big|a_n\big\rangle$. This is not an eigenstate of $\hat{B}$, defined by the appropriate EVE, $\hat{B}\big|b_n\big\rangle = b_n\big|b_n\big\rangle$, but rather a linear superposition of such eigenstates:

$$\big|a_n\big\rangle = \sum_n \alpha_n\big|b_n\big\rangle\,, \text{ with } \alpha_n = \langle b_n \,|\, a_n\rangle. \tag{6.14}$$

Hence, according to the 3$^{\rm rd}$ QMP, a measurement of $\hat{B}$ would give any of its eigenvalues b_n with the probability $P_{b_n} = |\alpha_n|^2$.

It might happen that $\big[\hat{A},\hat{B}\big]$, being nonzero in general, has zero expectation value for a particular choice of the state, *i.e.* for a certain $\big|\Psi_0\big\rangle$ we have $\big\langle\Psi_0\big|\big[\hat{A},\hat{B}\big]\big|\Psi_0\big\rangle = 0$. In this case the HUP allows a simultaneous measurement of incompatible observables. Let us look, as an example, at three components of angular momentum, $\hat{\boldsymbol{L}} = \hat{\boldsymbol{r}}\times\hat{\boldsymbol{p}} = (\hat{L}_x,\hat{L}_y,\hat{L}_z)$. The commutator

$$[\hat{L}_x,\hat{L}_y] = i\hbar\hat{L}_z\,, \tag{6.15}$$

vanishes for the state corresponding to zero eigenvalue of $\hat{L}_z$. Then there is a *possibility* to measure all the three components of angular momentum simultaneously but only in the case when the total angular momentum is also zero. Then all the three components of $\hat{\boldsymbol{L}}$ vanish.

6.2.1 A proof of the generalised HUP

Now we prove the generalised HUP of Eq. (6.12). First we shift operators $\hat{A}$ and $\hat{B}$ so that they have zero expectation values in the state $\big|\Psi\big\rangle$:

$$\hat{A}\to\hat{A}-\big\langle\Psi\big|\hat{A}\big|\Psi\big\rangle \qquad \hat{B}\to\hat{B}-\big\langle\Psi\big|\hat{B}\big|\Psi\big\rangle \tag{6.16}$$

Thus the uncertainties reduce to $(\Delta A)^2 = \big\langle\Psi\big|\hat{A}^2\big|\Psi\big\rangle$ and $(\Delta B)^2 = \big\langle\Psi\big|\hat{B}^2\big|\Psi\big\rangle$. Consider the state

$$\big|\Phi\big\rangle = \hat{A}\big|\Psi\big\rangle + i\lambda\hat{B}\big|\Psi\big\rangle = \big|\big(\hat{A}+i\lambda\hat{B}\big)\Psi\big\rangle\,, \tag{6.17}$$

where λ is a real number. Writing the inner product, $\langle\Phi\,|\,\Phi\rangle \geqslant 0$, using Eq. (6.17) and the fact that both operators are Hermitian, *i.e.* $\big\langle\Phi\big| = \big\langle\Psi\big(\hat{A}-i\lambda\hat{B}\big)\big|$, we find

$$\langle\Psi\big(\hat{A}-i\lambda\hat{B}\big)\,|\,\big(\hat{A}+i\lambda\hat{B}\big)\Psi\rangle = \big\langle\Psi\big|\hat{A}^2\big|\Psi\big\rangle + i\lambda\big\langle\Psi\big|[\hat{A},\hat{B}]\big|\Psi\big\rangle + \lambda^2\big\langle\Psi\big|\hat{B}^2\big|\Psi\big\rangle \geqslant 0 \tag{6.18}$$

The commutator of Hermitian operators is anti-Hermitian (hence its expectation value is purely imaginary) as $[\hat{A},\hat{B}]^\dagger = (\hat{A}\hat{B}-\hat{B}\hat{A})^\dagger = \hat{B}^\dagger\hat{A}^\dagger - \hat{A}^\dagger\hat{B}^\dagger = [\hat{B},\hat{A}] = -[\hat{A},\hat{B}]$. Therefore, one writes

$$(\Delta A)^2 - \lambda\Big|\big\langle\big[\hat{A},\hat{B}\big]\big\rangle\Big| + \lambda^2(\Delta B)^2 \equiv a\lambda^2 + b\lambda + c \geqslant 0\,. \tag{6.19}$$

The condition for the quadratic function of λ to be non-negative for all λ is to have no real roots, *i.e.* to have the negative discriminant

$$D = b^2 - 4ac = \Big|\big\langle\big[\hat{A},\hat{B}\big]\big\rangle\Big|^2 - 4\Delta A\Delta B \leqslant 0\,. \tag{6.20}$$

which is nothing else but the generalised HUP of Eq. (6.12).

Lecture 7. The 4th postulate: time evolution in QM.

The state change under time evolution is described by the evolution operator naturally related to the Schrödinger equation – in essence, this is the 4th QM Postulate.

Once we have understood the states of quantum mechanical systems as vectors in Hilbert space we can ask how these states change with time. Any change of a state vector is encoded in certain operator, so the time evolution is given by the

7.1 Time-evolution operator

Here we describe the reasoning that leads to the Schrödinger equation. This is by no means a derivation of it – this equation is actually the 4th postulate of QM, and a postulate (axiom) cannot be proved or derived, but its justification is still possible.

We define an operator $\hat{U}(t)$ which transforms the initial state $\big|\Psi(0)\big\rangle$ into the state $\big|\Psi(t)\big\rangle$ corresponding to the later time t,

$$\big|\Psi(t)\big\rangle = \hat{U}(t)\big|\Psi(0)\big\rangle \quad \Rightarrow \quad \big\langle\Psi(t)\big| = \big\langle\Psi(0)\big|\hat{U}^\dagger(t) \tag{7.1}$$

Such an operator must have the following properties:

(a) $\hat{U}(0) = \hat{I}$ – obviously the state cannot change momentarily.

(b) $\hat{U}(t_2)\hat{U}(t_1) = \hat{U}(t_1 + t_2)$ – combination of evolutions is also an evolution.

(c) $\langle\Psi(t) \mid \Psi(t)\rangle = \big\langle\Psi(0)\big|\hat{U}^\dagger(t)\hat{U}(t)\big|\Psi(0)\big\rangle = \langle\Psi(0) \mid \Psi(0)\rangle$ – the state must remain normalised under evolution. This leads to

$$\hat{U}^\dagger(t)\hat{U}(t) = \hat{I} \quad \Leftrightarrow \quad \hat{U}^\dagger(t) = \hat{U}^{-1}(t) \tag{7.2}$$

or *unitarity* of the time-evolution operator.

The requirement (b) is a defining characteristic of the exponential function: from $f(x+y) = f(x)f(y)$ follows that $f(x) = a\mathrm{e}^x$, and $f(0) = 1$ gives $a = 1$. This holds for operators (regular functions of which are defined by Taylor's expansion), so that we can write

$$\hat{U}(t) = \mathrm{e}^{-it\hat{h}} \tag{7.3}$$

of some operator $\hat{h}$ (where i is inserted for convenience). The requirement (c) gives

$$\hat{U}^\dagger\hat{U} = \mathrm{e}^{i\left(\hat{h}^\dagger - \hat{h}\right)t} = \hat{I}$$

that means that $\hat{h}^\dagger - \hat{h} = \hat{0}$, so that $\hat{h}$ is Hermitian. It has dimensions of inverse time, *i.e.* frequency, so that $\hat{H} \equiv \hbar\hat{h}$ has dimensions of energy, as $[\hbar h] = [\hbar/t] = \text{energy}$. Hence

$$\boxed{\hat{U}(t) = \mathrm{e}^{-it\hat{H}/\hbar}} \tag{7.4}$$

In these terms, the time-evolution equation (7.1) goes to[1]

$$\boxed{\big|\Psi(t)\big\rangle = \mathrm{e}^{-it\hat{H}/\hbar}\big|\Psi(0)\big\rangle} \tag{7.5}$$

Differentiating both sides of Eq. (7.5) with respect to time gives the Schrödinger equation written for time-dependent ket-states $\big|\Psi(t)\big\rangle$:

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\big|\Psi(t)\big\rangle = i\hbar(-i\hat{H}/\hbar)\mathrm{e}^{-it\hat{H}/\hbar}\big|\Psi(0)\big\rangle = \hat{H}\big|\Psi(t)\big\rangle\,. \tag{7.6}$$

Here we ***postulate*** that operator $\hat{H}$ is the Hamiltonian of the system, as all we have got from the evolution operator was the statement that $\hat{H}$ is Hermitian, and it has the dimensionality of energy.

If we want to obtain the standard time-dependent Schrödinger equation for wavefunctions, we take the inner product of both sides with $\big\langle x\big|$ which gives components of $\big|\Psi\big\rangle$ in the position eigenbasis, i.e. the Schrödinger equation for time-dependent wavefunctions $\Psi(x,t) = \langle x\,|\,\Psi(t)\rangle$:

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\langle x\,|\,\Psi(t)\rangle = \big\langle x\big|\hat{H}\big|\Psi(t)\big\rangle \quad\Leftrightarrow\quad i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\Psi(x,t) = \hat{H}\Psi(x,t)\,. \tag{7.7}$$

We can formulate now the 4th Postulate of Quantum Mechanics

The 4th QM Postulate: Time evolution of a state $\big|\Psi(t)\big\rangle$ is governed by the system's Hamiltonian and is described by the time-dependent Schrödinger equation.

The justification of the fact that $\hat{H}$ is the system Hamiltonian comes from the fact that its eigenvalues have the proper dimensionality and are ***conserved***.

7.2 Stationary states

The energy eigenbasis defined by the EVE for the Hamiltonian,

$$\hat{H}\big|\varepsilon_n\big\rangle = \varepsilon_n\big|\varepsilon_n\big\rangle, \tag{7.8}$$

plays a special role in describing time-evolution of quantum states. Indeed, let us expand the state $\big|\Psi(t)\big\rangle$ in this basis:

$$\big|\Psi(t)\big\rangle = \sum_n c_n(t)\big|\varepsilon_n\big\rangle\,, \qquad c_n(t) = \langle\varepsilon_n\,|\,\Psi(t)\rangle\,. \tag{7.9}$$

The time-dependence of the components $c_n(t)$ can be obtained from Eqs. (7.7) and (7.8):

$$\dot{c}_n(t) = \frac{\mathrm{d}}{\mathrm{d}t}\langle\varepsilon_n\,|\,\Psi(t)\rangle = \left\langle\varepsilon_n\,\middle|\,\frac{\mathrm{d}\Psi(t)}{\mathrm{d}t}\right\rangle = \frac{1}{i\hbar}\big\langle\varepsilon_n\big|\hat{H}\big|\Psi(t)\big\rangle = \frac{1}{i\hbar}\varepsilon_n\langle\varepsilon_n\,|\,\Psi(t)\rangle \equiv \frac{1}{i\hbar}\varepsilon_n c_n(t)\,. \tag{7.10}$$

[1] It is not a coincidence that this is similar to the translation operator $\hat{T}(a) = e^{-ia\hat{p}/\hbar}$. Indeed, momentum operator $\hat{p}$ induces (generates) displacements in space, while Hamiltonian generates displacements in time and the corresponding operators are the same under replacement of $\hat{p}$ by $\hat{H}$ and a by t.

The solution of this 1st order ODE gives a particularly simple time-dependence:

$$c_n(t) = \mathrm{e}^{-it\varepsilon_n/\hbar} c_n(0) \equiv \mathrm{e}^{-it\varepsilon_n/\hbar} c_n \,. \tag{7.11}$$

It means that the probabilities to find the system in any of these states *do not depend on time*:

$$P_n(t) = |\langle n \,|\, \Psi(t)\rangle|^2 = |c_n(t)|^2 = |\mathrm{e}^{-it\varepsilon_n/\hbar} c_n(0)|^2 = |c_n(0)|^2 = P_n(0)\,. \tag{7.12}$$

This justifies the name **stationary states** for

$$\big|\varepsilon_n(t)\big\rangle = \hat{U}(t)\big|\varepsilon_n\big\rangle = \mathrm{e}^{-it\varepsilon_n/\hbar}\big|\varepsilon_n\big\rangle\,, \tag{7.13}$$

since the probability $P_n(t)$ to find the system in one of this states is stationary, despite the eigenstates of $\hat{H}$ oscillating in time. The appropriate wavefunctions $\Psi_n(x,t) = \langle x \,|\, \varepsilon_n(t)\rangle$, corresponding to stationary states, oscillate in time in the same faction (QM2):

$$\Psi_n(x,t) = \big\langle x\big|\mathrm{e}^{-i\hat{H}/\hbar}\big|\varepsilon_n\big\rangle = e^{-it\varepsilon_n/\hbar}\langle x \,|\, n\rangle = e^{-it\varepsilon_n/\hbar}\Psi_n(x)\,. \tag{7.14}$$

Moreover, expectation values of observables in a stationary state do not depend on time:[2]

$$\langle A(t)\rangle \equiv \big\langle \varepsilon_n(t)\big|\hat{A}\big|\varepsilon_n(t)\big\rangle = \big\langle \varepsilon_n\big|\mathrm{e}^{+it\varepsilon_n/\hbar}\hat{A}\mathrm{e}^{-it\varepsilon_n/\hbar}\big|\varepsilon_n\big\rangle = \big\langle \varepsilon_n\big|\hat{A}\big|\varepsilon_n\big\rangle = \langle A(0)\rangle\ . \tag{7.15}$$

Example 7.1. *Consider a TLS with the Hamiltonian $\hat{H} = \sum_{n=0}^{1} \varepsilon_n \big|n\big\rangle\big\langle n\big|$, with $\varepsilon_0 < \varepsilon_1$. An operator $\hat{A}$ has the eigenkets $\big|a_1\big\rangle = \frac{1}{\sqrt{2}}\big(\big|0\big\rangle + \big|1\big\rangle\big)$ and $\big|a_2\big\rangle = \frac{1}{\sqrt{2}}\big(\big|0\big\rangle - \big|1\big\rangle\big)$ with the corresponding observables $a_{1,2}$. A measurement of $\hat{A}$ at $t = 0$ found the system in state $\big|a_1\big\rangle$. Find how its probability to remain in this state varies with time.*

After the measurement, the system is projected into $\big|a_1\big\rangle = \frac{1}{\sqrt{2}}\big(\big|0\big\rangle + \big|1\big\rangle\big)$, and then evolves in time according to Eqs. (7.5) and (7.13):

$$|a_1(t)\rangle = \hat{U}(t)\big|a_1\big\rangle = \frac{1}{\sqrt{2}}\Big[\big|\varepsilon_0(t)\big\rangle + \big|\varepsilon_1(t)\big\rangle\Big] = \frac{1}{\sqrt{2}}\left[\mathrm{e}^{-i\varepsilon_0 t/\hbar}|0\rangle + \mathrm{e}^{-i\varepsilon_1 t/\hbar}|1\rangle\right]\ .$$

The probability to remain in state $\big|a_1\big\rangle$ at time t is $P_1(t) = |\langle a_1(t) \,|\, a_1(0)\rangle|^2$. We have, using $\langle 0|\,0\rangle{=}\langle 1|\,1\rangle{=}1$ and $\langle 0 \,|\, 1\rangle{=}\langle 1 \,|\, 0\rangle{=}0$:

$$\langle a_1(t) \,|\, a_1(0)\rangle = \frac{1}{2}\left[\mathrm{e}^{i\varepsilon_0 t/\hbar}\big\langle 0\big| + \mathrm{e}^{i\varepsilon_1 t/\hbar}\big\langle 1\big|\right]\Big[|0\rangle + |1\rangle\Big] = \frac{1}{2}\left[\mathrm{e}^{i\varepsilon_0 t/\hbar} + \mathrm{e}^{i\varepsilon_1 t/\hbar}\right]$$

$$\Rightarrow \quad P_1(t) = \frac{1}{4}\left[\mathrm{e}^{i\varepsilon_0 t/\hbar} + \mathrm{e}^{i\varepsilon_1 t/\hbar}\right]\left[\mathrm{e}^{-i\varepsilon_0 t/\hbar} + \mathrm{e}^{-i\varepsilon_1 t/\hbar}\right] = \frac{1 + \cos\left[(\varepsilon_1 - \varepsilon_0)t\right]}{2}.$$

Starting with $P_1(t=0) = 1$, it periodically oscillates between 1 and 0.

[2] We assume $\hat{A}$ does not explicitly depend on t; stationarity would not work for operators like $\hat{x}(t) = \hat{x} + \hat{v}t$.

Lecture 8. Heisenberg picture. Correspondence principle.

Here we introduce the Heisenberg picture where time evolution can be described with *static* wavefunctions but *time-dependent* operators, in contrast to the Schrödinger picture which we have habitually used (albeit without attribution), where the time dependence is only in the wavefunctions, unless an operator is explicitly time-dependent.

8.1 Ehrenfest theorem

In the previous lecture we have seen that there was no time dependence in the expectation value of an operator $\hat{B}$ in a stationary state, the latter is a time-evolved eigenstate of a Hamiltonian, Eq. (7.13).

Now we shall find the expectation value of $\hat{B}$ in a general state, $\langle\hat{B}\rangle = \big\langle\Psi(t)\big|\hat{B}\big|\Psi(t)\big\rangle$, writing $\big|\Psi(t)\big\rangle$ as a linear superposition in the eigenbasis of Hamiltonian, $\big\{\big|\varepsilon_n\big\rangle\big\}$ (with $\big|\Psi\big\rangle \equiv \Psi(0)$), similar to Eq. (7.14)

$$\big|\Psi(t)\big\rangle = \sum_n c_n(t)\big|\varepsilon_n\big\rangle\,, \qquad c_n(t) = \langle\varepsilon_n \mid \Psi(t)\rangle = \mathrm{e}^{-i\varepsilon_n t}c_n\,. \tag{8.1}$$

where we used Eq. (7.11). Hence, we find the expectation value of the operator $\hat{B}$ as

$$\begin{aligned}\langle\hat{B}(t)\rangle \equiv \big\langle\Psi(t)\,\big|\,\hat{B}\,\big|\,\Psi(t)\big\rangle &= \sum_m \mathrm{e}^{i\varepsilon_m t}c_m\big\langle\varepsilon_m\big|\hat{B}\sum_n \mathrm{e}^{i\varepsilon_n t}c_n\big|\varepsilon_n\big\rangle c_n \\ &= \sum_{n,m}\mathrm{e}^{it(\varepsilon_n-\varepsilon_m)/\hbar}c_n^*c_m\big\langle\varepsilon_n\big|\hat{B}\big|\varepsilon_m\big\rangle\,.\end{aligned} \tag{8.2}$$

In general, $\langle\hat{B}(t)\rangle \neq \langle\hat{B}(0)\rangle$, so that the average oscillates in time, in contrast to the operator average in a stationary state, $\big\langle\varepsilon_n(t)\,\big|\,\hat{B}\,\big|\,\varepsilon_n(t)\big\rangle$. Nevertheless, under some conditions on $\hat{B}$, the expectation value in Eq. (8.2) can still be t-independent. To understand what are these conditions let us differentiate both sides of Eq. (8.2) (assuming, as usual, no explicit time-dependence of $\hat{B}$ and $\hat{H}$):

$$\begin{aligned}i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\big\langle\Psi(t)\big|\hat{B}\big|\Psi(t)\big\rangle &= i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\big\langle\Psi\big|\mathrm{e}^{+it\hat{H}/\hbar}\hat{B}\mathrm{e}^{-it\hat{H}/\hbar}\big|\Psi\big\rangle \\ &= i\hbar\Big\langle\Psi\,\Big|\,\frac{\mathrm{d}\mathrm{e}^{it\hat{H}/\hbar}}{\mathrm{d}t}\hat{B}\mathrm{e}^{-it\hat{H}/\hbar}\Big|\Psi\Big\rangle + i\hbar\Big\langle\Psi\,\Big|\,\mathrm{e}^{it\hat{H}/\hbar}\hat{B}\frac{\mathrm{d}\mathrm{e}^{-it\hat{H}/\hbar}}{\mathrm{d}t}\Big|\Psi\Big\rangle \\ &= -\underbrace{\big\langle\Psi\big|\mathrm{e}^{+it\hat{H}/\hbar}}_{\big\langle\Psi(t)\big|}\hat{H}\hat{B}\mathrm{e}^{-it\hat{H}/\hbar}\big|\Psi\big\rangle + \big\langle\Psi\big|\mathrm{e}^{+it\hat{H}/\hbar}\hat{B}\hat{H}\underbrace{\mathrm{e}^{-it\hat{H}/\hbar}\big|\Psi\big\rangle}_{\big|\Psi(t)\big\rangle} \\ &= \Big\langle\Psi(t)\,\Big|\,\big[\hat{B},\hat{H}\big]\,\Big|\,\Psi(t)\Big\rangle\,.\end{aligned} \tag{8.3}$$

The expectation value is time-independent when its time derivative vanishes which is therefore equivalent to the condition of compatibility, $\big[\hat{B},\hat{H}\big] = 0$, of the observable $\hat{B}$ and the Hamiltonian $\hat{H}$. This is equivalent to the statements that $\hat{B}$ is diagonal in the energy eigenbasis, so that only time-independent diagonal terms survive in the double sum over eigenstates we derived earlier.

If $\hat{B}$ is incompatible with $\hat{H}$, $\left[\hat{B},\hat{H}\right]\neq 0$, then we have just derived the **Ehrenfest theorem**:

$$\boxed{i\hbar\frac{\mathrm{d}\langle\hat{B}\rangle_t}{\mathrm{d}t}=\left\langle\left[\hat{B},\hat{H}\right]\right\rangle_t} \tag{8.4}$$

where the expectation value is $\langle\ldots\rangle_t=\left\langle\Psi(t)\right|\ldots\left|\Psi(t)\right\rangle$. We will return to this (very significant) theorem at the end of this lecture. Meanwhile, note that the time-dependence was contained in the time-dependent states $\left|\Psi(t)\right\rangle$ while the operators (typically some functions of $\hat{x}$ and $\hat{p}$) were static. This is called the **Schrödinger picture**. We can re-attribute time-dependence to operators, keeping all states Ψ static thus arriving at the **Heisenberg picture** of QM. Note in advance that both pictures are totally equivalent, but in different applications one could be more convenient than the other.

8.2 Heisenberg picture

In the expectation value $\langle\hat{B}\rangle_t$ we re-attribute the t-dependence:

$$\langle\hat{B}\rangle_t=\left\langle\Psi(t)\,\middle|\,\hat{B}\,\middle|\,\Psi(t)\right\rangle=\left\langle\Psi(0)\,\middle|\,\mathrm{e}^{+it\hat{H}/\hbar}\hat{B}\mathrm{e}^{-it\hat{H}/\hbar}\,\middle|\,\Psi(0)\right\rangle\equiv\left\langle\Psi\,\middle|\,\hat{B}_{\mathrm{H}}(t)\,\middle|\,\Psi\right\rangle,$$
$$\hat{B}_{\mathrm{H}}(t)\equiv\mathrm{e}^{+it\hat{H}/\hbar}\hat{B}\mathrm{e}^{-it\hat{H}/\hbar}\equiv\hat{U}^{\dagger}(t)\hat{B}\hat{U}(t). \tag{8.5}$$

Here we have defined the *time-dependent operator* $\hat{B}_{\mathrm{H}}(t)$, incorporating the time-dependence into the observable, while keeping the states $\left|\Psi\right\rangle=\left|\Psi(0)\right\rangle$ static. The label $_{\mathrm{H}}$ denotes Heisenberg picture to distinguish the time-dependent operator $B_{\mathrm{H}}(t)$ from the time-independent one, $\hat{B}$, in the Schrödinger picture.[1] The Hamiltonian remains the same in both pictures as $\mathrm{e}^{+it\hat{H}/\hbar}\hat{H}\mathrm{e}^{-it\hat{H}/\hbar}=\hat{H}$.

Both Heisenberg and Schrödinger pictures provide the same time dependence of expectation values of observables $\langle\hat{B}\rangle_t$ so that they are completely equivalent and their usage is a matter of convenience: one can use time-dependent operators or time-dependent states, but not *both* (there are mixed pictures as well but they will not be used in this course).

Now we re-attribute the time-dependence in the Ehrenfest theorem, Eq. (8.4). The l.h.s. is the expectation value of an operator which is the same in both pictures, $\langle\hat{B}_H\rangle_t=\langle\hat{B}\rangle_t$. In the r.h.s. we have

$$\left\langle\Psi\,\middle|\,i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\,\middle|\,\Psi\right\rangle=\left\langle\left[\hat{B},\hat{H}\right]\right\rangle_t\equiv\left\langle\Psi(t)\,\middle|\,\left[\hat{b},\hat{H}\right]\,\middle|\,\Psi(t)\right\rangle=\left\langle\Psi\,\middle|\,\mathrm{e}^{it\hat{H}/\hbar}\left[\hat{B},\hat{H}\right]\mathrm{e}^{-it\hat{H}/\hbar}\,\middle|\,\Psi\right\rangle$$
$$=\left\langle\Psi\,\middle|\,\left[\mathrm{e}^{it\hat{H}/\hbar}\hat{B}\mathrm{e}^{-it\hat{H}/\hbar},\hat{H}\right]\,\middle|\,\Psi\right\rangle\equiv\left\langle\Psi\,\middle|\,\left[\hat{B}_H,\hat{H}\right]\,\middle|\,\Psi\right\rangle$$

As this is valid in any state $\left|\Psi\right\rangle$, we arrive at the Heisenberg quantum equation of motion

$$\boxed{i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\hat{B}_{\mathrm{H}}(t)=\left[\hat{B}_{\mathrm{H}}(t),\hat{H}\right]} \tag{8.6}$$

which plays the same role in the Heisenberg picture as the Schrödinger equation in the Schrödinger picture. I will not use the Heisenberg picture in subsequent lectures but want to stress that it is very important in many areas, especially in quantum field theory.

[1] Note that time-dependent operators do exist in the Schrödinger picture – e.g., one corresponding to the vector-potential of the electromagnetic field, $\hat{\boldsymbol{A}}(t)$. Then the label is compulsory as $\hat{A}_{\mathrm{H}}(t)=\hat{U}^{\dagger}(t)\hat{A}(t)\hat{U}(t)$.

8.3 Correspondence Principle

Let us now return to the Ehrenfest Theorem, Eq. (8.4). Consider a **very important example** of the time evolution of operators $\hat{x}_{\rm H}$ and $\hat{p}_{\rm H}$ in one-particle system with the standard Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})\,. \tag{8.7}$$

We start with the time evolution of the expectation value of $\hat{x}$:

$$i\hbar\frac{\mathrm{d}\langle\hat{x}\rangle}{\mathrm{d}t} = \left\langle\left[\hat{x},\hat{H}\right]\right\rangle = \frac{1}{2m}\left\langle\left[\hat{x},\hat{p}\right]^2\right\rangle, \tag{8.8}$$

since $\left[\hat{x},V(\hat{x})\right]=0$. We calculate the commutator

$$\left[\hat{x},\hat{p}^2\right] = \left(\hat{x}\hat{p}\hat{p}-\hat{p}\hat{p}\hat{x}\right) = \left(\hat{x}\hat{p}\hat{p}-\hat{p}\hat{x}\hat{p}+\hat{p}\hat{x}\hat{p}-\hat{p}\hat{p}\hat{x}\right) = \Big(\underbrace{\left[\hat{x},\hat{p}\right]}_{=i\hbar}\hat{p}+\hat{p}\underbrace{\left[\hat{x},\hat{p}\right]}_{=i\hbar}\Big) = 2i\hbar\hat{p}. \tag{8.9}$$

Substituting this into Eq. (8.8), we find the same relation between the expectation values of the time derivative of position and the velocity as in classical mechanics:

$$\left\langle\frac{\mathrm{d}\hat{x}}{\mathrm{d}t}\right\rangle = \left\langle\frac{\hat{p}}{m}\right\rangle \equiv \langle\hat{v}\rangle \tag{8.10}$$

The find the time derivative of the expectation value of momentum we need the average commutator $\langle[\hat{p},V(\hat{x})]\rangle$, which is straightforward in the x-representation (using the resolution of unity and the locality of $\hat{V}(x)$, i.e. $\left\langle x\,\middle|\,\hat{V}(x)\,\middle|\,x'\right\rangle = \left\langle x\,\middle|\,\hat{V}(x)\,\middle|\,x\right\rangle\delta(x-x')$):

$$\begin{aligned}
\left\langle\Psi(t)\,\middle|\,\left[\hat{p},V(\hat{x})\right]\,\middle|\,\Psi(t)\right\rangle &= \int\limits_{-\infty}^{\infty}\mathrm{d}x\int\limits_{-\infty}^{\infty}\mathrm{d}x'\left\langle\Psi(t)\,\middle|\,x\right\rangle\left\langle x\,\middle|\,\left[\hat{p},V(\hat{x})\right]\,\middle|\,x'\right\rangle\left\langle x'\middle|\Psi(t)\right\rangle\\
&= -i\hbar\int\limits_{-\infty}^{\infty}\mathrm{d}x\Psi^*(x,t)\left(\frac{\mathrm{d}}{\mathrm{d}x}V(x)-V(x)\frac{\mathrm{d}}{\mathrm{d}x}\right)\Psi(x,t)\\
&= -i\hbar\int\limits_{-\infty}^{\infty}\mathrm{d}x\Psi^*(x,t)\left\{\frac{\mathrm{d}V(x)}{\mathrm{d}x}\Psi(x,t)+V(x)\frac{\mathrm{d}\Psi(x,t)}{\mathrm{d}x}-V(x)\frac{\mathrm{d}\Psi(x,t)}{\mathrm{d}x}\right\}\\
&= -i\hbar\int\limits_{-\infty}^{\infty}\mathrm{d}x\Psi^*(x,t)\frac{\mathrm{d}V(x)}{\mathrm{d}x}\Psi(x,t) = -i\hbar\int\limits_{-\infty}^{\infty}\mathrm{d}x\left\langle\Psi(t)\middle|x\right\rangle\left\langle x\,\middle|\,\frac{\mathrm{d}V(x)}{\mathrm{d}x}\Psi(t)\right\rangle\\
&= -i\hbar\left\langle\Psi(t)\,\middle|\,\frac{\mathrm{d}V(x)}{\mathrm{d}x}\,\middle|\,\Psi(t)\right\rangle
\end{aligned}$$

Substituting this into the Ehrenfest theorem for $\hat{p}$ and cancelling $i\hbar$, we arrive at

$$\left\langle\Psi(t)\,\middle|\,\frac{\mathrm{d}\langle\hat{p}\rangle}{\mathrm{d}t}\,\middle|\,\Psi(t)\right\rangle = -\left\langle\Psi(t)\,\middle|\,\frac{\mathrm{d}V(x)}{\mathrm{d}x}\,\middle|\,\Psi(t)\right\rangle \quad\Rightarrow\quad \left\langle\frac{\mathrm{d}\hat{p}}{\mathrm{d}t}\right\rangle = -\left\langle\frac{\mathrm{d}\hat{V}(x)}{\mathrm{d}x}\right\rangle. \tag{8.11}$$

In classical mechanics, $-\,\mathrm{d}V/\,\mathrm{d}x$ is the potential force. Hence, the expectation values of quantum operators obey classical laws – the classical counterpart of Eq. (8.11) is the 2$^{\rm nd}$ Newton's law.

Although we have used the Ehrenfest theorem for the expectation values of position and momentum only, we generalise this to any operators that have classical counterpart to formulate

The correspondence principle: Relations between expectation values of quantum operators are the same as classical relations between the corresponding classical variables (if they exist

Of course, this by no means reduces QM to classical mechanics, as we are talking only about expectation values, *i.e.* averages. Even leaving aside the fact that QM is inherently probabilistic and nonlocal (entanglement!), the HUP has already taught us that quantum fluctuations (*i.e.* deviations from the average) are totally non-classical. Note that this means, *e.g.*, that $\langle V(\hat{x})\rangle \neq \hat{V}(\langle\hat{x}\rangle)$. But when the fluctuations are important?

For an illustration, let us use a very well known example from QM1 and QM2 – a quantum particle in a box, i.e. in the potential $V(x) = 0$ for $|x| < a/2$ and $V(x) = \infty$ for $|x| \geqslant a/2$. Then the uncertainty in the position, $\Delta x \leqslant a$, as the particle cannot be outside the box. Hence, $\Delta p \gtrsim \hbar/a$ (we cannot use the exact minimum $\hbar/2$, which is only good for Gaussian wavefunction packet). Thus the uncertainty in the kinetic energy (caused by quantum fluctuations) $\Delta\mathcal{E} \gtrsim (\Delta p)^2/m \sim \hbar^2/ma^2$.
How large is this? For a spec of dust, $m \sim 10^{-6}\,\mathrm{kg}$ in a box of size of $1\mu = 10^{-6}\mathrm{m}$, we find $\Delta\mathcal{E} \sim 10^{-68}/10^{-18}\mathrm{J} \sim 10^{-50}\mathrm{J}$ so that one can suspect that QM is not relevant at a human scale (or even a dust scale). This is not the case, however, as such a naïve estimate only works for one-particle problems. Long-range quantum correlations (not to be confused with entanglement) in many-body systems lead QM to be relevant at macroscopic scales, as it governs such effects as superconductivity, superfluidity, etc.

Example 8.1. *Consider a TLS where $\hat{H} = \varepsilon_0|0\rangle\langle 0| + \varepsilon_1|1\rangle\langle 1|$, and $\hat{B}|\pm\rangle = \pm|\pm\rangle$ with $|\pm\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle \pm |1\rangle\big)$. At $t = 0$, the measurement of $\hat{B}$ returned $a_+ = 1$. Find how $\langle\hat{B}\rangle$ changes with time.*

First, we express $\hat{B}$ in the energy basis:

$$\hat{B} = \sum_{a_n=\pm} a_n|a_n\rangle\langle a_n| = |+\rangle\langle +| - |-\rangle\langle -|$$
$$= \tfrac{1}{2}\Big\{\Big(|0\rangle + |1\rangle\Big)\Big(\langle 0| + \langle 1|\Big) - \Big(|0\rangle - |1\rangle\Big)\Big(\langle 0| - \langle 1|\Big)\Big\} = |0\rangle\langle 1| + |1\rangle\langle 0|\,.$$

Then

$$\hat{H}\hat{B} = \Big[\varepsilon_0|0\rangle\langle 0| + \varepsilon_1|1\rangle\langle 1|\Big]\Big[|0\rangle\langle 1| + |1\rangle\langle 0|\Big] = \varepsilon_0|0\rangle\langle 1| + \varepsilon_1|1\rangle\langle 0|$$
$$\hat{B}\hat{H} = \Big[|0\rangle\langle 1| + |1\rangle\langle 0|\Big]\Big[\varepsilon_0|0\rangle\langle 0| + \varepsilon_1|1\rangle\langle 1|\Big] = \varepsilon_1|0\rangle\langle 1| + \varepsilon_0|1\rangle\langle 0|$$
$$\Rightarrow \quad [\hat{H},\hat{B}] = (\varepsilon_0 - \varepsilon_1)\,\big[|0\rangle\langle 1| - |1\rangle\langle 0|\big]$$

Now

$$|+\rangle(t) = \hat{U}|+\rangle = \tfrac{1}{\sqrt{2}}\mathrm{e}^{it\hat{H}/\hbar}\Big(|0\rangle + |1\rangle\Big) = \tfrac{1}{\sqrt{2}}\Big(\mathrm{e}^{-i\varepsilon_0 t/\hbar}|0\rangle + \mathrm{e}^{-i\varepsilon_1 t/\hbar}|1\rangle\Big)$$

so that

$$\Big\langle[\hat{H},\hat{B}]\Big\rangle_t = \tfrac{1}{2}(\varepsilon_0 - \varepsilon_1)\Big(\mathrm{e}^{i\varepsilon_0 t/\hbar}\langle 0| + \mathrm{e}^{i\varepsilon_1 t/\hbar}\langle 1|\Big)\Big[|0\rangle\langle 1| - |1\rangle\langle 0|\Big]\Big(\mathrm{e}^{-i\varepsilon_0 t/\hbar}|0\rangle + \mathrm{e}^{-i\varepsilon_1 t/\hbar}|1\rangle\Big)$$
$$= \tfrac{1}{2}(\varepsilon_0 - \varepsilon_1)\Big(\mathrm{e}^{i\varepsilon_0 t/\hbar}\langle 0| + \mathrm{e}^{i\varepsilon_1 t/\hbar}\langle 1|\Big)\Big(\mathrm{e}^{-i\varepsilon_1 t/\hbar}|0\rangle - \mathrm{e}^{-i\varepsilon_0 t/\hbar}|1\rangle\Big) = i(\varepsilon_0 - \varepsilon_1)\sin\frac{(\varepsilon_0 - \varepsilon_1)t}{\hbar}.$$

Now use the Ehrenfest theorem (noticing that $[\hat{H},\hat{B}] = -[\hat{B},\hat{H}]$):

$$i\hbar\frac{\mathrm{d}\langle\hat{B}\rangle_t}{\mathrm{d}t} = -i(\varepsilon_0 - \varepsilon_1)\sin\frac{(\varepsilon_0 - \varepsilon_1)t}{\hbar} \;\Rightarrow\; \mathrm{d}\langle\hat{B}\rangle_t = -\sin\frac{(\varepsilon_0 - \varepsilon_1)t}{\hbar}\,\mathrm{d}\left[\frac{(\varepsilon_0 - \varepsilon_1)t}{\hbar}\right]$$
$$\Rightarrow\; \langle\hat{B}\rangle_t = \cos\frac{(\varepsilon_0 - \varepsilon_1)t}{\hbar} + \mathrm{const}$$

As $\langle\hat{B}\rangle = 1$ at $t = 0$, the constant vanishes.

Lecture 9. Harmonic Oscillator. Ladder Operators

We have up to now applied techniques based on Dirac notations on simple TLS or similar. From now on, we shall apply this to multi-level system starting with describing the quantum simple harmonic oscillator (QHO) without solving the Schrödinger equation. Instead, we introduce raising and lowering operators – precursors of second quantisation techniques you learn in Y4 – and use their commutators to find the QHO eigenvalues and eigenvectors.

9.1 Dimensionless variables.

Our goal is to apply the knowledge accumulated so far to diagonalise the Hamiltonian of the QHO,

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2 \,. \tag{9.1}$$

i.e. to find its eigenvalues and eigenstates by solving the EVE *without reference to any particular basis* like the position representation used in QM2.

$$\hat{H}\big|n\big\rangle = \varepsilon_n\big|n\big\rangle \quad \Leftrightarrow \quad \big\langle n \,\big|\, \hat{H} \,\big|\, m\big\rangle = \varepsilon_n \delta^n{}_m \,. \tag{9.2}$$

Instead, we will work directly with states $\big|n\big\rangle$ and operators $\hat{p}$, $\hat{x}$ acting on them.

To simplify calculations, let us define the length unit ℓ (to be specified below) and hence dimensionless position and momentum operators:

$$\hat{X} \equiv \frac{\hat{x}}{\ell}\,, \quad \hat{x} = \ell\hat{X}\,; \qquad\qquad \hat{P} \equiv \frac{\ell}{\hbar}\hat{p} = -i\ell\frac{\mathrm{d}}{\mathrm{d}x} = -i\frac{\mathrm{d}}{\mathrm{d}X}\,, \quad \hat{p} = \frac{\hbar}{\ell}\hat{P}\,. \tag{9.3}$$

This means that we measure all lengths in units of ℓ and momenta in units of $\hbar/\ell$. Substituting this into Eq. (9.1), we rewrite the Hamiltonian as

$$\hat{H} = \frac{\hbar^2}{2m\ell^2}\hat{P}^2 + \frac{m\omega\ell^2}{2}\hat{X}^2, \tag{9.4}$$

and then choose ℓ in such a way as to have equal coefficients attached to $\hat{P}^2$ and $\hat{X}^2$, *i.e.*, $\hbar^2/m\ell^2 = m\omega^2\ell^2$, which results in

$$\ell^4 = \frac{\hbar^2}{m^2\omega^2} \quad \Rightarrow \quad \ell = \sqrt{\frac{\hbar}{m\omega}}\,. \tag{9.5}$$

We recognise in ℓ the characteristic length of QHO, which defines a typical size of its ground state wave-function.[1] Substituting ℓ into the Hamiltonian (9.4) we arrive at

$$\hat{H} = \frac{\hbar\omega}{2}\left(\hat{P}^2 + \hat{X}^2\right) \tag{9.6}$$

Here $\hbar\omega$ is the characteristic energy of the QHO, while $\frac{1}{2}\left(\hat{P}^2 + \hat{X}^2\right)$ is a dimensionless combination. Had P and X been numbers rather than operators, one would be able to factorise such a combination into $\frac{1}{2}(X + iP)(X - iP)$.

[1] This is known from QM2 but we will find the groundstate WF later without referring to any earlier knowledge.

9.2 Ladder operators

The factorisation above inspires one to define mutually adjoint **ladder operators**:

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\hat{X} + i\hat{P}\right), \qquad \hat{a}^\dagger = \frac{1}{\sqrt{2}}\left(\hat{X} - i\hat{P}\right). \tag{9.7}$$

The operator $\hat{a}$ is called ***lowering*** (or ***annihilation***) operator and its Hermitian conjugate, $\hat{a}^\dagger$, is called ***raising*** (or ***creation***) operator. The meaning of these names will become clear soon. Since $\hat{a}$ and $\hat{a}^\dagger$ are non-Hermitian, they do not correspond to any physical observables. However, we express Hermitian $\hat{H}$ and $\hat{P}$ as their combinations:

$$\hat{x} \equiv \ell\hat{X} = \frac{\ell}{\sqrt{2}}\left(\hat{a} + \hat{a}^\dagger\right), \qquad \hat{p} = \frac{\hbar}{\ell}\hat{P} = \frac{\hbar}{i\ell\sqrt{2}}\left(\hat{a} - \hat{a}^\dagger\right). \tag{9.8}$$

Then, from our exercise with factorising $X^2 + P^2$, we expect that the products $\hat{a}^\dagger\hat{a}$ and $\hat{a}\hat{a}^\dagger$ should be related to the Hamiltonian of Eq. (9.6). Using the commutator

$$\left[\hat{X}, \hat{P}\right] = \left[\frac{\hat{x}}{\ell}, \frac{\ell\hat{p}}{\hbar}\right] = \frac{1}{\hbar}\left[\hat{x}, \hat{p}\right] = i,$$

we find

$$\hat{a}^\dagger\hat{a} = \frac{1}{2}\left(\hat{X} - i\hat{P}\right)\left(\hat{X} + i\hat{P}\right) = \frac{1}{2}\left(\hat{X}^2 + \hat{P}^2 + i[\hat{X}, \hat{P}\right) = \frac{1}{2}\left(\hat{X}^2 + \hat{P}^2 - 1\right). \tag{9.9}$$

Comparing this with Eq. (9.6), we rewrite the QHO Hamiltonian as

$$\hat{H} = \frac{\hbar\omega}{2}\left(\hat{P}^2 + \hat{X}^2\right) = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\right) \equiv \hbar\omega\left(\hat{N} + \tfrac{1}{2}\right). \tag{9.10}$$

The Hermitian combination $\hat{N} = \hat{a}^\dagger\hat{a}$ plays an important role and is called the **number operator**. Finding its eigenstates and eigenvalues from the EVE,

$$\hat{N}\big|n\big\rangle = n\big|n\big\rangle, \tag{9.11}$$

will obviously solve the QHO problem giving the energy levels as[2]

$$\boxed{\varepsilon_n = \hbar\omega(n + \tfrac{1}{2})}. \tag{9.12}$$

It looks exactly what we learned in QM2, except from the fact that we still are to prove that n is a non-negative integer number. Non-negativity is straightforward: if we call $\hat{a}\big|n\big\rangle \equiv \big|\Psi\big\rangle$, we take the inner product on both sides of Eq. (9.11), noticing that $\langle\Psi| = \langle n|\,\hat{a}^\dagger$:

$$n = n\langle n\,|\,n\rangle = \left\langle n\,\middle|\,\hat{N}\,\middle|\,n\right\rangle \equiv \left\langle n\,\middle|\,\hat{a}^\dagger\hat{a}\,\middle|\,n\right\rangle \equiv \langle\Psi\,|\,\Psi\rangle \geqslant 0. \tag{9.13}$$

To prove that n is integer we need to formulate commutation relations.

[2]The eigenkets of $\hat{H}$ and $\hat{N}$ are obviously the same which justifies the simplification of notations, $\big|\varepsilon_n\big\rangle \to \big|n\big\rangle$. As usual, we assume the basis $\big\{\big|n\big\rangle\big\}$ to be orthonormal, $\langle n\,|\,m\rangle = \delta_{nm}$.

9.3 Commutation relations and ladders of states.

Similarly to Eq. (9.9), we calculate $\hat{a}\hat{a}^\dagger$:

$$\hat{a}\hat{a}^\dagger = \frac{1}{2}\Big(\hat{X} + i\hat{P}\Big)\Big(\hat{X} - i\hat{P}\Big) = \frac{1}{2}\left(\hat{X}^2 + \hat{P}^2 + i[\hat{P}, \hat{X}]\right) = \frac{1}{2}\Big(\hat{X}^2 + \hat{P}^2 + 1\Big)\,.$$

Subtracting from this $\hat{a}^\dagger\hat{a}$, Eq. (9.9), we arrive at the following commutator:

$$\boxed{\big[\hat{a}, \hat{a}^\dagger\big] = 1\,.} \tag{9.14}$$

We use it to calculate commutation relations of $\hat{N}$ with the ladder operators, which would allow us, amongst other things, to prove that n are integers.

$$[\hat{N}, \hat{a}^\dagger] = [\hat{a}^\dagger\hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}^\dagger\hat{a} = \hat{a}^\dagger\big[\hat{a}, \hat{a}^\dagger\big] = \hat{a}^\dagger \tag{9.15a}$$

$$[\hat{N}, \hat{a}\ ] = [\hat{a}^\dagger\hat{a}, \hat{a}] = \hat{a}^\dagger\hat{a}\hat{a} - \hat{a}\hat{a}^\dagger\hat{a} = \big[\hat{a}^\dagger, \hat{a}\big]\hat{a} = -\hat{a}\,. \tag{9.15b}$$

Let us now act on by $\hat{a}$ on the eigenket $\big|n\big\rangle$ to produce a new state vector, $\hat{a}\big|n\big\rangle$. Next, we apply the number operator to this state and note that from Eq. (9.15b) follows $\hat{N}\hat{a} = \hat{a}\hat{N} - \hat{a}$. Then

$$\hat{N}\hat{a}\big|n\big\rangle = \hat{a}\hat{N}\big|n\big\rangle - \hat{a}\big|n\big\rangle = n\hat{a}\big|n\big\rangle - \hat{a}\big|n\big\rangle = (n{-}1)\hat{a}\big|n\big\rangle\,. \tag{9.16}$$

We see that $\hat{a}\big|n\big\rangle$ is also an eigenket of $\hat{N}$ with the eigenvalue $n{-}1$. Hence it is proportional (rather than equal) to the normalised eigenket, $\big|n{-}1\big\rangle$, corresponding to the eigenvalue $n{-}1$ *i.e.* $\hat{a}\big|n\big\rangle = c_n\big|n{-}1\big\rangle$. To find the normalisation coefficient c_n, we bracket the r.h.s. and l.h.s. with the corresponding r.h.s. and l.h.s. of the bra eigenvector, $\big\langle n\big|\hat{a}^\dagger = \big\langle n{-}1\big|c_n^*$, to get the following inner product:

$$c_n^* c_n \underbrace{\big\langle n{-}1\,\big|\,n{-}1\big\rangle}_{=1} = |c_n|^2 = \big\langle n\big|\hat{a}^\dagger\hat{a}\big|n\big\rangle \equiv \big\langle n\big|\hat{N}\big|n\big\rangle = n\underbrace{\big\langle n\,\big|\,n\big\rangle}_{=1} = n \quad\Rightarrow\quad |c_n|^2 = n\,. \tag{9.17}$$

Hence $|c_n| = \sqrt{n}$ and we can choose a trivial phase factor, putting $c_n = \sqrt{n}$. We have thus established the action of the lowering operator,

$$\hat{a}\big|n\big\rangle = \sqrt{n}\big|n{-}1\big\rangle\,. \tag{9.18}$$

Therefore, starting from an arbitrary state $\big|n\big\rangle$ we create a ***descending ladder*** of states, $\big|n{-}1\big\rangle$, $\big|n-2\big\rangle$, ... by repeatedly acting with operator $\hat{a}$. As n must remain non-negative, Eq. (9.13), while the eigenvalues decrease by integer steps, n must be integer so that the ladder naturally stops at $n = 0$. The penultimate step in the descending ladder is hence $\hat{a}\big|1\big\rangle = \big|0\big\rangle$, while the ultimate one is $\hat{a}\big|0\big\rangle = 0$. Hence the eigenket $\big|0\big\rangle$ corresponds to the groundstate with the energy $\frac{1}{2}\hbar\omega$, as seen from Eq. (9.12) which, with non-negative integer n, gives the entire eigenvalue spectrum.

The operator $\hat{a}^\dagger$ allows to build the ***ascending ladder*** of eigenstates. To see this we use Eq. (9.15a) as $\hat{N}\hat{a}^\dagger = \hat{a}^\dagger\hat{N} + \hat{a}^\dagger$ to calculate

$$\hat{N}\hat{a}^\dagger\big|n\big\rangle = \hat{a}^\dagger\hat{N}\big|n\big\rangle + \hat{a}^\dagger\big|n\big\rangle = n\hat{a}^\dagger\big|n\big\rangle + \hat{a}^\dagger\big|n\big\rangle = (n{+}1)\hat{a}^\dagger\big|n\big\rangle\,. \tag{9.19}$$

and thus $\hat{a}^\dagger|n\rangle = d_n|n{+}1\rangle$, where d_n is a normalisation constant obtained, similarly to c_n, from

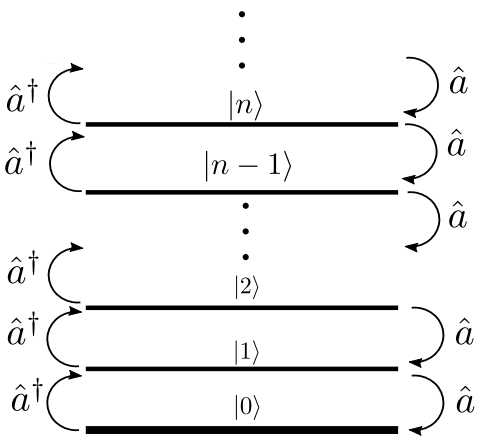
$$d_n^* d_n \underbrace{\langle n{+}1 \,|\, n{+}1\rangle}_{=1} = |d_n|^2 = \langle n|\hat{a}\hat{a}^\dagger|n\rangle = \langle n|\hat{a}^\dagger\hat{a}+1|n\rangle = \langle n|\hat{N}+1|n\rangle = (n{+}1)\underbrace{\langle n \,|\, n\rangle}_{=1} = n$$
$$\Rightarrow \quad |d_n|^2 = n{+}1 \mapsto d_n = \sqrt{n{+}1}.$$

Hence we have derived the action of the raising operator $\hat{a}^\dagger$ on state $|n\rangle$:

$$\hat{a}^\dagger|n\rangle = \sqrt{n{+}1}|n{+}1\rangle. \tag{9.20}$$

Therefore, starting from the ground state $|0\rangle$ we climb up the ladder of states:

$$\hat{a}^\dagger|0\rangle = |1\rangle, \qquad \hat{a}^\dagger|1\rangle = \sqrt{2}|2\rangle, \qquad \ldots \qquad \hat{a}^\dagger|n{-}1\rangle = \sqrt{n}|n\rangle, \ldots \tag{9.21}$$



In other words we can obtain any state $|n\rangle$ as a result of repeated action of the creation operator:

$$|n\rangle = \frac{\hat{a}^\dagger}{\sqrt{n}}|n{-}1\rangle = \frac{\hat{a}^\dagger}{\sqrt{n}}\frac{\hat{a}^\dagger}{\sqrt{n{-}1}}|n-2\rangle = \ldots = \frac{\hat{a}^\dagger}{\sqrt{n}}\frac{\hat{a}^\dagger}{\sqrt{n{-}1}}\ldots\frac{\hat{a}^\dagger}{\sqrt{2}}\frac{\hat{a}^\dagger}{\sqrt{1}}|0\rangle = \frac{\left(\hat{a}^\dagger\right)^n}{\sqrt{n!}}|0\rangle\,. \tag{9.22}$$

The state $|n\rangle$ can be described as a state in which n quanta of oscillations, each contributing energy $\hbar\omega$ on top of the ground state energy $\varepsilon_0 = \hbar\omega/2$, which is called the energy of zero-point oscillations. These quanta are *created* by the *creation* operator $\hat{a}^\dagger$ and *annihilated* (or destroyed) by the *annihilation* operator $\hat{a}$ – the other names for the ladder operators which are ubiquitous in modern physics. The QHO is fundamental in the high-energy or many-body physics where its eigenstates represent quanta of different quantum fields interpreted as particles, like photons or mesons. In condensed matter physics, different modes of crystal vibrations are also modelled by harmonic oscillators.

To summarise, the commutation relations of $\hat{a}$ and $\hat{a}^\dagger$ allowed us to construct the eigenstates of the QHO and to find their energies. The ladder states we have obtained are states in the abstract Hilbert space. To relate them to the wavefunctions $\Psi_n(x)$ obtained by solving the Schrödinger equation, we need to take usual projections onto eigenstates of position operator,[3] *i.e.* $\Psi_n(x) = \langle x \,|\, n\rangle$. In the next lecture we will obtain these wave function by using position representation of the ladder operators $\hat{a}$, $\hat{a}^\dagger$. The advantage is that we will be solving the 1$^{\rm st}$ order differential equations (which are always solvable), in contrast to the 2$^{\rm nd}$ order Schrödinger equation.

[3]One can also ask what are these states in the momentum representation, $\tilde{\Psi}_n(p) = \langle p \,|\, n\rangle$.

Lecture 10. QHO wavefunctions and operator matrix elements

In this lecture we recover wave functions of the Harmonic Oscillator without solving the Schrödinger equation, and calculate matrix elements of position and momentum operators.

10.1 Wavefunctions of QHO

In the previous lecture we have constructed the eigenstates of the Harmonic Oscillator Hamiltonian in terms of the creation operator $\hat{a}^\dagger$ acting on the vacuum state:

$$\big|n\big\rangle = \frac{\left(\hat{a}^\dagger\right)^n}{\sqrt{n!}}\big|0\big\rangle\,. \tag{10.1}$$

To obtain wave functions corresponding to these states we project onto position eigenstates

$$\Psi_n(x) = \langle x\,|\,n\rangle = \frac{1}{\sqrt{n!}}\big\langle x\big|\left(\hat{a}^\dagger\right)^n\big|0\big\rangle\,. \tag{10.2}$$

Let us begin with the ground (vacuum state). It is annihilated by the action of the lowering operator,

$$\hat{a}\big|0\big\rangle = 0 \quad\Rightarrow\quad \big\langle X\big|\hat{a}\big|0\big\rangle = 0\,. \tag{10.3}$$

Here we use the rescaled position representation, with dimensionless $X \equiv x/\ell$ being eigenvalues of the operator $\hat{X}$ defined by Eq. (9.3). The action of the dimensionless $\hat{P}$ on $\big|X\big\rangle \equiv \big|x/\ell\big\rangle$ is straightforwardly obtained by scaling as

$$\hat{P}\big|X\big\rangle = \frac{\ell\hat{p}}{\hbar}\left|\frac{x}{\ell}\right\rangle = \frac{1}{\hbar}\hat{p}\big|x\big\rangle = \frac{1}{\hbar}\left(-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\right)\big|x\big\rangle = -i\frac{\mathrm{d}}{\mathrm{d}X}\big|X\big\rangle.$$

Then the 2nd part of Eq. (10.3) can be rewritten as a 1st-order differential equation by substituting the definition of $\hat{a}$, Eq. (9.7), in terms of $\hat{X}$ and $\hat{P}$:

$$0 = \big\langle X\big|\hat{a}\big|0\big\rangle = \frac{1}{\sqrt{2}}\left\langle X\,\middle|\,\hat{X}+i\hat{P}\,\middle|\,0\right\rangle = \frac{1}{\sqrt{2}}\left(X+\frac{\mathrm{d}}{\mathrm{d}X}\right)\langle X\,|\,0\rangle \;\Rightarrow\; \frac{\mathrm{d}\Psi_0(X)}{\mathrm{d}X} = -X\Psi_0(X)\,.$$

The solution is straightforward:

$$\frac{\mathrm{d}\Psi_0(X)}{\Psi_0(X)} = -X\,\mathrm{d}X \quad\Rightarrow\quad \ln|\Psi_0(X)| = -\tfrac{1}{2}X^2 + \mathrm{const} \quad\Rightarrow\quad \Psi_0(X) = A\mathrm{e}^{-\frac{1}{2}X^2}, \tag{10.4}$$

where A is a normalisation constant. To normalise, we return to dimensional variable $x \equiv \ell X$ so that $\Psi_0(x) = \alpha\exp\left[-\frac{x^2}{2\ell^2}\right]$ where $\ell^2 \equiv \hbar/m\omega$, Eq. (9.5) (and the dimensionless $A = \alpha\sqrt{\ell}$). The normalisation condition gives

$$1 = \alpha^2\int\limits_{-\infty}^{\infty}\mathrm{d}x\,\mathrm{e}^{-x^2/\ell^2} = \alpha^2\ell\int\limits_{-\infty}^{\infty}\mathrm{d}X\,\mathrm{e}^{-X^2} = \alpha^2\ell\sqrt{\pi} \;\Rightarrow\; \alpha = \frac{1}{\sqrt[4]{\ell^2\pi}} \equiv \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}\,. \tag{10.5}$$

Hence we have calculated the groundstate wavefunction (originally obtained by solving the 2nd order ODE – the Schrödinger equation):

$$\Psi_0(x) \equiv \langle x\,|\,0\rangle = \left(\pi\ell^2\right)^{-\frac{1}{4}}\mathrm{e}^{-x^2/2\ell^2} \equiv \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}\exp\left[-\frac{m\omega x^2}{2\hbar}\right]. \tag{10.6}$$

Once the wavefunction is constructed for the ground state, all wavefunctions for excited states, Eq. (10.2) are obtained algebraically by using representation of $\hat{a}^\dagger$ via $\hat{X}$ and $\hat{P}$, Eq. (9.7). We again start with the dimensionless variables:

$$\Psi_n(X) = \frac{1}{\sqrt{n!}}\Big\langle X \,\Big|\, \left(\hat{a}^\dagger\right)^n \,\Big|\, 0\Big\rangle = \frac{1}{\sqrt{2^n n!}}\Big\langle X \,\Big|\, \left(\hat{X}-i\hat{P}\right)^n \,\Big|\, 0\Big\rangle$$
$$= \frac{1}{\sqrt{2^n n!}}\left(X-\frac{\mathrm{d}}{\mathrm{d}X}\right)^n \Psi_0(X) = \frac{A}{\sqrt{2^n n!}}\left(X-\frac{\mathrm{d}}{\mathrm{d}X}\right)^n \mathrm{e}^{-X^2/2}\,. \tag{10.7}$$

For example, the wavefunction of the 1st excited state is

$$\Psi_1(X) = \frac{A}{\sqrt{2}}(X+X)\mathrm{e}^{-X^2/2} = AX\sqrt{2}\mathrm{e}^{-X^2/2} \;\Rightarrow\; \Psi_1(x) = \frac{\alpha x\sqrt{2}}{\ell}\mathrm{e}^{-x^2/2\ell^2},$$

with the same normalisation coefficient, $A = \alpha\sqrt{\ell}$, given by Eq. (10.5).

Repeating a similar calculation in Eq. (10.7) n times will result in the same Gaussian multiplied by a certain polynomial of degree n, $H_n(x)$, which is called the Hermite polynomial:

$$\Psi_n(X) = \frac{A}{\sqrt{2^n n!}}H_n(X)\mathrm{e}^{-X^2/2} \quad\Rightarrow\quad \Psi_n(x) = \frac{\alpha}{\sqrt{2^n n!}}H_n\left(\frac{x}{\ell}\right)\mathrm{e}^{-x^2/2\ell^2}\,. \tag{10.8}$$

It is possible to show that the polynomial $H_n(X)$ satisfies the following Rodrigues relation:

$$H_n(X) = \mathrm{e}^{X^2/2}\left(X-\frac{\mathrm{d}}{\mathrm{d}X}\right)^n e^{-X^2/2} = (-1)^n e^{X^2}\frac{\mathrm{d}^n}{\mathrm{d}X^n}e^{-X^2}\,. \tag{10.9}$$

The first part of the above definition directly follows from Eq. (10.7), while the second could be proved in many ways, e.g., using the mathematical induction method.

10.2 Matrix elements and expectation values of operators in QHO eigenstates

Given we now know the wave functions $\Psi_n(x) = \langle x \,|\, n\rangle$ of the QHO we can use them to calculate expectation values of any observable by the standard Schrödinger prescription:

$$\big\langle n\big|\hat{A}\big|n\big\rangle = \int \mathrm{d}x\Psi_n^*(x)\hat{A}\Psi_n(x)\,. \tag{10.10}$$

However, we do not need to do this: one of the main advantages of introducing the ladder operators is that we find matrix elements of operators without performing integrals. Instead, we shall express any operator that is a function of the position and momentum operators, $\hat{x}$ and $\hat{p}$, via the ladder operators $\hat{a}$, $\hat{a}^\dagger$, using Eq. (9.8),

$$\hat{X} \equiv \frac{\hat{x}}{\ell} = \frac{1}{\sqrt{2}}\left(\hat{a}+\hat{a}^\dagger\right)\,, \qquad\qquad \hat{P} \equiv \frac{\ell\hat{p}}{\hbar} = \frac{i}{\sqrt{2}}\left(\hat{a}^\dagger-\hat{a}\right)\,. \tag{10.11}$$

and use the action of the ladder operators on the eigenkets, Eqs. (9.18) and (9.20):

$$\hat{a}\big|n\big\rangle = \sqrt{n}\big|n-1\big\rangle\,, \qquad \hat{a}^\dagger\big|n\big\rangle = \sqrt{n+1}\big|n+1\big\rangle \tag{10.12}$$

Thus to get matrix elements of any $\hat{A}(\hat{x},\hat{p})$, we only need to know the matrix elements of the ladder operators, which immediately follow from Eq. (10.12) and the orthonormality of the basis $\{|n\rangle\}$:

$$\langle m|\hat{a}|n\rangle = \sqrt{n}\langle m\,|\,n-1\rangle = \sqrt{n}\,\delta_{m,n-1} = \sqrt{m+1}\,\delta_{n,m+1}\,, \tag{10.13}$$

$$\langle m|\hat{a}^\dagger|n\rangle = \sqrt{n+1}\langle m\,|\,n+1\rangle = \sqrt{n+1}\,\delta_{m,n+1} = \sqrt{m}\,\delta_{n,m-1}\,. \tag{10.14}$$

For illustration purposes, we can write the appropriate matrices, although for all practical applications we only need to know matrix elements. Note that for both ladder operators these matrices contain only one non-zero diagonal which is next to the main one:

$$\{\hat{a}\}_{nm} = \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & 0 & \dots \\ 0 & 0 & \sqrt{2} & 0 & 0 & \dots \\ 0 & 0 & 0 & \sqrt{3} & 0 & \dots \\ 0 & 0 & 0 & 0 & \sqrt{4} & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad \{\hat{a}^\dagger\}_{nm} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \dots \\ \sqrt{1} & 0 & 0 & 0 & 0 & \dots \\ 0 & \sqrt{2} & 0 & 0 & 0 & \dots \\ 0 & 0 & \sqrt{3} & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{10.15}$$

Obviously, they are Hermitian conjugates of each other.

Using Eq. (10.13) one can calculate matrix elements of products of $\hat{a}$ and $\hat{a}^\dagger$. Consider, for example, the number operator,

$$\langle m|\hat{N}|n\rangle = \langle m|\hat{a}^\dagger\hat{a}|n\rangle = \sum_{l=0}^{\infty}\langle m|\hat{a}^\dagger|l\rangle\langle l|\hat{a}|n\rangle = \sum_{l=0}^{\infty}\sqrt{m}\,\delta_{l,m-1}\sqrt{l+1}\,\delta_{l,n-1}\,. \tag{10.16}$$

To perform the summation we notice that Krönecker δ's imply that l must be equal to $m-1$ and $n-1$ at the same time. This is only possible if $n=m$ and we have

$$\langle m|\hat{N}|n\rangle = \sum_{l=0}^{\infty}\sqrt{m}\,\delta_{l,m-1}\sqrt{l+1}\,\delta_{l,n-1} = \sqrt{n}\sqrt{n}\delta_{n,m} = n\delta_{n,m}\,. \tag{10.17}$$

as it should be, as $\hat{N}|n\rangle = n|n\rangle$. As a less trivial example of matrix elements consider a product of operators

$$\langle m|\hat{a}\hat{a}|n\rangle = \sum_{l=0}^{\infty}\langle m|\hat{a}|l\rangle\langle l|\hat{a}|m\rangle = \sum_{l=0}^{\infty}\sqrt{m+1}\,\delta_{l,m+1}\sqrt{l+1}\,\delta_{n,l+1} \tag{10.18}$$

$$= \sqrt{m+1}\sqrt{m+2}\delta_{n,m+2} = \sqrt{n(n-1)}\delta_{n,m+2}\,. \tag{10.19}$$

where we have used the first Kronecker delta to perform the summation over intermediate state l. In fact it would be easier to simply act with the annihilation operator on $|n\rangle$ and then on the state $|n-1\rangle$:

$$\langle m|\hat{a}\hat{a}|n\rangle = \langle m|\hat{a}\sqrt{n}|n-1\rangle = \sqrt{n(n-1)}\langle m\,|\,n-2\rangle = \sqrt{n(n-1)}\delta_{m,n-2}\,,$$

getting the same result as before. Of course in these formulas for $n=0,1$ the answer is 0 as the ground state is annihilated by $\hat{a}$. Similarly,

$$\langle m|\hat{a}^\dagger\hat{a}^\dagger|n\rangle = \langle m|\hat{a}^\dagger\sqrt{n+1}|n+1\rangle = \sqrt{(n+2)(n+1)}\langle m\,|\,n+2\rangle = \sqrt{(n+2)(n+1)}\delta_{m,n+2}\,,$$

The matrix elements of creation and annihilation operators allow us to calculate matrix elements of various combinations of position and momentum operators by simply expanding them in $\hat{a}$, $\hat{a}^\dagger$. Here are some important examples:

$$\langle n|\hat{x}|n\rangle = \frac{\ell}{\sqrt{2}}\left(\langle n|\hat{a}|n\rangle + \langle n|\hat{a}^\dagger|n\rangle\right) = 0$$

as diagonal matrix elements of $\hat{a}$, $\hat{a}^\dagger$ are zero. For the same reason $\langle n|\hat{p}|n\rangle = 0$.[1] The diagonal matrix elements of $\hat{x}^2$, $\hat{p}^2$ are non zero:

$$\begin{aligned}
\langle n|\hat{x}^2|n\rangle &= \frac{\ell^2}{2}\langle n|(\hat{a}+\hat{a}^\dagger)(\hat{a}+\hat{a}^\dagger)|n\rangle \\
&= \frac{\ell^2}{2}\langle n|\hat{a}\hat{a}+\hat{a}\hat{a}^\dagger+\hat{a}^\dagger\hat{a}+\hat{a}^\dagger\hat{a}^\dagger|n\rangle = \frac{\ell^2}{2}\langle n|\hat{a}\hat{a}^\dagger+\hat{a}^\dagger\hat{a}|n\rangle \\
&= \frac{\ell^2}{2}\langle n|2\hat{a}^\dagger\hat{a}+1|n\rangle = \ell^2\left(n+\tfrac{1}{2}\right) \qquad (10.20)\\
\langle n|\hat{p}^2|n\rangle &= -\frac{\hbar^2}{2\ell^2}\langle n|(\hat{a}-\hat{a}^\dagger)(\hat{a}-\hat{a}^\dagger)|n\rangle \\
&= -\frac{\hbar^2}{2\ell^2}\langle n|\hat{a}\hat{a}-\hat{a}\hat{a}^\dagger-\hat{a}^\dagger\hat{a}+\hat{a}^\dagger\hat{a}^\dagger|n\rangle = \frac{\hbar^2}{2\ell^2}\langle n|\hat{a}\hat{a}^\dagger+\hat{a}^\dagger\hat{a}|n\rangle \\
&= -\frac{\hbar^2}{2\ell^2}\langle n|2\hat{a}^\dagger\hat{a}+1|n\rangle = \frac{\hbar^2}{\ell^2}\left(n+\tfrac{1}{2}\right)\,. \qquad (10.21)
\end{aligned}$$

As $\langle\hat{x}\rangle = \langle\hat{p}\rangle = 0$ for these states we have obtained the Heisenberg uncertainty relations,

$$\Delta x\Delta p = \sqrt{\langle n|\hat{x}^2|n\rangle\langle n|\hat{p}^2|n\rangle} = \hbar\left(n+\tfrac{1}{2}\right) \qquad (10.22)$$

for the eigenstates $|n\rangle$. We see that for the groundstate the uncertainty has the minimal value $\hbar/2$ allowed by the Heisenberg Uncertainty Principle, but grows for the higher excited states. One can ask whether by taking linear superpositions of these states one can minimise the uncertainty and make the system behave as classically as possible. The answer is yes: these close-to-classical states are called *coherent states*, which will be considered in one of the supervision sessions.

10.3 Time evolution.

We have learned that there is no time evolution in a stationary state, $|n\rangle$. But if the QHO is in any other state, it will change with time. So we need to consider a state which is in a quantum superposition of stationary states. As an example let us take

$$|\Psi\rangle_+ = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\,, \qquad (10.23a)$$

where (at a time $t=0$) the probabilities to be found in either $|0\rangle$ or $|1\rangle$ are equal to $\frac{1}{2}$. The average position and momentum do not vanish in the superposition state, as we have, *e.g.*

$$\langle\hat{x}\rangle = \frac{\ell}{2\sqrt{2}}\left(\langle 0| + \langle 1|\right)\left(\hat{a}+\hat{a}^\dagger\right)\left(|0\rangle + |1\rangle\right) = \frac{\ell}{2\sqrt{2}}\left(\langle 0\,|\,\hat{a}\,|\,1\rangle + \langle 1\,|\,\hat{a}^\dagger\,|\,0\rangle\right) = \frac{\ell}{\sqrt{2}}\left|\langle 0\,|\,\hat{a}\,|\,1\rangle\right| = \frac{\ell}{\sqrt{2}}\,.$$

[1] Both results are the physical consequences of the symmetry of the QHO: its Hamiltonian is even in both x and p. See, however, on the next page that the symmetry does not always help.

Notice how strange and non-classical is such a behaviour: $\hat{x} = 0$ either in the groundstate $\big|0\big\rangle$ or in the first excited state, $\big|1\big\rangle$, but not in the mixed state $\big|\Psi\big\rangle_+$, albeit the system in it is equally probable to appear after the appropriate measurement in either $\big|0\big\rangle$ or $\big|1\big\rangle$. The spatial symmetry seems to be broken in this state: the oscillator potential is even but the average position is shifted to the right. To restore (in some way) the symmetry, let us consider another state equally spread between $\big|0\big\rangle$ and $\big|1\big\rangle$, different only by the sign from that in Eq. (10.23a):

$$\big|\Psi\big\rangle_- = \frac{1}{\sqrt{2}}\big|0\big\rangle - \frac{1}{\sqrt{2}}\big|1\big\rangle\,, \tag{10.23b}$$

It is easy to check that $\big\langle\hat{x}\big\rangle = -\ell/\sqrt{2}$ in this state.

Now we want to see how the expectation value $\big\langle\Psi_+\,\big|\,\hat{x}\,\big|\,\Psi_+\big\rangle$ changes in time. Using the Schrödinger picture, we have both stationary states with energies $\varepsilon_0 = \hbar\omega/2$ and $\varepsilon_1 = 3\hbar\omega/2$, Eq. (9.12), oscillating in time, Eq. (7.13), so that (omitting the $+$ index)

$$\big|\Psi(t)\big\rangle = \frac{1}{\sqrt{2}}\big(\mathrm{e}^{-i\varepsilon_0 t/\hbar}\big|0\big\rangle + \mathrm{e}^{-i\varepsilon_1 t/\hbar}\big|1\big\rangle\big) = \frac{1}{\sqrt{2}}\big(\mathrm{e}^{-i\omega t/2}\big|0\big\rangle + \mathrm{e}^{-3i\omega t/2}\big|1\big\rangle\big)\,. \tag{10.24}$$

Hence

$$\hat{x}\big|\Psi(t)\big\rangle = \tfrac{\ell}{2}\big(\hat{a} + \hat{a}^\dagger\big)\big(\mathrm{e}^{-i\omega t/2}\big|0\big\rangle + \mathrm{e}^{-3i\omega t/2}\big|1\big\rangle\big) = \tfrac{\ell}{2}\Big(\mathrm{e}^{-i\omega t/2}\big|1\big\rangle + \mathrm{e}^{-3i\omega t/2}\big|0\big\rangle + \sqrt{2}\mathrm{e}^{-3i\omega t/2}\big|2\big\rangle\Big)$$

Therefore, attempting to measure $\hat{x}$ might raise the QHO to the 2nd excited state. However, it will not contribute into the t-dependent expectation value $\langle\hat{x}\rangle_t$, as we need to take the inner product with $\big\langle\Psi(x)\big|$, which is the conjugate of $\big|\Psi(x)\big\rangle$, Eq. (10.24), that does not contain $\big\langle 2\big|$. Hence we obtain

$$\begin{aligned}\langle\hat{x}\rangle_t &= \Big\langle\Psi(t)\,\Big|\,\hat{x}\Psi(t)\Big\rangle = \frac{\ell}{2\sqrt{2}}\big(\mathrm{e}^{i\omega t/2}\big\langle 0\big| + \mathrm{e}^{3i\omega t/2}\big\langle 1\big|\big)\big(\mathrm{e}^{-i\omega t/2}\big|1\big\rangle + \mathrm{e}^{-3i\omega t/2}\big|0\big\rangle\big)\\ &= \frac{\ell}{2\sqrt{2}}\big(\mathrm{e}^{i\omega t} + \mathrm{e}^{-i\omega t}\big) = \frac{\ell}{\sqrt{2}}\cos\omega t\,.\end{aligned} \tag{10.25}$$

It oscillates between the initial value, $\ell/\sqrt{2}$, and its opposite – in agreement with the Correspondence Principle for the QHO. A similar calculation (which will be asked in the appropriate non-assessed Problem Sheet) for the average of momentum would give $\langle\hat{p}\rangle_t = \big(\hbar/\ell\sqrt{2}\big)\sin\omega t$. Note that $\hbar\omega = \varepsilon_1 - \varepsilon_0$ – a typical transition frequency between the levels.

We see that the expectation values in the eigenstates $\big|n\big\rangle$ *do not behave classically* and one has to consider their superpositions to see the Correspondence Principle in work. A very special superposition (a wave-packet) which imitates most closely the classical oscillator, ie bounces back and forth without spreading in space is given by a *coherent state* which is an eigenstate of the non-Hermitian annihilation operator,

$$\hat{a}\big|z\big\rangle = z\big|z\big\rangle\,, \tag{10.26}$$

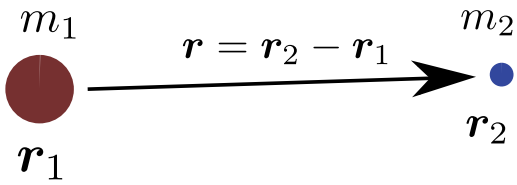
with a *complex eigenvalue* z. The coherent states and their quite remarkable properties will be discussed in one of the supervision sessions.

Lecture 11. Three-dimensional Schrödinger equation. Angular momentum.

In this lecture we switch to QM in three dimensions (3D) and recall that rotational symmetry is natural for a two-body problem. Hence it is described by the Schrödinger equation for one body with an effective *central* potential which depends on the angular momentum. The focus here is on a separation of radial and angular motion based on the conservation of angular momentum. A good part of this is a revision of QM2 material – which is given here for completeness but will be only briefly touched in the class.

11.1 Two-body problem and 3D Schrödinger equation with rotationally-symmetric potential

In classical mechanics, describing the motion of two particles with masses $m_{1,2}$ and positions $\boldsymbol{r}_{1,2}$ interacting via the central potential $V(|\boldsymbol{r}_2 - \boldsymbol{r}_1|$



can be reduced to a problem for an effective "relative" particle with the *reduced mass* $\mu = m_1 m_2/(m_1 + m_2)$ and position $\boldsymbol{r} = \boldsymbol{r}_2 - \boldsymbol{r}_1$ moving in the potential $V(r) = V(|\boldsymbol{r}|)$. This is done by defining the relative and centre of mass positions:

$$\boldsymbol{r} = \boldsymbol{r}_2 - \boldsymbol{r}_1 \,, \qquad \boldsymbol{R} = \frac{m_1 \boldsymbol{r}_1 + m_2 \boldsymbol{r}_2}{m_1 + m_2} \tag{11.1}$$

and momenta

$$\frac{m_1 + m_2}{m_1 m_2} \boldsymbol{p} = \frac{\boldsymbol{p}_1}{m_1} - \frac{\boldsymbol{p}_2}{m_2} \,, \qquad \boldsymbol{P} = \boldsymbol{p}_1 + \boldsymbol{p}_2 \,. \tag{11.2}$$

In these variables the Hamiltonian separates into the Hamiltonians for the centre of mass and relative motion

$$H = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(|\boldsymbol{r}_2 - \boldsymbol{r}_1|) = \frac{P^2}{2(m_1 + m_2)} + \frac{p^2}{2\mu} + V(r) = H_{\rm CM} + H_{\rm rel} \tag{11.3}$$

Since the centre of mass Hamiltonian $\hat{H}_{\rm CM}$ is that of a free particles with mass $M = m_1 + m_2$, its dynamics is trivial – one can always assume the centre of mass is at rest by choosing an appropriate frame. Then we are dealing with the effective particle of reduced mass μ moving in the central potential $V(r)$. In what follows we will concentrate on the quantum Hamiltonian of relative motion which we denote simply $\hat{H}$ from now on.

Turning to QM we replace the relative position $\boldsymbol{r} = (x, y, z)$ and momentum $\boldsymbol{p} = (p_x, p_y, p_z)$ by the corresponding quantum operators to obtain the quantum Hamiltonian of the relative motion. Due to the rotational symmetry, it is natural to write it in spherical polar coordinates:

$$\hat{H} = \frac{\hat{\boldsymbol{p}}^2}{2\mu} + V(\hat{r}) \,, \tag{11.4}$$

where $x = r\sin\theta\cos\phi,\ \ y = r\sin\theta\sin\phi,\ \ z = r\cos\theta$. The potential energy is a function of r only. The kinetic energy dependence on angular variables θ and ϕ enters via the expression for the Laplace operator in polar coordinates:

$$\frac{\hat{\boldsymbol{p}}^2}{2\mu} = -\frac{\hbar^2}{2\mu}\nabla^2 = -\frac{\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}}{\mathrm{d}r}\right) + \frac{1}{r^2\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}}{\mathrm{d}\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\mathrm{d}^2}{\mathrm{d}\phi^2}\right] \tag{11.5}$$

$$= -\frac{\hbar^2}{2\mu r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}}{\mathrm{d}r}\right) + \frac{1}{2\mu r^2}\hat{L}^2. \tag{11.6}$$

where $\hat{\boldsymbol{L}} = \boldsymbol{r}\times\boldsymbol{p} = \left(\hat{L}_x,\ \hat{L}_y,\ \hat{L}_z\right)$ is the angular momentum operator.[1] Hence the Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2\mu r^2}\frac{\mathrm{d}}{\mathrm{d}r}r^2\frac{\mathrm{d}}{\mathrm{d}r} + \frac{\hat{L}^2}{2\mu r^2} + V(r)\,. \tag{11.7}$$

You know from QM2 that all components of $\hat{\boldsymbol{L}}$, as well as $\hat{L}^2$, commute with a rotationally symmetric $\hat{H}$ but do not commute between themselves. Hence, one can simultaneously measure $\hat{H}$ (i.e. energy of the system), $\hat{L}^2$ and one of its components, traditionally chosen as $\hat{L}_z$. Independently of $\hat{H}$, the EVEs for $\hat{L}^2$ and $\hat{L}_z$ can be written as follows:[2]

$$\hat{L}^2\big|l,m\big\rangle = \hbar^2 l(l+1)\big|l,m\big\rangle\,, \qquad \hat{L}_z\big|l,m\big\rangle = \hbar m\big|l,m\big\rangle, \qquad 0\leqslant|m|\leqslant l, \tag{11.8}$$

where l is integer and m can take all integer values between $-l$ and l.

The eigenstates of $\hat{H}$ depend on both quantum numbers l and m as they correspond to the conserving quantities. In addition, they depend on a *radial* quantum number, n (as angular momentum is independent of r). Then the energy eigenvalue equation in the Hilbert space of Hamiltonian Eq. (11.7) can be written for *simultaneous eigenvectors* of $\hat{H}$, $\hat{L}^2$ and $\hat{L}_z$ as

$$\hat{H}\big|n,\,l,\,m\big\rangle = \varepsilon_{nl}\big|n,\,l,\,m\big\rangle\,. \tag{11.9}$$

The Hamiltonian depends on $\hat{\boldsymbol{L}}^2$ but not on $\hat{L}_z$. Hence all the states with the same l, n and different m have the same energy, i.e. QM states in central potentials are $2l+1$-degenerate in m.

If rotational symmetry is broken this degeneracy is lifted and the $2l+1$ energy levels split. This is the situation in the presence of magnetic field which interacts with orbiting charged particles, explaining why the quantum number m is called *magnetic quantum number*.

Now we bracket Eq. (11.9) with $\big\langle\boldsymbol{r}\big| = \big\langle r,\,\theta,\,\phi\big|$ to project it onto the real space, where $\hat{H}$ is defined, thus arriving at the standard Schrödinger equation:

$$\hat{H}\Psi_{nlm}(r,\theta,\phi) = \varepsilon_{nl}\Psi_{nlm}(r,\theta,\phi)\,, \qquad \Psi_{nlm}(r,\theta,\phi) \equiv \langle r,\theta,\phi\,|\,n,l,m\rangle\,. \tag{11.10}$$

The angular momentum part is n-independent, while the energy-dependent radial part is m-independent due to this degeneracy. Hence, we expect the factorisation

$$\big|n,l,m\big\rangle = \big|n,l\big\rangle\big|l,m\big\rangle\,. \tag{11.11}$$

[1] The expression of the angular part of ∇^2 via angular momentum $\hat{L}^2$ is proved in the next section, Eq. (11.20).

[2] We shall re-derive these equations algebraically in the next lecture; now I use the results known from QM2.

Projecting the eigenket of Eq. (11.11) into position representation, we have

$$\Psi_{n,l,m}(r,\theta,\phi) \equiv \langle r,\theta,\phi \,|\, n,l,m\rangle = \langle r \,|\, n,l\rangle\langle \theta,\phi \,|\, l,m\rangle \equiv R_{nl}(r)Y_{lm}(\theta,\phi)\,. \tag{11.12}$$

The functions $Y_{lm}(\theta,\phi) = \langle \theta,\phi \,|\, l,m\rangle$ absorb all angular dependence. They are called **spherical harmonics** and they are eigenfunctions of the angular momentum operators, Eq. (11.8), in the position representation:

$$\hat{L}^2 Y_{l,m}(\theta,\phi) = \hbar^2 l(l+1)Y_{l,m}(\theta,\phi), \qquad \hat{L}_z Y_{l,m}(\theta,\phi) = \hbar m Y_{l,m}(\theta,\phi) \tag{11.13}$$

Substituting this into the Schrödinger equation (11.10), and using the fact that Y_{lm} is the eigenfunction of $\hat{\boldsymbol{L}}^2$, we arrive at the equation for the radial part of the wavefunctions:

$$\left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}}{\mathrm{d}r}\right) + \frac{\hbar^2 l(l+1)}{2\mu r^2} + V(r)\right] R_{nl} = E_{nl}R_{nl}\,. \tag{11.14}$$

Here we have replaced $\hat{L}^2$ by its eigenvalue in the subspace of given l. Thus we have reduced the full three-dimensional problem to an effective one-dimensional one. We will deal with this equation, which does not require any knowledge of spherical harmonics, in Section 13.2.

11.2 Spherical Harmonics.

To find these explicitly, we need to write the angular momentum operators in the position basis, switching from Cartesian to polar coordinate. To this end, knowing the answer, we act backwards and express $\frac{\mathrm{d}}{\mathrm{d}\phi}$ via Cartesian derivatives using the chain rule as follows:

$$\begin{aligned}\frac{\partial F(x,y,z)}{\partial\phi} &= \frac{\partial F}{\partial x}\underbrace{\frac{\partial x}{\partial\phi}}_{=-y} + \underbrace{\frac{\partial F}{\partial y}}_{=x}\frac{\partial y}{\partial\phi} + \underbrace{\frac{\partial F}{\partial z}}_{=0}\frac{\partial z}{\partial\phi} = \left(x\frac{\mathrm{d}}{\mathrm{d}y} - y\frac{\mathrm{d}}{\mathrm{d}x}\right)F = -\frac{1}{i\hbar}(x\hat{p}_y - y\hat{p}_x)F \\ &= -\frac{1}{i\hbar}\hat{L}_z F \quad\Rightarrow\quad \left\langle r,\theta,\phi \,\middle|\, \hat{L}_z \,\middle|\, F\right\rangle = -i\hbar\frac{\partial F}{\partial\phi} \text{ or } \hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}.\end{aligned} \tag{11.15}$$

The analogy with $\hat{p}$ in the position representation, Eq. (5.9), is obvious.

Similarly to showing that $\hat{T}(a) \equiv \mathrm{e}^{-ia\hat{p}\hbar}$, Eq. (6.1), generates translation by a, we can show that $\mathrm{e}^{-i\varphi\hat{L}_z/\hbar}$ generates a rotation through angle φ in polar coordinates. There is a crucial difference, which becomes obvious if we substitute $\hat{L}_z$, Eq. (11.15), into Eq. (11.13):

$$\begin{aligned}\left\langle \boldsymbol{r} \,\middle|\, \hat{L}_z \,\middle|\, n,l,m\right\rangle = \hbar m\langle \boldsymbol{r} \,|\, n,l,m\rangle \quad\Longrightarrow\quad R_{n,l}\hat{L}_z Y_{l,m}(\theta,\phi) = mR_{n,l}(r)Y_{l,m}(\theta,\phi) \\ \Rightarrow\ -i\hbar\frac{\partial Y_{l,m}(\boldsymbol{r})}{\partial\phi} = \hbar m Y_{l,m}(\boldsymbol{r}),\end{aligned} \tag{11.16}$$

as the radial part, R_{nl}, cancels on both sides of this equation. The solution is straightforward:

$$Y_{l,m}(\theta,\phi) = c_{l,m}(\theta)\mathrm{e}^{im\phi} \tag{11.17}$$

As Eq. (11.16) contains only differentiation w.r.t. ϕ, an arbitrary (as yet) coefficient $c_{l,m}(\theta)$ is present here to be calculated later using the fact that $Y_{l,m}$ is also an eigenfunction of $\hat{L}^2$. This looks similar

to eigenfunctions of $\hat{p}$ with a crucial difference: angles ϕ and $\phi + 2\pi$ are fully equivalent. Hence we must have

$$\mathrm{e}^{im\phi} = \mathrm{e}^{im(\phi+2\pi)} \quad \Rightarrow \quad m \text{ is integer.} \tag{11.18}$$

Hence we have illustrated why the eigenvalues of the orbital angular momentum are integers.

To find how $c_{n,l}(r,\theta)$ in Eq. (11.17) depends on θ, we need other components of $\hat{\boldsymbol{L}}$ in polar coordinates. It is done in a way similar to what we have used for $\hat{L}_z$, but requires more algebra. I omit here the appropriate derivations which can be found in any book from the reading list. The result is:

$$\hat{L}_x = i\hbar\left(\sin\phi\frac{\mathrm{d}}{\mathrm{d}\theta} + \cot\theta\cos\phi\frac{\mathrm{d}}{\mathrm{d}\phi}\right), \qquad \hat{L}_y = i\hbar\left(-\cos\phi\frac{\mathrm{d}}{\mathrm{d}\theta} + \cot\theta\sin\phi\frac{\mathrm{d}}{\mathrm{d}\phi}\right). \tag{11.19}$$

Summing squares of each of them with $\hat{L}_z^2$, Eq. (11.15) gives

$$\hat{L}^2 F(\theta,\phi) \equiv \big\langle\theta,\phi\big|\hat{L}^2\big|F\big\rangle = -\hbar^2\left\{\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}}{\mathrm{d}\theta}\right) + \frac{1}{\sin^2\theta}\frac{\mathrm{d}^2}{\mathrm{d}\phi^2}\right\}F(\theta,\phi)\,. \tag{11.20}$$

This finally justifies Eq. (11.10): substituting there $\hat{\boldsymbol{L}}^2$ by the expression above and taking out the factor of $\hbar^2/2\mu$ would give (omitting the potential part) the standard expression for the Laplacian operator in polar coordinates.

Solving the eigenvalue equation for $\hat{\boldsymbol{L}}^2$ in the position representation is not an easy task. As in the case of the QHO, the ladder operators would allow one to reduce solving the 2nd order differential equation to the 1st order ones. The spherical harmonics, $Y_{lm}(\theta,\phi) = \langle\theta,\phi\,|\,l,m\rangle$, introduced in Eq. (11.12), are the eigenfunctions of both $\hat{L}^2$ and $\hat{L}_z$. As we have solved the EVE for $\hat{L}_z$, Eq. (11.16), we know that the ϕ-dependence is simply exponential, Eq. (11.17). The θ dependence is found using, e.g., $\hat{L}_+\big|l,l\big\rangle = 0$. The corresponding wavefunction is found, using Eq. (11.17), from

$$\begin{aligned} 0 = \big\langle\theta,\phi\big|\hat{L}_+\big|l,l\big\rangle &= \hbar\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right)Y_{ll}(\theta,\phi) = \hbar\left(\frac{\mathrm{d}}{\mathrm{d}\theta} - l\cot\theta\right)c_{ll}(\theta) = 0 \\ \Rightarrow\ \mathrm{d}c_{ll}(\theta) &= l\cot\theta\,\mathrm{d}\theta = l\frac{\cos\theta\,\mathrm{d}\theta}{\sin\theta} = l\ln|\sin\theta| \end{aligned} \tag{11.21}$$

Hence we obtain

$$\langle\theta,\phi\,|\,l,l\rangle = Y_{ll}(\theta,\phi) = a_l\mathrm{e}^{il\phi}\sin^l\theta\,. \tag{11.22}$$

The constant a_l can be obtained from normalisation. Starting with $Y_{ll}(\theta,\phi)$ we can find spherical harmonics $Y_{l,l-1}(\theta,\phi)$, $Y_{l,l-2}(\theta,\phi)$, ... by successive action of $\hat{L}_-$ until we reach the lowest ladder state $Y_{l,-l}(\theta,\phi)$. These functions are tabulated in any QM book and we will not do it here.

11.3 Angular momentum commutation relations

We have use the **angular momentum conservation** in the previous lecture, where we have also employed EVEs for $\hat{L}^2$ and $\hat{L}_z$, known from QM2, Eq. (11.8). Our task is to re-derive this purely algebraically, introducing the ladder operators similarly to those for the QHO, and understand the physical meaning of this.

In this section, I present for your convenience a re-derivation (not to be shown in the class) of commutation relations between components of $\hat{\boldsymbol{L}}$, $\hat{\boldsymbol{p}}$ and $\hat{\boldsymbol{r}}$, also known from QM2.

The angular momentum operator in QM is defined by replacing Cartesian position and momentum components by the corresponding operators,[3] $\boldsymbol{L} \to \hat{\boldsymbol{L}} = \hat{\boldsymbol{r}} \times \hat{\boldsymbol{p}}$. The conservation of angular momentum (or any other observable, Eq. (8.3)) in quantum mechanics means that $\hat{\boldsymbol{L}}$ commutes with Hamiltonian, since in any state $\big|\Psi\big\rangle$

$$i\hbar\Big\langle \frac{\mathrm{d}\hat{\boldsymbol{L}}}{\mathrm{d}t}\Big\rangle_t = \Big\langle [\hat{\boldsymbol{L}}, \hat{H}]\Big\rangle_t = 0\,. \tag{11.23}$$

Hence the conservation requires each component of $\hat{\boldsymbol{L}}$ to commute with $\hat{H}$. Let us check this explicitly for one of the components, say $\hat{L}_z$. It commutes with the kinetic energy since

$$\begin{aligned}[\hat{L}_z, (p_x^2 + p_y^2 + p_z^2)] &= [\hat{p}_y\hat{x} - \hat{p}_x\hat{y}, \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2] = \hat{p}_y[\hat{x}, \hat{p}_x^2] - \hat{p}_x[\hat{y}, \hat{p}_y^2] \\ &= \hat{p}_y 2i\hbar\hat{p}_x - \hat{p}_x 2i\hbar\hat{p}_y = 0\,,\end{aligned} \tag{11.24}$$

where we have used as earlier in Lecture 8, Eq. (8.9):

$$\big[\hat{x}, \hat{p}_x^2\big] = \big(\hat{x}\hat{p}_x\hat{p}_x - \hat{p}_x\hat{p}_x\hat{x}\big) = \big(\big[\hat{x}, \hat{p}_x\big]\hat{p}_x + \hat{p}_x\big[\hat{x}, \hat{p}_x\big]\big) = i\hbar\big(\hat{p}_x + \hat{p}_x\big) = 2i\hbar\hat{p}_x.$$

and similarly for $[\hat{y}, \hat{p}_y^2] = 2i\hbar\hat{p}_y$.

$\hat{L}_z$ also commutes with the ***rotationally-symmetric potential***, $V(\hat{\boldsymbol{r}}) = V(\hat{r})$. In this case, using the chain rule (and $r = \sqrt{x^2 + y^2 + z^2}$) we find[4]

$$\frac{\partial V}{\partial x} = \frac{\mathrm{d}V}{\mathrm{d}r}\frac{\partial r}{\partial x} = \frac{xV_r'}{r}\,, \qquad \frac{\partial V}{\partial y} = \frac{\mathrm{d}V}{\mathrm{d}r}\frac{\partial r}{\partial y} = \frac{yV_r'}{r}\,, \qquad \frac{\partial V}{\partial z} = \frac{\mathrm{d}V}{\mathrm{d}r}\frac{\partial r}{\partial z} = \frac{zV_r'}{r} \tag{11.25}$$

Hence we find

$$[\hat{L}_z, \hat{V}] = \hat{x}[\hat{p}_y, \hat{V}] - \hat{y}[\hat{p}_x, \hat{V}] = -i\hbar\left(\hat{x}\frac{\mathrm{d}\hat{V}}{\mathrm{d}\hat{y}} - \hat{y}\frac{\mathrm{d}\hat{V}}{\mathrm{d}\hat{x}}\right) = -i\hbar V_r'\frac{\hat{x}\hat{y} - \hat{y}\hat{x}}{r} = 0\,. \tag{11.26}$$

As there is nothing special about the z-axis, the commutator of $\hat{L}_x$ and $\hat{L}_y$ with the Hamiltonian also vanishes. Thus, like in classical mechanics, angular momentum $\hat{L}$ is conserved or is a *constant of motion*. However, operators of the angular momentum components do not commute, and hence cannot be measured together. For example

$$[\hat{L}_x, \hat{L}_y] = [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] = \hat{y}\hat{p}_x[\hat{p}_z, \hat{z}] + \hat{x}\hat{p}_y[\hat{z}, \hat{p}_z] = i\hbar(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x) = i\hbar\hat{L}_z\,. \tag{11.27a}$$

[3] Although $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{p}}$ do not commute, there are no problems for $\hat{\boldsymbol{L}}$ as $\hat{\boldsymbol{r}} \times \hat{\boldsymbol{p}} = -\hat{\boldsymbol{p}} \times \hat{\boldsymbol{r}}$. The reason is that the vector product only includes product of components on different axes which commute: e.g., $\big[\hat{p}_x, \hat{y}\big] = 0$.

[4] As coordinate operators commute between themselve, we have omitted all hats in this expression.

Similarly,[5]

$$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x \qquad [\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y\,. \tag{11.27b}$$

So, what is the meaning of the angular momentum conservation when we cannot measure $\langle\hat{\boldsymbol{L}}\rangle$ in any state apart from that when all the components are zero? Firstly, as each component commute with $\hat{H}$, one can measure the projection of $\boldsymbol{L}$ on any direction. Secondly, although $\hat{\boldsymbol{L}}$ is not measurable, $\hat{\boldsymbol{L}}^2$ is. It is clearly conserved, as the square of the conserved quantity, and it commutes with any component operator,[6] *e.g.* $\hat{L}_z$. The appropriate operator is

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \tag{11.28}$$

which commutes with $\hat{L}_z$. Indeed, substituting below $\hat{L}_x\hat{L}_z = \hat{L}_z\hat{L}_x - i\hbar\hat{L}_y$, *etc*, we find

$$\begin{aligned}[\hat{L}^2, \hat{L}_z] &= [\hat{L}_x^2, \hat{L}_z] + [\hat{L}_y^2, \hat{L}_z] = \hat{L}_x\hat{L}_x\hat{L}_z - \hat{L}_z\hat{L}_x\hat{L}_x + \hat{L}_y\hat{L}_y\hat{L}_z - \hat{L}_z\hat{L}_y\hat{L}_y \\ &= \hat{L}_x\hat{L}_z\hat{L}_z - i\hbar\hat{L}_x\hat{L}_y - \hat{L}_x\hat{L}_z\hat{L}_x - i\hbar\hat{L}_y\hat{L}_x + \hat{L}_y\hat{L}_z\hat{L}_y + i\hbar\hat{L}_y\hat{L}_x - \hat{L}_y\hat{L}_z\hat{L}_y + i\hbar\hat{L}_x\hat{L}_y \\ &= 0\,,\end{aligned}$$

so it can be diagonalised together with $\hat{L}_z$ (or measured simultaneously). Then the eigenvalues of $\hat{L}^2$ and $\hat{L}_z$ can be used to classify the states of a rotationally-symmetric system.

Before going to this task, let us notice useful commutation relations between components of $\hat{\boldsymbol{L}}$, $\hat{\boldsymbol{x}}$, and $\hat{\boldsymbol{p}}$, leaving a proof to the (non-assessed) problem sheets,[7] and including those of Eqs. (11.27)

$$\left[\hat{L}_n, \hat{L}_m\right] = \pm i\hbar\hat{L}_k\,, \qquad \left[\hat{L}_n, \hat{x}_m\right] = \pm i\hbar\hat{x}_k\,, \qquad \left[\hat{L}_n, \hat{p}_m\right] = \pm i\hbar\hat{p}_k\,. \tag{11.29}$$

Here the component notations are $\left\{\hat{L}_n\right\} \equiv \left\{\hat{L}_x,\ \hat{L}_y,\ \hat{L}_z\right\}$, *etc*. The sign in all $\pm$ above is $+$, if nmk is a cyclic (even) permutation of xyz, *i.e.* yzx and zxy, and $-$ if the permutation is anti-cyclic (odd), *i.e.* yxz, xzy or zyx, in obvious correspondence with Eqs. (11.27).

11.4 Eigenvectors and eigenvalues of angular momentum

Since $\left[\hat{L}^2, \hat{L}_z\right] = 0$, there exists their joint eigenbasis which we denote $\big|\lambda, m\big\rangle$. We need to have here two "quantum numbers" as we are have two angular degrees of freedom, and we assume these quantum numbers to be the eigenvalues of the two operators. The corresponding EVEs are[8]

$$\hat{L}^2\big|\lambda, m\big\rangle = \hbar^2\lambda\big|\lambda, m\big\rangle \qquad \hat{L}_z\big|\lambda, m\big\rangle = \hbar m\big|\lambda, m\big\rangle \tag{11.30}$$

[5] A way to remember the rules is to memorise that of Eq. (11.27a) and use cyclic permutations $(xyz) \to (yzx) \to (zxy)$ to generate the two remaining commutators. For a different order of operators, use the antisymmetry of the commutator, e.g. $[\hat{L}_z, \hat{L}_y] = -[\hat{L}_y, \hat{L}_z] = -i\hbar\hat{L}_x$.

[6] One can simultaneously measure the magnitude of the angular momentum and its projection on any direction - which can be called z.

[7] All these relations follow from the definitions of operators in the position basis and the canonical commutation relations, $\left[\hat{x}_n, \hat{p}_m\right] = i\hbar\delta_{nm}$, where $n, m = 1, 2, 3$, and $x_1 \equiv x$, $x_2 \equiv y$, $x_3 \equiv z$, $\hat{p}_1 \equiv \hat{p}_x$. *etc*.

[8] We do not assume that $\lambda = l(l{+}1)$ as in Eq. (11.8) but we shall prove this in this section.

where λ and m are dimensionless numbers, so that the eigenvalues of angular momentum are dimensionally correct.

As in the case of the QHO we will use the commutation relations of angular momentum operators to determine possible values of λ and m. We start by introducing **ladder operators**

$$\hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y \qquad \Rightarrow \qquad \hat{L}_x = \frac{1}{2}\big(\hat{L}_+ + \hat{L}_-\big)\,, \quad \hat{L}_y = -\frac{i}{2}\big(\hat{L}_+ - \hat{L}_-\big)\,. \tag{11.31}$$

Neither $\hat{L}_+$ nor $\hat{L}_-$ is Hermitian, but they are mutually adjoint:

$$\big(\hat{L}_+\big)^\dagger = \hat{L}_-\,, \qquad \big(\hat{L}_-\big)^\dagger = \hat{L}_+\,. \tag{11.32}$$

Their commutation relations are obtained from the commutation relations of $\hat{L}_x$, $\hat{L}_y$,

$$\begin{aligned}
[\hat{L}_+, \hat{L}_-] &= [\hat{L}_x + i\hat{L}_y, \hat{L}_x - i\hat{L}_y] = -i\underbrace{[\hat{L}_x, \hat{L}_y]}_{i\hbar\hat{L}_z} + i\underbrace{[\hat{L}_y, \hat{L}_x]}_{-i\hbar\hat{L}_z} = 2\hbar\hat{L}_z \\
[\hat{L}_z, \hat{L}_\pm] &= [\hat{L}_z, \hat{L}_x \pm i\hat{L}_y] = \underbrace{[\hat{L}_z, \hat{L}_x]}_{i\hbar\hat{L}_y} \pm i\underbrace{[\hat{L}_z, \hat{L}_y]}_{-i\hbar\hat{L}_x} = i\hbar L_y \pm \hbar\hat{L}_x = \pm\hbar\hat{L}_\pm
\end{aligned} \tag{11.33}$$

In addition, as $[\hat{L}^2, \hat{L}_x] = [\hat{L}^2, \hat{L}_y] = 0$, the ladder operators $\hat{L}_\pm$ commute with $\hat{L}^2$,

$$[\hat{L}^2, \hat{L}_\pm] = 0\,. \tag{11.34}$$

This means that applying $\hat{L}_\pm$ to an eigenstate $\big|\lambda, m\big\rangle$ does not affect the eigenvalue of $\hat{L}^2$:

$$\hat{L}^2\big(\hat{L}_\pm\big|\lambda, m\big\rangle\big) = \hat{L}_\pm\hat{L}^2\big|\lambda, m\big\rangle = \hbar^2\lambda\big(\hat{L}_\pm\big|\lambda, m\big\rangle\big)\,. \tag{11.35}$$

However, it does change the eigenvalue of $\hat{L}_z$, as we see using the commutator $[\hat{L}_z, \hat{L}_\pm]$, Eq. (11.33):

$$\hat{L}_z\big(\hat{L}_\pm\big|\lambda, m\big\rangle\big) = \big(\hat{L}_\pm\hat{L}_z \pm \hbar L_\pm\big)\big|\lambda, m\big\rangle = \hbar(m \pm 1)\big(\hat{L}_\pm\big|\lambda, m\big\rangle\big)\,. \tag{11.36}$$

Hence $\hat{L}_\pm$ are eigenvectors of $\hat{L}_z$ corresponding to the eigenvalues $m\pm1$ so that $\hat{L}_+$ is the raising ladder operator and $\hat{L}_-$ is the lowering one:

$$\hat{L}_\pm\big|\lambda, m\big\rangle = c^\pm_{\lambda m}\big|\lambda, m \pm 1\big\rangle \qquad c^\pm_{\lambda m} \text{ – complex numbers.} \tag{11.37}$$

For the QHO the process of applying ladder operators was shown to terminate at the lowest state - the ground state. In the case of angular momentum the ladder terminates as well, both at the lower and at the upper end. To see this, one notes that in any state

$$\Big\langle \hat{\boldsymbol{L}}^2 \Big\rangle = \Big\langle \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \Big\rangle \geqslant \Big\langle \hat{L}_z^2 \Big\rangle\,. \tag{11.38}$$

Moreover, the '=' sign is only applicable when all the three components are zero, since even when $\langle\hat{L}_x\rangle = \langle\hat{L}_y\rangle = 0$, we still have $\langle\hat{\boldsymbol{L}}^2\rangle > \langle\hat{L}_z^2\rangle$ due to the HUP. Indeed, using Eq. (6.12) we find

$$\Delta L_x \Delta L_y \geqslant \frac{1}{2}\Big|\Big\langle\big[\hat{L}_x, \hat{L}_y\big]\Big\rangle\Big| = \frac{\hbar}{2}\Big\langle\hat{L}_z\Big\rangle \neq 0.$$

Hence, $\langle \hat{L}_x^2 \rangle$ and $\langle \hat{L}_y^2 \rangle$ cannot simultaneously vanish.

The inequality 11.38 is valid in any state, including the eigenstates $\big|\lambda, m\big\rangle$, which immediately leads to $\langle \hat{\boldsymbol{L}}^2 \rangle = \hbar^2\lambda \geqslant \hbar^2 m^2 = \langle \hat{L}_z^2 \rangle$, again with '=' sign only for zero angular momentum. Otherwise, $0 < |m| < \sqrt{\lambda}$. Let us denote the maximal value of the projection by l:

$$l \equiv m_{\max} = -m_{\min}\,. \tag{11.39}$$

Then, similar to $\hat{a}\big|0\big\rangle = 0$ for the QHO, we have

$$\hat{L}_+\big|\lambda,\, l\big\rangle = 0 \text{ and } \hat{L}_-\big|\lambda,\, -l\big\rangle = 0, \tag{11.40}$$

as there is no way to further raise l in a state with the max value of the projection or to lower one with the min value. So we have both descending and ascending ladders. Acting by $\hat{L}_-$ on $\big|\lambda,\, l\big\rangle$ we have an eigenket proportional to $\big|\lambda,\, l-1\big\rangle$. We can repeat this action no more than $2l+1$ times when we arrive at the eigenket with the min possible $m = -l$.

The number of steps, $2l{+}1$, is integer, so that l must be either integer, or half-integer. We have shown, Eq. (11.18), that magnetic quantum number m is integer, and thus l for orbital momentum must be integer. The algebra would have predicted the existence of non-orbital angular momentum, ***spin***, with half-integer l, had it not been discovered prior to this formulation.

Now we find λ by acting with $\hat{L}_-$ on the first of Eq. (11.40):

$$\hat{L}_-\hat{L}_+\big|\lambda,\, l\big\rangle = 0 \tag{11.41}$$

The product $\hat{L}_-\hat{L}_+$ is found using the definitions, (11.31), and the commutation relations, (11.33):

$$\big(\hat{L}_x{+}i\hat{L}_y\big)\big(\hat{L}_x - i\hat{L}_y\big)\big|\lambda,\, l\big\rangle = \Big(\hat{L}_x^2{+}\hat{L}_y^2{+}i\big[\hat{L}_x, \hat{L}_y\big]\Big)\big|\lambda,\, l\big\rangle = \Big(\hat{\boldsymbol{L}}^2{-}\hat{L}_z^2{-}\hbar\hat{L}_z\Big)\big|\lambda,\, l\big\rangle \tag{11.42}$$

$$= \hbar^2\big(\lambda - l^2 - l\big)\big|\lambda,\, l\big\rangle = 0 \quad \Rightarrow \quad \boxed{\lambda = l(l{+}1)}\,. \tag{11.43}$$

So we can rename the eigenket for any projection m as

$$\big|\lambda,\, m\big\rangle \mapsto \big|l,\, m\big\rangle, \text{ with } m = 0, \pm 1, \cdots \pm l. \tag{11.44}$$

Hence, we arrive at the EVEs for $\hat{L}^2$ and $\hat{L}_z$ given in the previous lecture, Eq. (11.8). Using this, we can determine the matrix elements of these operators.

Lecture 12. Ladder operators for angular momentum

We develop here the ladder operators algebra similar to that for the QHO. We will look at the matrix elements of $\hat{\boldsymbol{L}}$ and $\hat{L}_{x,y,z}$ and then generalise this for spin – the intrinsic angular momentum – to be considered in Lec.14.

12.1 Algebra of ladder operators

We start with the EVEs (11.8) for $\hat{\boldsymbol{L}}^2$ and $\hat{L}_z$.

$$\hat{L}^2\big|l,m\big\rangle = \hbar^2 l(l{+}1)\big|l,m\big\rangle, \qquad\qquad \hat{L}_z\big|l,m\big\rangle = \hbar m\big|l,m\big\rangle$$

where the quantum numbers l and m take integer or half-integer values[1], and m numbering "ladder states" change by integer steps from $-l$ to l, $m = -l, -l{+}1, \ldots, l-1, l$. These states are orthonormal, $\langle l,m \,|\, l',m'\rangle = \delta_{ll'}\delta_{mm'}$. To move up and down the ladder, we use the mutually adjoint operators of Eq. (11.31), $\hat{L}_\pm$, and rewrite the EVE (11.37) in terms of l instead of λ:

$$\hat{L}_\pm\big|l,m\big\rangle = \hbar c^\pm_{l,m}\big|l,m\pm 1\big\rangle\,. \tag{12.1}$$

To determine the coefficients $c^\pm_{l,m}$, consider the inner product of $\hat{L}_+\big|l,m\big\rangle$ and the corresponding bra-state , $\big\langle l,m\big|\hat{L}_- = (c^+_{l,m})^*\hbar\big\langle l{+}1,m\big|$. Since the ladder states are normalised, we get

$$\big\langle l,m\big|\hat{L}_-\hat{L}_+\big|l,m\big\rangle = \hbar^2|c^+_{l,m}|^2\,\underbrace{\langle l,m{+}1\,|\,l,m{+}1\rangle}_{=1} = \hbar^2|c^+_{l,m}|^2\,. \tag{12.2a}$$

On the other hand, using $\hat{L}_-\hat{L}_+ = \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z$, Eq. (11.42), we find the l.h.s. of Eq. (12.2a) as

$$\big\langle l,m\big|\hat{L}_-\hat{L}_+\big|l,m\big\rangle = \big\langle l,m\big|\hat{L}^2{-}\hat{L}_z^2{-}\hbar\hat{L}_z\big|l,m\big\rangle = \hbar^2\big\langle l,m\big|\Big(l(l{+}1){-}m^2{-}m\Big)\big|l,m\big\rangle\,. \tag{12.2b}$$

Comparing both Eqs. (12.2), we find

$$c^+_{l,m} = \sqrt{l(l{+}1)-m(m+1)} = \sqrt{(l-m)(l+m+1)}\,. \tag{12.2c}$$

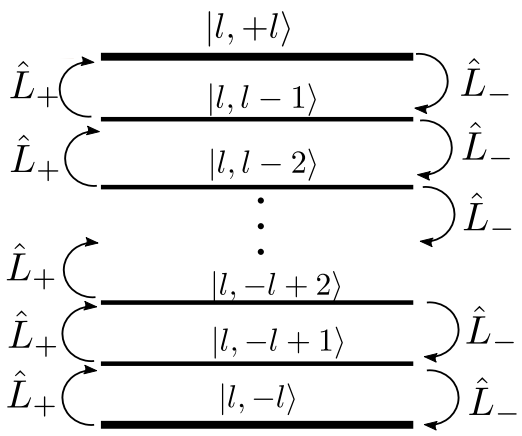
Similarly, by calculating the inner product of $\hat{L}_-\big|l,m\big\rangle = \hbar c^-_{l,m}\big|l-1,m\big\rangle$ and the corresponding bra-vector $\big\langle l,m\big|\hat{L}_+ = \hbar(c^-_{l,m})^*\big\langle l-1,m\big|$ we get

$$c^-_{l,m} = \sqrt{l(l{+}1)-m(m-1)} = \sqrt{(l+m)(l-m+1)}\,. \tag{12.3}$$

As expected $c^+_{l,l} = c^-_{l,-l} = 0$, ensuring the termination of the state ladder on both ends: $\hat{L}_+\big|l,l\big\rangle = 0$, $\hat{L}_-\big|l,-l\big\rangle = 0$.

We refer to $l = 0, 1, 2, \ldots$ as the *orbital angular momentum of a particle*. It cannot be changed by the action of any of components $\hat{\boldsymbol{L}}$ thus defining a subspace of a full Hilbert space. In each such a subspace there are $2l{+}1$ basis states corresponding to $2l{+}1$ integer values the angular momentum projection, $m = 0, \pm1, \cdots, \pm l$. To move between these states one uses the ladder operators $\hat{L}_\pm = \hat{L}_x \pm \hat{L}_y$ as shown in the figure below. The orbital angular momentum is not an ***intrinsic*** property of the particle, and can be changed by the action of other operators.

[1] We postpone the explanation of why only integer values are possible for the quantum numbers l and m of the orbital angular momentum, i.e. the one due to particle's motion. As you already know, half-integer values are possible for the intrinsic angular momentum – spin – of fermionic particles



12.2 Matrix elements of angular momentum operators

We are now ready to calculate these matrix elements in each Hilbert subspace with a given l. The simplest are those for the operator $\hat{L}_z$:

$$\langle l,m'|\hat{L}_z|l,m\rangle = \hbar m\langle l,m' \mid l,m\rangle = \hbar m\delta_{m,m'} \tag{12.4}$$

In matrix form for any l, $\left\{\hat{L}_z\right\}_{m'm} = \mathrm{diag}\,\{l, l{-}1, \cdots - l{+}1, l\}$. To find matrix elements of the other two components of $\hat{\boldsymbol{L}}$ we write them in terms of the ladder operators, Eq. (11.31), and then perform calculations in several subspaces:

$$\hat{L}_x = \frac{1}{2}\big(\hat{L}_+ + \hat{L}_-\big)\,, \qquad \hat{L}_y = -\frac{i}{2}\big(\hat{L}_+ - \hat{L}_-\big)\,. \tag{12.5}$$

For $l=0$, there is only one state $|0,0\rangle$. The matrix elements of *all components* are zero:

$$\langle 0,0|\hat{L}_x|0,0\rangle = \langle 0,0|\hat{L}_x|0,0\rangle = \langle 0,0|\hat{L}_x|0,0\rangle = 0\,.$$

For $l=1$, there are three states: $|1,-1\rangle$, $|1,0\rangle$, $|1,1\rangle$. To calculate matrix elements of $\hat{L}_{x,y}$ we use Eq. (12.2) for the particular values of l and m:

$$\begin{aligned}\hat{L}_+|1,-1\rangle &= \hbar c^+_{1,-1}|1,0\rangle = \hbar\sqrt{1(1+1)-(-1)(-1+1)}|1,0\rangle = \hbar\sqrt{2}|1,0\rangle\\ \hat{L}_+|1,0\rangle &= \hbar c^+_{1,0}|1,1\rangle = \hbar\sqrt{1(1+1)-0(0+1)}|1,1\rangle = \hbar\sqrt{2}|1,1\rangle\end{aligned}$$

so that there are only two nonzero matrix elements,

$$\left\langle 1,0\,\middle|\,\hat{L}_+\,\middle|\,1,-1\right\rangle = \left\langle 1,1\,\middle|\,\hat{L}_+\,\middle|\,1,0\right\rangle = \hbar\sqrt{2}.$$

We can represent this in matrix form:

$$\left\{\hat{L}_+\right\}_{m'm} \equiv \left\{\left\langle 1,m'\,\middle|\,\hat{L}_+\,\middle|\,1,m\right\rangle\right\} = \hbar\sqrt{2}\begin{pmatrix}0&1&0\\0&0&1\\0&0&0\end{pmatrix}. \tag{12.6}$$

Dirac's representation of $\hat{L}_+$ is
$\hat{L}_+ = \sum_{m=-1}^{1} \hat{L}_+ |m\rangle\langle m| = \hbar \sum_{m=-1}^{1} c^+_{1,m} |m\rangle\langle m| = \hbar\sqrt{2}\big(|1\rangle\langle 0| + |0\rangle\langle -1|\big)$. Since $\hat{L}_-$ is the Hermitian conjugate of $\hat{L}_+$, the appropriate matrix would be just the transpose of that in the above equations. Hence, using Eq. (12.5) we find

$$\left\{\hat{L}_x\right\}_{m'm} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \left\{\hat{L}_y\right\}_{m'm} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}. \tag{12.7}$$

For $l = 2$, there are five states: $|2,-2\rangle$, $|2,-1\rangle$, $|2,0\rangle$, $|2,1\rangle$, $|2,2\rangle$. Again, using Eq. (12.2) for $l = 2$ and the appropriate m, and knowing that the only nonvanishing matrix elements of $\hat{L}_+$ are $\left\langle l, m+1 \middle| \hat{L}_+ \middle| l, m \right\rangle$, we find

$$\begin{aligned}
\langle 2,-1|\hat{L}_+|2,-2\rangle &= \hbar c^+_{2,-2} = \hbar\sqrt{2(2+1)-(-2)(-2+1)} &&= 2\hbar \\
\langle 2,0|\hat{L}_+|2,-1\rangle &= \hbar c^+_{2,-1} = \hbar\sqrt{2(2+1)-(-1)(-1+1)} &&= \hbar\sqrt{6} \\
\langle 2,1|\hat{L}_+|2,0\rangle &= \hbar c^+_{2,0} = \hbar\sqrt{2(2+1)-0(0+1)} &&= \hbar\sqrt{6} \\
\langle 2,2|\hat{L}_+|2,1\rangle &= \hbar c^+_{2,1} = \hbar\sqrt{2(2+1)-1(1+1)} &&= 2\hbar
\end{aligned}$$

Using again Eq. (12.5) and the fact that $\left\{\hat{L}_-\right\}_{m'm}$ is transpose of $\left\{\hat{L}_+\right\}_{m'm}$ above, we find

$$\left\{\hat{L}_x\right\} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & \sqrt{2} & 0 & 0 & 0 \\ \sqrt{2} & 0 & \sqrt{3} & 0 & 0 \\ 0 & \sqrt{3} & 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{3} & 0 & \sqrt{2} \\ 0 & 0 & 0 & \sqrt{2} & 0 \end{pmatrix}, \quad \left\{\hat{L}_y\right\} = \frac{i\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -\sqrt{2} & 0 & 0 & 0 \\ \sqrt{2} & 0 & -\sqrt{3} & 0 & 0 \\ 0 & \sqrt{3} & 0 & -\sqrt{3} & 0 \\ 0 & 0 & \sqrt{3} & 0 & -\sqrt{2} \\ 0 & 0 & 0 & \sqrt{2} & 0 \end{pmatrix} \tag{12.8}$$

Using such a procedure one can construct matrices for angular momentum $l = 3, 4, \ldots$ in a similar way. These calculations are quite algorithmic and can be easily programmed.

Let us find eigenstates of $\hat{L}_{x,y}$ for the simplest nontrivial case of $l = 1$, writing the appropriate EVEs in matrix form:

$$\hat{L}_x|m_x\rangle = \hbar m_x|m_x\rangle \quad \Leftrightarrow \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_0 \\ a_{-1} \end{pmatrix} = m_x \begin{pmatrix} a_1 \\ a_0 \\ a_{-1} \end{pmatrix}. \tag{12.9}$$

To find possible values of m_x we solve the characteristic equation:

$$\begin{aligned}
0 = \det\left(\hat{L}_x/\hbar - m_x\hat{I}\right) &= \begin{vmatrix} -m_x & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -m_x & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -m_x \end{vmatrix} = -m_x \begin{vmatrix} -m_x & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -m_x \end{vmatrix} - \frac{1}{\sqrt{2}} \begin{vmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -m_x \end{vmatrix} \\
&= -m_x({m_x}^2 - 1/2) + m_x/2 = m_x(1 - {m_x}^2)
\end{aligned}$$

giving a non-surprising result $m_x = -1, 0, 1$. Naturally, the eigenvalues of any component of $\hat{\boldsymbol{L}}$ must be the same. The eigenstates is a different matter: we have chosen the joint eigenbasis of $\hat{\boldsymbol{L}}$ and $\hat{L}_z$, where both are diagonal, but we have in these basis different non-diagonal representations of $\hat{L}_{x,y}$.

Now we find the eigenvectors corresponding to each of these eigenvalues. For $m_x = 1$ we have (remembering that only two of the equations below are independent and using the simplest ones):

$$\begin{pmatrix} -1 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_0 \\ a_{-1} \end{pmatrix} = 0 \qquad \Rightarrow \qquad a_1 = a_0/\sqrt{2} = a_{-1}\,.$$

Hence the corresponding eigenvector is

$$\big|m_x = 1\big\rangle = a \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \mapsto \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \tag{12.10a}$$

where the constant a has been found from the normalisation, $1 = \langle 1 \,|\, 1\rangle = a^2 + 2a^2 + a^2 = 4a^2 \quad \Rightarrow$ $a = 1/2$. Similarly,

$$\big|m_x = 0\big\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad \big|m_x = -1\big\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}. \tag{12.10b}$$

The orthogonality can be easily checked. In Dirac notations we have $\big|m_x\big\rangle \equiv \big|m\big\rangle_x$ as superpositions of states $\big|l,\, m_z\big\rangle$:

$$\big|1\big\rangle_x = \frac{1}{2}\big|1,1\big\rangle + \frac{1}{\sqrt{2}}\big|1,0\big\rangle + \frac{1}{2}\big|1,-1\big\rangle \tag{12.11a}$$

$$\big|0\big\rangle_x = \frac{1}{\sqrt{2}}\big|1,1\big\rangle - \frac{1}{\sqrt{2}}\big|1,-1\big\rangle \tag{12.11b}$$

$$\big|-1\big\rangle_x = \frac{1}{2}\big|1,1\big\rangle - \frac{1}{\sqrt{2}}\big|1,0\big\rangle + \frac{1}{2}\big|1,-1\big\rangle\,. \tag{12.11c}$$

This expression allows one to predict results of measuring $\hat{L}_z$ in a given eigenstate of $\hat{L}_x$ (and vice versa). For example, let the particle be in the eigenstate of $\hat{L}_x$ corresponding to eigenvalue $m_x = 1$, Eq. (12.11a). The measurement of $\hat{L}_z$ will give the following results:

$\hbar m_z = \hbar$ with probability $\qquad P_+ = |\langle 1,1 \,|\, 1\rangle_x|^2 = \left|\frac{1}{2}\langle 1,1 \,|\, 1,1\rangle\right|^2 = \frac{1}{4}$

$\hbar m_z = 0$ with probability $\qquad P_0 = |\langle 1,0 \,|\, 1\rangle_x|^2 = \left|\frac{1}{\sqrt{2}}\langle 1,1 \,|\, 1,1\rangle\right|^2 = \frac{1}{2}$

$\hbar m_z = -\hbar$ with probability $\qquad P_- = |\langle 1,-1 \,|\, 1\rangle_x|^2 = \left|\frac{1}{2}\langle 1,-1 \,|\, 1,-1\rangle\right|^2 = \frac{1}{4}.$

Our next task is to see how to embed the knowledge of the angular momenta algebra into a 3D quantum mechanics in rotationally symmetric systems. For this, we find it useful to project Dirac states into the position representation. But before doing this, we shall find solution for the radial part of the WF, described by Eq. (11.14).

Lecture 13. Radial equation for a 3D central potential. Virial theorem

Here we solve the equation for the radial motion and establish an important relation between the expectation values of kinetic and potential energies in a stationary state known as the ***Virial Theorem***.

Let us rewrite radial equation (11.14) as follows:

$$\left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}}{\mathrm{d}r}\right)+V_{\text{eff}}(r)\right]R_{nl}=E_{nl}R_{nl}\,, \tag{13.1}$$

where where we have defined the *effective potential*

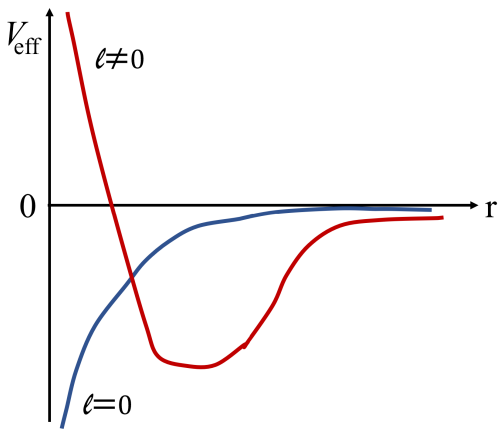
$$V_{\text{eff}}(r)\equiv V(r)+\frac{\hbar^2 l(l+1)}{2\mu r^2}\,. \tag{13.2}$$

combining the potential energy $V(r)$ with the *centrifugal barrier*. The latter depends on the value l of angular momentum.

For $l>0$ the barrier increases like $1/r^2$ for small r and prevents particles to approach the centre $r=0$. However, the particle is protected from the fall even when $l=0$ and V is an attractive potential, like the Coulomb one, $V\propto -1/r$, in an atom. The reason is the HUP:

$$\Delta p_r\Delta r\gtrsim\hbar\quad\Rightarrow\quad\langle\hat{p}_r\rangle\gtrsim\Delta p_r\gtrsim\frac{\hbar}{r}\quad\Rightarrow\quad\left\langle\hat{T}\right\rangle=\frac{\langle\hat{p}_r\rangle^2}{2\mu}\gtrsim\frac{\hbar^2}{\mu r^2} \tag{13.3}$$

This acts exactly like the centrifugal barrier protecting against the fall in any physical potential.[1]



It is convenient to look for the solution of Eq. (13.1) using the substitution

$$R_{nl}(r)=u_{nl}(r)/r\,. \tag{13.4}$$

Since the spherical harmonics, Y_{lm} are normalised, the normalisation condition for R_{nl} reduces to

$$1=\langle\Psi\,|\,\Psi\rangle=\int\mathrm{d}^3r\langle\Psi\,|\,r\rangle\langle r\,|\,\Psi\rangle=\int_0^\infty r^2\,\mathrm{d}r|R_{nl}(r)|^2\underbrace{\int\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi|Y_{lm}(\theta,\phi)|^2}_{=1}$$

[1] Potentials that decrease faster than $-r^{-2}$ would act as a QM "black hole" and so do not exist.

$$= \int_0^\infty \mathrm{d}r |u_{nl}(r)|^2 \,. \tag{13.5}$$

Hence the radial Schrödinger equation for R_{nl} is reduced to the following one for $u_{nl}(r)$:

$$\left[-\frac{\hbar^2}{2\mu}\frac{\partial^2}{\partial r^2} + V_{\text{eff}}(r)\right] u_{nl}(r) = E_{nl} u_{nl}(r)\,. \tag{13.6}$$

This is like a 1D Schrödinger equation but on the line $0 < r < \infty$ instead of the entire x-axis as earlier. Nevertheless, many relations established $1D$ work for $3D$ problems reduced to the above form. We introduce the effective Hamiltonian,

$$\hat{H}_{\text{eff}} = \frac{\hat{p}_r^2}{2\mu} + V_{\text{eff}}(r) \text{ with } \hat{p}_r \equiv -i\hbar\frac{\mathrm{d}}{\mathrm{d}r} \text{ and canonical commutator } \left[\hat{r}, \hat{p}_r\right] = i\hbar, \tag{13.7}$$

and note that the expectation values (or other matrix elements) of any rotationally-invariant (*i.e.* angle-independent) operator can be calculated using the Hamiltonian (13.7) and its eigenfunctions $u_{nl}(r)$.

As we are able to bring the three-dimensional problem to a one-dimensional one using symmetry and separation of variables the rest will depend on the form of the potential $V(r)$. The exact solutions exist for $V(r) = \kappa/r$ (Hydrogen atom), and $V(r) = \frac{1}{2}\mu\omega^2 r^2$ – a 3D QHO (describing, e.g., a wide class of two-atomic molecules). Each such problem can be treated by methods developed in Maths2B and/or QM2 and we will not be repeating this here. However, certain properties can be investigated without knowing an exact solution but using

13.1 Virial Theorem

It allows us to find a robust relation between the expectation values of kinetic and potential energies in a stationary state, provided that the potential is rotationally invariant. Let us apply the Ehrenfest theorem, Eq. (8.4), to the expectation value of the "virial operator", $\hat{\boldsymbol{p}} \cdot \hat{\boldsymbol{r}}$, in a ***stationary state***:

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\langle \hat{\boldsymbol{p}} \cdot \hat{\boldsymbol{r}} \rangle = \left\langle [\hat{\boldsymbol{p}} \cdot \hat{\boldsymbol{r}}, \hat{H}] \right\rangle = 0\,, \tag{13.8}$$

since the expectation value of any Hermitian operator in a stationary state is time-independent and the time-derivative on the l.h.s. vanishes.

As there is no angular dependence in $\hat{\boldsymbol{p}} \cdot \hat{\boldsymbol{r}} = \hat{p}_r \hat{r}$, we evaluate the commutator above using the 1D Hamiltonian, $\hat{H}_{\text{eff}} = \hat{T} + \hat{V}$, and the canonical commutation relation, Eq. (13.7),

$$\left[\hat{p}_r\hat{r}, \hat{T}\right] = \left[\hat{p}_r\hat{r}, \frac{\hat{p}_r^2}{2\mu}\right] = \frac{\hat{p}_r}{2\mu}\left[\hat{r}, \hat{p}_r^2\right] = \frac{\hat{p}_r}{2\mu}\left(\left[\hat{r}, \hat{p}_r\right]p_r + p_r\left[\hat{r}, \hat{p}_r\right]\right) = \frac{i\hbar\hat{p}_r^2}{\mu} = 2i\hbar\hat{T}; \tag{13.9}$$

$$\left[\hat{p}_r\hat{r}, \hat{V}\right] = \left[\hat{p}_r, \hat{V}\right]\hat{r} = -i\hbar\left\{\frac{\mathrm{d}V(r)r}{\mathrm{d}r} - v(r)\frac{\mathrm{d}}{\mathrm{d}r}r\right\} = -i\hbar r\frac{\mathrm{d}V(r)}{\mathrm{d}r}. \tag{13.10}$$

As the sum of these two commutators vanishes, Eq. (13.8), we arrive at the Virial Theorem (VT):

$$\boxed{2\left\langle \hat{T} \right\rangle = \left\langle r\frac{\mathrm{d}\hat{V}}{\mathrm{d}r} \right\rangle} \tag{13.11}$$

For a power-like potential, $V(r) = ar^n$, the right hand side is $r\,\mathrm{d}V/\,\mathrm{d}r = nV(r)$ and we have the following relation between the expectation values of kinetic and potential energy in a stationary state:

$$\left\langle \hat{T} \right\rangle = \frac{n}{2}\left\langle \hat{V} \right\rangle \text{ for } \hat{V} \propto r^n\,,\ n \geqslant -2. \tag{13.12}$$

Harmonic potential, $n = 2$**:** $\left\langle \hat{T} \right\rangle = \left\langle \hat{V} \right\rangle$.

Coulomb potential, $n = -1$**:** $\left\langle \hat{T} \right\rangle = -\frac{1}{2}\left\langle \hat{V} \right\rangle$.

13.2 Groundstate energy of Hydrogen atom

Let us use VT to evaluate the ground state energy of Hydrogen atom. The potential energy is[2]

$$V(\hat{r}) = -\frac{e^2}{4\pi\varepsilon_0 r} \equiv -\kappa\frac{e^2}{r}. \tag{13.13}$$

Clearly, the groundstate wavefunction must be fully rotationally symmetric,[3] so that $V_{\text{eff}}(r) = V(r)$. Then the VT gives

$$2\left\langle \hat{T} \right\rangle = -\left\langle \hat{V} \right\rangle \quad \Rightarrow \quad E_0 = \left\langle \hat{T} \right\rangle + \left\langle \hat{V} \right\rangle = -\left\langle \hat{T} \right\rangle = \frac{1}{2}\left\langle \hat{V} \right\rangle. \tag{13.14}$$

On the other hand, the Heisenberg uncertainty principle, $\Delta p_r \Delta r \geq \hbar$, allows one to estimate the kinetic energy in the ground state:

$$\left\langle \hat{T} \right\rangle \sim \frac{p_r^2}{2\mu} \gtrsim \frac{1}{2\mu}\left(\frac{\hbar}{\Delta r}\right)^2 \sim \frac{1}{2\mu}\left(\frac{\hbar}{a_0}\right)^2, \tag{13.15}$$

where we have introduced typical size of the groundstate wavefunction $a_0 \sim \Delta r$. In this terms, $\left\langle \hat{V} \right\rangle \sim -\kappa e^2/a_0$. Then we find a_0 using Eq. (13.14):

$$\frac{\hbar}{\mu a_0^2} \sim \frac{\kappa e^2}{a_0} \quad \Rightarrow \quad a_0 \sim \frac{\hbar^2}{\mu\kappa e^2} \quad \text{and} \quad E_0 = -\left\langle \hat{T} \right\rangle = \frac{1}{2}\left\langle \hat{V} \right\rangle \sim -\frac{\kappa e^2}{2a_0} = -\frac{\mu\kappa^2 e^4}{2\hbar^2}\,.$$

We know that Bohr radius $a_{\text{Bohr}} = \hbar^2\varepsilon_0/2\mu e^2$ and $E_0 = \mu\kappa^2 e^4/2\hbar^2 =$ Ry $= 13.6$eV so our estimate gives an exact answer. However, it is a serendipity as we cannot keep a numerical coefficient in such an estimate. It is still good as it gives a correct analytic dependence, if not a coefficient.

[2]Using κ is convenient as $\kappa \equiv (4\pi\varepsilon_0)^{-1}$ in SI units, and $\kappa = 1$ in Gaussian (cgs) units.

[3]The angular momentum contribution, $\left(\hbar^2/2\mu\right)l(l+1)$ is positive so that the states with $l \neq 0$ have higher energy.

Lecture 14. Orbital motion in magnetic field. Spin

Here we again return to QM2 material describing orbital motion of charged particles in a magnetic field – briefly – the Stern–Gerlach experiment which discovered intrinsic angular momentum – spin. Then we look at how the ladder operator algebra works for spin operators.[1]

14.1 Orbital motion of a charged particle in magnetic field.

In the previous lectures we have seen that energy levels of a QM particle in a rotationally symmetric potential $V(\boldsymbol{r}) = V(r)$ are are independent of the z component of angular momentum and hence $(2l+1)$-degenerate.[2] If a uniform magnetic field $\boldsymbol{B}$ is present, the rotational symmetry is broken and this degeneracy is lifted.

To see this, we start with deriving the potential energy of a charged ***classical*** particle moving along a closed loop in the field, which is assumed to be a flat circle of radius r. From EM2, you know that such a particle creates the magnetic moment

$$\boldsymbol{\mu} = \mathcal{I}\boldsymbol{\mathcal{A}}. \tag{14.1}$$

Here $\boldsymbol{\mathcal{A}} = \pi r^2 \boldsymbol{n}$ is the oriented area of the loop with the unit vector $\boldsymbol{n}$ perpendicular to it, and $\mathcal{I}$ is the current created by a particle of charge q and mass m_0 moving around the centre:

$$\mathcal{I} = \frac{\mathrm{d}q}{\mathrm{d}t} = \frac{q}{T} = \frac{qv}{2\pi r}\,, \tag{14.2}$$

where $T = 2\pi/\omega = 2\pi r/v$ is the period of the motion around the loop and $v = \omega r$ is the velocity. Hence the magnetic moment of the particle can be represented as

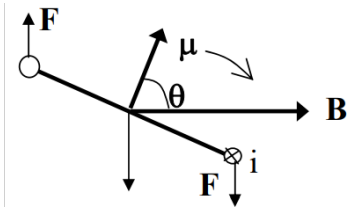
$$\boldsymbol{\mu} = \frac{qv}{2\pi r}\pi r^2 \boldsymbol{n} = \frac{q}{2m_0} r(m_0 v)\boldsymbol{n} = \frac{q}{2m_0}\boldsymbol{r}\times\boldsymbol{p} \equiv \frac{q}{2m_0}\boldsymbol{L}, \tag{14.3}$$

where we used the definition of the classical angular momentum and the fact that $\boldsymbol{r}\times\boldsymbol{p} = rp\boldsymbol{n}$, as $\boldsymbol{n}$ is perpendicular to both $\boldsymbol{r}$ and $\boldsymbol{p}$.

Such a current loop in the field $\boldsymbol{B}$ experiences a torque,

$$\boldsymbol{\tau} \equiv \boldsymbol{r}\times\boldsymbol{F} = \boldsymbol{\mu}\times\boldsymbol{B}, \tag{14.4}$$

that tries to line up $\boldsymbol{\mu}$ with $\boldsymbol{B}$. As there is no torque when $\boldsymbol{\mu}\|\boldsymbol{B}$, there exists potential energy which is minimal in this case. To find it one calculates the work done by rotating the loop. As usual, we choose the z-axis to be along the direction of the field, $\boldsymbol{B} = B\mathbf{k}$. Let $\boldsymbol{F}$ to be the force required for such a rotation, and $\boldsymbol{\mu}\cdot\boldsymbol{B} = |\boldsymbol{\mu}|B\cos\theta$. Then, noting that $\boldsymbol{F}\cdot\mathrm{d}\boldsymbol{s} = |\boldsymbol{r}||\boldsymbol{F}|\sin\theta\,\mathrm{d}\theta$ the work is



$$W = \int \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{s} = \int |\boldsymbol{r}||\boldsymbol{F}|\sin\theta\,\mathrm{d}\theta = \int |\boldsymbol{r}\times\boldsymbol{F}|\,\mathrm{d}\theta \equiv \int |\boldsymbol{\tau}|\,\mathrm{d}\theta. \tag{14.5}$$

[1] **The first two sections of this lecture represent a QM2 revision which will be only briefly shown in the class**

[2] There is no degeneracy for a particle moving along a straight line when $l{=}0$ and we are back to the 1D QM.

The potential energy $U = -W$. Substituting Eq. (14.3) for $\boldsymbol{\mu}$we find[3]

$$U = \int |\boldsymbol{\mu} \times \boldsymbol{B}| \, \mathrm{d}\theta = \mu B \int \sin\theta \, \mathrm{d}\theta = -\mu B \cos\theta = -\boldsymbol{\mu} \cdot \boldsymbol{B} = -\frac{q}{2m_0} \boldsymbol{L} \cdot \boldsymbol{B} \,. \tag{14.6}$$

For motion of a quantum particle in a magnetic field[4] we add this term to Hamiltonian (11.7):

$$\Delta \hat{H} = -\hat{\boldsymbol{\mu}} \cdot \boldsymbol{B} = -\frac{q}{2m_0} \hat{\boldsymbol{L}} \cdot \boldsymbol{B} \,. \tag{14.7}$$

When considering motion of one electron, $m_0 \to m_\mathrm{e}$ and $q \to e$, we rewrite this using Eq. (14.3) as

$$\Delta \hat{H} = -\frac{\mu_\mathrm{B}}{\hbar} B \hat{L}_z \tag{14.8}$$

The quantity $\mu_\mathrm{B} = e\hbar/2m_\mathrm{e}$ is called *Bohr magneton*.

Thus, for magnetic field in z-direction the quantum Hamiltonian acquires an additional term describing interaction of the electron's magnetic moment with the magnetic field. It is diagonal in the eigenbasis of angular momentum:

$$\Delta \hat{H} \big| l, m \big\rangle = -\mu_\mathrm{B} B \, m \big| l, m \big\rangle \tag{14.9}$$

and produces dependence of the energy level on the magnetic quantum number m splitting the energy E_{nl} into $2l + 1$ sub-levels. This additional quantisation has been discovered in the seminal Stern-Gerlach experiments.

14.2 Stern-Gerlach experiment.

This splitting of levels in the magnetic field was studied in the early years of quantum mechanics in the celebrated experiment by Stern and Gerlach. The main idea was to send silver atoms through a region of inhomogeneous magnetic field as shown in Figure 1. In the region with inhomogeneous

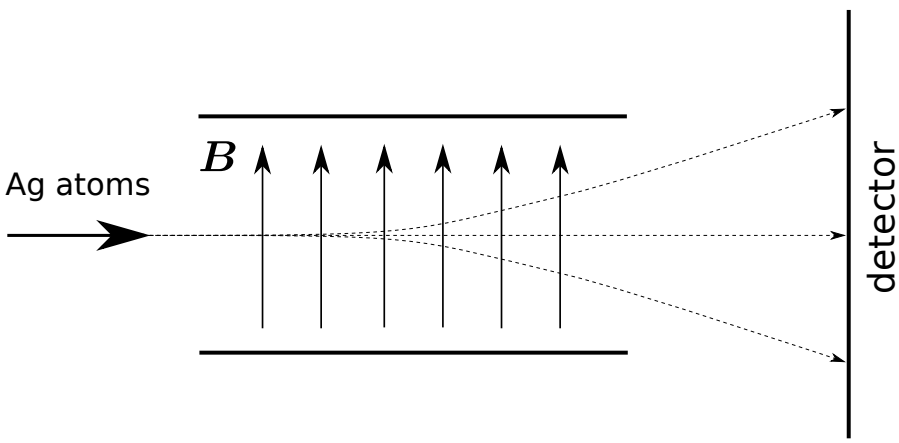


Figure 3: Stern-Gerlach experiment

[3]As the zero point of U is arbitrary, we choose an arbitrary constant in the integral of Eq. (14.6) to be 0

[4]We do not consider here a more subtle effect due to the changing of the kinetic energy in a magnetic field which results in discrete energy levels (so called Landau levels) of a *free* particle in the field.

magnetic field $\boldsymbol{B} = B(z)\mathbf{k}$, the atoms feel a vertical force

$$F_z = -\frac{\partial}{\partial z}\Delta H = \frac{\mu_{\rm B}}{\hbar}\frac{\partial B(z)}{\partial z}\hat{L}_z \tag{14.10}$$

proportional to the value of $\hat{L}_z$. Thus the atoms were expected to be deflected in the vertical direction according to their angular momentum component $\hat{L}_z$. The resulting deflection can be detected once the atoms hit the detector.[5]

For atoms having total angular momentum l there should be $2l+1$-fold splitting and $2l+1$ peaks detected. The silver atoms have one electron in the outer shell which may contribute to l, but its ground state value is $l = 0$. Therefore *only one* peak was expected to be observed. Instead, the beam of atoms was split into *two*! This suggested the existence of intrinsic angular momentum with two possible values of its z-component. Since the number of possible values of the magnetic quantum number m is $2l+1$ this means that the total angular momentum quantum number must be $l = 1/2$.

The algebra of angular momentum operators taught us that both half-integer and integer values of quantum numbers l and m are allowed. After the Stern-Gerlach discovery, existence of the ***intrinsic*** angular momentum of a particle – **spin** – has been confirmed. The magnetic quantum number for spin can be either half-integer (fermions) or integer (bosons) (including $l = 0$ – Bosons without spin). The algebra is the same but spin operators cannot be realised as the vector product of position and momentum operators. Thus one have to work with abstract operators of half-integer angular momentum which do not have classical counterparts.

14.3 Spin operators

Although spin $\boldsymbol{S}$ is not a usual vector (you need to rotate it through angle 4π rather than 2π to return to the same value), it still has three components $\hat{S}_x$, $\hat{S}_y$, $\hat{S}_z$. Their commutation relations are the same as for the orbital angular momentum,

$$[\hat{S}_x, \hat{S}_y] = \mathrm{i}\hbar\hat{S}_z\,, \quad [\hat{S}_y, \hat{S}_z] = \mathrm{i}\hbar\hat{S}_x\,, \quad [\hat{S}_z, \hat{S}_x] = \mathrm{i}\hbar\hat{S}_y\,. \tag{14.11}$$

As algebra is the same, $\hat{\boldsymbol{S}}^2$ and $\hat{S}_z$ commute and have joint eigenstates, i.e. can be diagonalised simultaneously. By analogy with $\big|l, m\big\rangle$ we call these eigenstates $\big|s, \sigma\big\rangle$ with

$$\hat{S}^2\big|s,\sigma\big\rangle = \hbar^2 s(s+1)\big|s,\sigma\big\rangle\,, \qquad \hat{S}_z\big|s,\sigma\big\rangle = \hbar\sigma\big|s,\sigma\big\rangle\,. \tag{14.12}$$

In case of fermions, a half-integer $s = 1/2, 3/2, 5/2, \ldots$ replacing the total angular momentum quantum number l and the quantum number σ takes $2s+1$ values from $-s$ to s. The letters s, σ will be used to distinguish quantum numbers of spin operators from letters l, m used traditionally for orbital angular momentum. In case of bosons, s takes integer values, $0, 1, 2\ldots$ but the spin operators still must be distinguished from the orbital angular momentum operators, despite having exactly the same algebra.

[5]Reminder: Stern-Gerlach device can be oriented along any axis and using a set of them with different orientations is crucial for experiments with entanglement described in Lecture 4.

As before, the spin algebra is simpler for the ladder operators $\hat{S}_\pm = \hat{S}_x \pm \mathrm{i}\hat{S}_y$. Their action on the eigenstates $\big|s,\sigma\big\rangle$ is governed by the same formulae with replacing l by s and m by σ:

$$\hat{S}_\pm\big|s,\sigma\big\rangle = \hbar c^\pm_{s,\sigma}\big|s,\sigma\pm 1\big\rangle\,, \qquad c^\pm_{s,\sigma} = \sqrt{(s\mp\sigma)(s\pm\sigma+1)}\,. \tag{14.13}$$

The ladder operators allow to construct matrix representation for a given fixed s of the spin operators $\hat{S}_x$ and $\hat{S}_y$ as well as any function of them, while by definition $\hat{S}_z$ is diagonal in its own basis:

$$\Big\{\hat{S}_z\Big\}_{\sigma'\sigma} \equiv \Big\langle s,\sigma' \,\Big|\, \hat{S}_z \,\Big|\, s,\sigma\Big\rangle = \hbar\,\mathrm{diag}\,\{s,\ s-1,\ \cdots - s+1,\ -s\}. \tag{14.14}$$

Let us start with the most important case of $s = 1/2$.

14.4 Pauli Matrices for spin ½.

Among all possible half-integer values spin the *lowest value*, $s = 1/2$ plays a distinguished role. First it is the spin of many elementary particles such as electron, proton, neutron *etc.*. Secondly, as we shall see later, any angular momentum (including an integer one) can be obtained by adding sufficient number of spins with $s = 1/2$.

We have already used spin ½ when describing quantum entanglement and two-level system which also can be represented using spin operators. We employ the same notation as in Lecture 3:

$$\big|s{=}\tfrac{1}{2},\sigma{=}+\tfrac{1}{2}\big\rangle \equiv \big|\uparrow\big\rangle \equiv \big|+\big\rangle, \qquad\qquad \big|s{=}\tfrac{1}{2},\sigma{=}-\tfrac{1}{2}\big\rangle \equiv \big|\downarrow\big\rangle \equiv \big|-\big\rangle.$$

The action of the ladder operators $\hat{S}_\pm = \hat{S}_x \pm \mathrm{i}\hat{S}_y$ on these basis vectors we find, using Eq. (14.13):

$$\hat{S}_+\big|\downarrow\big\rangle = \hbar\big|\uparrow\big\rangle\,, \qquad \hat{S}_-\big|\uparrow\big\rangle = \hbar\big|\downarrow\big\rangle\,, \qquad \hat{S}_+\big|\uparrow\big\rangle = \hat{S}_-\big|\downarrow\big\rangle = 0\,. \tag{14.15}$$

Hence each has only one nonzero matrix element:

$$\Big\langle\uparrow\,\Big|\,\hat{S}_+\,\Big|\,\downarrow\Big\rangle = \Big\langle\downarrow\,\Big|\,\hat{S}_-\,\Big|\,\uparrow\Big\rangle = \hbar. \tag{14.16}$$

Expressing $\hat{S}_{x,y}$ via $\hat{S}_\pm$ and using Eq. (14.14) for $\hat{S}_z$, we find

$$\begin{aligned}
\Big\langle\sigma'\,\Big|\,\hat{S}_z\,\Big|\,\sigma\Big\rangle &\equiv \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} = \frac{\hbar}{2}\hat{\sigma}_z\\
\Big\langle\sigma'\,\Big|\,\hat{S}_x\,\Big|\,\sigma\Big\rangle &\equiv \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = \frac{\hbar}{2}\hat{\sigma}_x\\
\Big\langle\sigma'\,\Big|\,\hat{S}_y\,\Big|\,\sigma\Big\rangle &\equiv \frac{\hbar}{2}\begin{pmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{pmatrix} = \frac{\hbar}{2}\hat{\sigma}_y\,.
\end{aligned} \tag{14.17}$$

The dimensionless matrices $\hat{\boldsymbol{\sigma}} = (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$ are called Pauli matrices. Their main properties:

Hermiticity: $\hat{\sigma}_i^\dagger = \hat{\sigma}_i$ for $i = x, y, z$.

Unit square $\hat{\sigma}_i^2 = \mathbb{1}_{\,2\times 2}$.

Thus $\hat{S}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2 = \frac{\hbar^2}{4}(\sigma_x^2 + \sigma_y^2 + \sigma_z^2) = \frac{3}{4}\hbar^2 = \frac{1}{2}(\frac{1}{2}+1)\hbar$ as expected.

Commutators $[\hat{\sigma}_i, \hat{\sigma}_j] = 2\mathrm{i}\sigma_k$ where ijk is a cyclic permutation of (xyz).

Anti-commutators $\{\hat{\sigma}_i, \hat{\sigma}_j\} \equiv \hat{\sigma}_i\hat{\sigma}_j + \hat{\sigma}_j\hat{\sigma}_i = 0$ for $i \neq j$.

14.5 Matrix representation for spin $s = \frac{3}{2}$ operators in the standard basis.

We shall do this as an exercise and for future references. The number of the projections σ is $2s+1 = 4$. The expectation $\langle \hat{S}^2 \rangle = \frac{15}{4}\hbar^2$ in any state. $\hat{S}_z$ is a diagonal: $\left\langle \sigma' \,\middle|\, \hat{S}_z \,\middle|\, \sigma \right\rangle = \frac{1}{2}\hbar \,\mathrm{diag}\,(3,\, 1,\, -1,\, -3)$, traditionally counted from the largest eigenvalue. Writing this as a full matrix gives

$$\left\langle \sigma' \,\middle|\, \hat{S}_z \,\middle|\, \sigma \right\rangle = \hbar \begin{pmatrix} \frac{3}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & -\frac{3}{2} \end{pmatrix}$$

I stress again: the numbering of both rows and columns starts with $\frac{3}{2}$ and goes down to $-\frac{3}{2}$, instead of usual numbering from 1 to n in a $n \times n$ matrix.[6]

The we have to deal with $\hat{S}_+ \equiv \hat{S}_x = i\hat{S}_y$, using Eq. (14.13), $\hat{S}^+\big|\sigma\big\rangle = \hbar c^+_{\frac{3}{2},\sigma}\big|\sigma+1\big\rangle$ with

$$c^+_{\frac{3}{2},\sigma} = \sqrt{\left(\tfrac{3}{2}-\sigma\right)\left(\tfrac{3}{2}+\sigma+1\right)} = \left\{0, \sqrt{3}, 2, \sqrt{3}\right\}.$$

Hence the matrix representations of $\hat{S}_+$ and its Hermitian conjugate $\hat{S}_-$are

$$\left\langle \sigma' \,\middle|\, \hat{S}_+ \,\middle|\, \sigma \right\rangle = \hbar \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \left\langle \sigma' \,\middle|\, \hat{S}_- \,\middle|\, \sigma \right\rangle = \hbar \begin{pmatrix} 0 & 0 & 0 & 0 \\ \sqrt{3} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \tag{14.18}$$

As $\hat{S}_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-)$ and $\hat{S}_y = -\frac{i}{2}(\hat{S}_+ - \hat{S}_-)$, we find

$$\left\langle \sigma' \,\middle|\, \hat{S}_x \,\middle|\, \sigma \right\rangle = \frac{\hbar}{2} \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}, \quad \left\langle \sigma' \,\middle|\, \hat{S}_y \,\middle|\, \sigma \right\rangle = \frac{i\hbar}{2} \begin{pmatrix} 0 & -\sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & -2 & 0 \\ 0 & 2 & 0 & -\sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}$$

[6] The same was done – implicitly – for matrix elements of operators $\hat{\boldsymbol{L}}$.

Lecture 15. Spin ½ . Pauli Matrices and spinors

We have established that in the standard basis spin-½ operators are represented by Pauli matrices. As there is nothing special about the z-axis, we look for a representation in an arbitrary oriented axes.

15.1 Eigenstates of spin ½ in arbitrary direction. Spinors.

Now we return to this most important case. So far we dealt with basis eigenvectors of spin ½ directed along z-axis, *i.e.* eigenstates of $\hat{S}_z$,

$$\hat{S}_z|\uparrow\rangle = \frac{\hbar}{2}|\uparrow\rangle\,, \qquad \hat{S}_z|\downarrow\rangle = -\frac{\hbar}{2}|\downarrow\rangle, \tag{15.1}$$

and have represented matrix elements of $\hat{S}_{x,y,z}$ in terms of Pauli's matrices, $\langle\sigma'\,|\,\hat{\boldsymbol{S}}\,|\,\sigma\rangle = \frac{1}{2}\hbar\hat{\boldsymbol{\sigma}}$:

$$\hat{\boldsymbol{\sigma}} = (\hat{\sigma}_x,\,\hat{\sigma}_y,\,\hat{\sigma}_z) \quad\Rightarrow\quad \hat{\sigma}_z \equiv \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \quad \hat{\sigma}_x \equiv \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \quad \hat{\sigma}_y \equiv \begin{pmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{pmatrix}. \tag{15.2}$$

Reminder: commutators and anticommutators of Pauli's matrices:

$$\begin{aligned}&\big[\hat{\sigma}_j,\hat{\sigma}_k\big] = 2i\hat{\sigma}_l \quad\Leftrightarrow\quad \big[\hat{S}_j,\hat{S}_k\big] = \frac{i\hbar}{2}\hat{S}_l \text{ where } jkl \text{ is an even (cyclic) permutation of } xyz\\ &\{\hat{\sigma}_j,\hat{\sigma}_k\} \equiv \hat{\sigma}_j\hat{\sigma}_k + \hat{\sigma}_k\hat{\sigma}_j = \delta_{kj}\end{aligned} \tag{15.3}$$

Now we want to find eigenvectors of spin projections along arbitrary direction $\boldsymbol{n}$ in space where the unit vector $\boldsymbol{n}$ is given in spherical coordinates by

$$\hat{\boldsymbol{n}} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta). \tag{15.4}$$

The eigenvectors of $\hat{\boldsymbol{S}}\cdot\boldsymbol{n}$, which is the spin projection along $\boldsymbol{n}$, are linear combinations[1]

$$|\pm\rangle = a_\pm|\uparrow\rangle + b_\pm|\downarrow\rangle, \qquad \text{or in matrix form } \langle\sigma\,|\,\pm\rangle = \begin{pmatrix}a_\pm\\ b_\pm\end{pmatrix}, \tag{15.5}$$

satisfying the eigenvalue problem where we know eigenvalues are $\pm\frac{\hbar}{2}$ even without solving the appropriate characteristic equation:[2]

$$\hat{\boldsymbol{S}}\cdot\hat{\boldsymbol{n}}\,|\pm\rangle = (n_x\hat{S}_x + n_y\hat{S}_y + n_z\hat{S}_z)\,|\pm\rangle = \pm\frac{\hbar}{2}|\pm\rangle\,. \tag{15.6}$$

Noticing that $\hat{\boldsymbol{S}}\cdot\boldsymbol{n} = \frac{\hbar}{2}\hat{\boldsymbol{\sigma}}\cdot\boldsymbol{n}$, we arrive at the EVEs $\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{\sigma}}|\pm\rangle = \pm|\pm\rangle$, where in the standard basis

$$\begin{aligned}\boldsymbol{n}\cdot\hat{\boldsymbol{\sigma}} &= \begin{pmatrix}0 & \sin\theta\cos\phi\\ \sin\theta\cos\phi & 0\end{pmatrix} + \begin{pmatrix}0 & -\mathrm{i}\sin\theta\sin\phi\\ \mathrm{i}\sin\theta\sin\phi & 0\end{pmatrix} + \begin{pmatrix}\cos\theta & 0\\ 0 & -\cos\theta\end{pmatrix}\\ &= \begin{pmatrix}\cos\theta & \sin\theta\,\mathrm{e}^{-\mathrm{i}\phi}\\ \sin\theta\,\mathrm{e}^{\mathrm{i}\phi} & -\cos\theta\end{pmatrix}.\end{aligned} \tag{15.7}$$

[1] We reserve the notations $|\sigma\rangle = |\uparrow,\downarrow\rangle$ for the standard basis and use $|\pm\rangle$ for arbitrarily oriented axes.
[2] Check this by writing and solving the characteristic equation.

Writing the EVEs for both $\big|+\big\rangle$ and $\big|-\big\rangle$ with eigenvalues ± 1, we have

$$\begin{pmatrix} \cos\theta - 1 & \sin\theta\, \mathrm{e}^{-\mathrm{i}\phi} \\ \sin\theta\, \mathrm{e}^{\mathrm{i}\phi} & -\cos\theta - 1 \end{pmatrix} \begin{pmatrix} a_+ \\ b_+ \end{pmatrix} = 0, \qquad \begin{pmatrix} \cos\theta + 1 & \sin\theta\, \mathrm{e}^{-\mathrm{i}\phi} \\ \sin\theta\, \mathrm{e}^{\mathrm{i}\phi} & -\cos\theta + 1 \end{pmatrix} \begin{pmatrix} a_- \\ b_- \end{pmatrix} = 0, \tag{15.8}$$

and remembering that only one equation for the coefficient is relevant, we find

$$(\cos\theta \mp 1) a_\pm + \sin\theta\, \mathrm{e}^{-\mathrm{i}\phi} b_\pm = 0\,. \tag{15.9}$$

This can be rewritten, using $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$, $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$, $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$, as

$$b_+ = \tan\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi} a_+\,, \qquad b_- = -\cot\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi} a_-\,. \tag{15.10}$$

Using the normalisation conditions $|a_\pm|^2 + |b_\pm|^2 = 1$ which amount to

$$|a_+|^2\left(1 + \tan^2\frac{\theta}{2}\right) = \frac{|a_+|^2}{\cos^2\frac{\theta}{2}} = 1, \quad |a_-|^2\left(1 + \cot^2\frac{\theta}{2}\right) = \frac{|a_-|^2}{\sin^2\frac{\theta}{2}} = 1,$$

we choose[3] $a_+ = \cos\frac{\theta}{2}$, $a_- = \sin\frac{\theta}{2}$ and find the normalised eigenvectors

$$\langle\theta,\phi\,|\,+\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ \mathrm{e}^{\mathrm{i}\phi}\sin\frac{\theta}{2} \end{pmatrix} \qquad \langle\theta,\phi\,|\,-\rangle = \begin{pmatrix} \sin\frac{\theta}{2} \\ -\mathrm{e}^{\mathrm{i}\phi}\cos\frac{\theta}{2} \end{pmatrix}. \tag{15.11}$$

The column vectors of components we found are called **spinors**. Like usual three-dimensional vectors they depend on angles and change if the coordinate system is rotated. However they have only two components, are complex and do not obey the vector transformation rules. You can see that changing $\theta \to \theta + 2\pi$ does not keep eigenvectors intact – you need to rotate through 4π. The origin of this could only be understood in the framework of relativistic QM.

A general spin state $\big|\psi\big\rangle$ can be represented as a spinor:

$$\langle\sigma\,|\,\psi\rangle = \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix} \tag{15.12}$$

using the components $\psi_\uparrow = \langle\uparrow\,|\,\psi\rangle$, $\psi_\downarrow = \langle\downarrow\,|\,\psi\rangle$ in the eigenbasis of $\hat{S}_z$. Comparing them with the states $\big|\pm\big\rangle$ one can always find the angles θ, ϕ such that (up to a overall phase) $\big|\psi\big\rangle$ is an eigenstate of spin with $s = 1/2$ in the direction specified by these angles. This is not true for higher values of s.

As an exercise, let us repeat the calculations we have done for a two-level system. Suppose we have two Stern-Gerlach (SG) spin polarisers in a row. The first puts all coming through electrons into state $\big|\downarrow\big\rangle$, while the second is oriented along $\boldsymbol{n}$, Eq. (15.4). What are the probabilities to have projections $\big|\pm\big\rangle$ along $\boldsymbol{n}$? According to the 3rd QMP, they are given by $P_\pm = |\langle\downarrow\,|\,\pm\rangle|^2$. In the standard basis, $\big\langle\downarrow\big| \mapsto \begin{pmatrix} 0 & 1 \end{pmatrix}$, while $\big|\pm\big\rangle$ are given in this representation by matrix columns (15.11) so that

$$\langle\downarrow\,|\,+\rangle = \begin{pmatrix} 0 & 1 \end{pmatrix}\begin{pmatrix} \cos\frac{\theta}{2} \\ \mathrm{e}^{\mathrm{i}\phi}\sin\frac{\theta}{2} \end{pmatrix} = \mathrm{e}^{i\phi}\sin\frac{\theta}{2} \qquad \Rightarrow \qquad P_+ = \sin^2\frac{\theta}{2}, \tag{15.13}$$

$$\langle\downarrow\,|\,-\rangle = \begin{pmatrix} 0 & 1 \end{pmatrix}\begin{pmatrix} \sin\frac{\theta}{2} \\ -\mathrm{e}^{\mathrm{i}\phi}\cos\frac{\theta}{2} \end{pmatrix} = \mathrm{e}^{i\phi}\cos\frac{\theta}{2} \qquad \Rightarrow \qquad P_- = \cos^2\frac{\theta}{2}. \tag{15.14}$$

Note that there is no dependence on the angle ϕ.

[3] Remember that phase factors are arbitrary so that, e.g., $\langle\theta,\phi\,|\,+\rangle = \begin{pmatrix} \mathrm{e}^{-\mathrm{i}\phi/2}\cos\frac{\theta}{2} \\ \mathrm{e}^{\mathrm{i}\phi/2}\sin\frac{\theta}{2} \end{pmatrix}$ is equally good. However, the **relative** phase factors are fixed, so that changing them in $\langle\theta,\phi\,|\,+\rangle$ we must simultaneously change $\langle\theta,\phi\,|\,-\rangle$ to guarantee $\langle +\,|\,-\rangle = 0$.

15.2 Spin and orbital degrees of freedom

So far we have discussed spin degrees of freedom in separation from orbital motion. Let us combine this, first limiting ourselves to the rotationally symmetric situation. In this case, for a fixed spin s the standard basis in the Hilbert space becomes

$$\left\{\left|n,l,m\right\rangle \otimes \left|S\right\rangle\right\} \text{ with } \left|S\right\rangle \equiv \left|s,\,\sigma\right\rangle . \tag{15.15}$$

The direct product sign, $\otimes$, means that the spin components and spatial components do not affect each other – they belong to different Hilbert subspaces. Consider spin ½, when $\left|\sigma\right\rangle = \left|\uparrow\right\rangle$ or $\left|\downarrow\right\rangle$. The wavefunction corresponding to an arbitrary state $\left|\psi\right\rangle$ becomes

$$\left\langle \boldsymbol{r},\sigma \,\middle|\, \psi \right\rangle \equiv \psi_\sigma(\boldsymbol{r}) \to \begin{pmatrix} \psi_\uparrow(\boldsymbol{r}) \\ \psi_\downarrow(\boldsymbol{r}) \end{pmatrix}. \tag{15.16}$$

In this way particles with spin ½ are described by a two-component wave function[4] as above. This wavefunction is a component of the state $\left|\psi\right\rangle$ in the simultaneous eigenbasis of position and spin operators. The wave function written as a two-component column, Eq. (15.16), is also referred to as a spinor, Eq. (15.12), even when it has a position dependence.

We write the spin component as a label to stress that σ is discrete and takes only two values. The resolution of unity takes this into account by adding summation over spin indices:

$$\hat{I} = \sum_{\sigma=\uparrow,\downarrow} \int \mathrm{d}^3 r \left|\boldsymbol{r},\sigma\right\rangle\left\langle \boldsymbol{r},\sigma\right| . \tag{15.17}$$

Using the resolution of unity, Eq. (15.17), one finds the normalisation of spinors:

$$1 = \langle \psi \,|\, \psi \rangle = \sum_{\sigma=\uparrow,\downarrow} \int \mathrm{d}\boldsymbol{r}\, \langle \psi \,|\, \boldsymbol{r},\sigma\rangle\langle \boldsymbol{r},\sigma \,|\, \psi\rangle = \int \mathrm{d}^3 r\, \left(|\psi_\uparrow(\boldsymbol{r})|^2 + |\psi_\downarrow(\boldsymbol{r})|^2\right) . \tag{15.18}$$

If there is no interaction between spin and orbital (position) degrees of freedom, then $\psi_\uparrow(\boldsymbol{r})$ and $\psi_\downarrow(\boldsymbol{r})$ have the same spatial dependence and the normalised spinor of Eq. (15.16) can be represented as

$$\left\langle \boldsymbol{r},\sigma \,\middle|\, \psi \right\rangle = \psi(\boldsymbol{r}) \begin{pmatrix} a_\uparrow \\ a_\downarrow \end{pmatrix} \text{ with } |a_\uparrow|^2 + |a_\downarrow|^2 = 1 \text{ and } \int \mathrm{d}^3 r\, \psi^*(\boldsymbol{r})\psi(\boldsymbol{r}) = 1 . \tag{15.19}$$

This situation is referred to as a spin-degeneracy: summing over two values of spin adds a factor of 2 in the resolution of unity.[5]

In this representation operators independent of spin act in the same way on both components of the wavefunction, *i.e.* they are represented by diagonal (operator-valued) matrices, like

$$\hat{\boldsymbol{p}}\psi_\sigma(\boldsymbol{r}) = \left\langle \boldsymbol{r},\sigma\middle|\hat{\boldsymbol{p}}\middle|\psi\right\rangle = -i\hbar\boldsymbol{\nabla}\psi_\sigma(\boldsymbol{r}) \qquad \text{or} \qquad \begin{pmatrix} -i\hbar\boldsymbol{\nabla} & 0 \\ 0 & -i\hbar\boldsymbol{\nabla} \end{pmatrix} \begin{pmatrix} \psi_\uparrow(\boldsymbol{r}) \\ \psi_\downarrow(\boldsymbol{r}) \end{pmatrix} . \tag{15.20}$$

[4] and by a $2s+1$-component wavefunction for spin s.

[5] It is important in statistical physics when the extra factor of 2 due to spin degeneracy emerges, e.g., in calculating the density of state.

15.3 Spin-orbit coupling

The factorisation (direct product) of orbital and spin degrees of freedom, as in Eqs. (15.15) and (15.19), breaks down when there is an interaction between spin and position operators. The most important example is the operator proportional to $\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{S}}$ known as **spin-orbit coupling**. Such a coupling inevitably exists for charged particles as both $\hat{L}$ and $\hat{S}$ create, independently of each other, magnetic moments which interact.

In a rotationally-symmetric system with states $|n, l, m; s, \sigma\rangle$ the energy depends only on n and l, *i.e.* there is degeneracy with respect to magnetic quantum number, m, and spin quantum numbers s, σ. As you might remember from QM2, in the hydrogen atom there is an additional, "incidental" degeneracy[6]: the energy levels do not depend also on l for $\hat{H}_0 = \hat{\boldsymbol{p}}^2/2\mu + \kappa e^2/r$, and are given by

$$E_n = -\frac{\mathrm{Ry}}{n^2}, \qquad \mathrm{Ry} \equiv \frac{m\kappa^2 e^4}{2\hbar^2} \approx 13.6\, eV. \tag{15.21}$$

The existence of electron spin is one of the factors that lifts this degeneracy. As we have learned, Eq. (14.3), for the electron orbital motion we have

$$\boldsymbol{\mu} = \frac{\mu_\mathrm{B}}{\hbar}\boldsymbol{L} \text{ where Bohr's magneton } \mu_\mathrm{B} \equiv \frac{e\hbar}{2m_e}. \tag{15.22}$$

The intrinsic angular moment – spin – produces the magnetic moment in a similar way, albeit the coefficient is different by a factor[7] of $g \approx 2$,

$$\hat{\boldsymbol{\mu}}_\mathrm{S} = \frac{g\mu_\mathrm{B}}{\hbar}\hat{\boldsymbol{S}}\,. \tag{15.23}$$

When place in a magnetic field, its energy changes similarly to the change caused by the orbital momentum, Eq. (14.7):

$$\Delta\hat{H}_S = -\hat{\boldsymbol{\mu}}_s \cdot \boldsymbol{B} = -\frac{g\mu_B}{\hbar}\hat{\boldsymbol{S}} \cdot \boldsymbol{B}$$

This results in an additional level splitting in a magnetic field. Moreover, the two magnetic moments interact – in QM as well as in the classical physics, which leads to the level splitting in the H-atom even without the field. This interaction is traditionally called ***spin-orbit coupling.*** The appropriate addition to the Hamiltonian $\hat{H}_0$ is[8]

$$H_\mathrm{so} = -A(r)\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{S}}\,. \tag{15.24}$$

It is one of the relativistic corrections to the hydrogen spectrum and its contribution is small by the factor $\mathrm{Ry}/m_\mathrm{e}c^2 \sim \alpha^2$ where $\alpha \equiv \kappa e^2/\hbar c \approx 1/137$ is the so-called fine-structure constant. This can be established using the perturbation theory to be introduced in the last four lectures. However, our interest at the moment is to see how this correction destroys the factorisation of orbital and spin degrees of freedom as in Eqs. (15.15) and (15.19). We shall consider this in the next lecture.

[6]Nothing is "incidental" in QM, and this additional degeneracy is due to existence of an additional conservation law, *i.e.* some operator commuting with the Hamiltonian. However, we will not be considering this here.

[7]The origin of the *gyromagnetic factor* g in Eq. (15.23) is relativistic, and it is of no impact on what follows.

[8]The r-dependence of $A(r)$ is irrelevant for what follows but – for curious – it is given by $A(r) = \dfrac{\kappa e^2}{2m_e^2 c^2 r^3}$.

Lecture 16. Spin-orbit interaction. Addition of angular momenta.

Here we learn how to add orbital angular momentum and spin to diagonalise the spin-orbit interaction, representing states via quantum numbers of the operator of the total angular momentum. This is generalised to addition of arbitrary angular momenta, either orbital or spin.

16.1 Spin-orbit operator

We have seen in the previous lecture that the magnetic moments, one created by spin and the other by orbital motion, interact adding the term described by Eq. (15.24) to the Hamiltonian of the Hydrogen atom[1]:

$$H_{\rm so} = -A(r)\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{S}}\,.$$

Using Pauli matrices, $\hat{\boldsymbol{S}} = \frac{1}{2}\hbar\hat{\boldsymbol{\sigma}}$ and omitting (for now) $A(r)$ we represent operator (15.24) as

$$\frac{\hbar}{2}\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{\sigma}} = \frac{\hbar}{2}\Big(\hat{L}_x\hat{\sigma}_x + \hat{L}_y\hat{\sigma}_y + \hat{L}_z\hat{\sigma}_z\Big) = \frac{\hbar}{2}\begin{pmatrix}\hat{L}_z & \hat{L}_x - \mathrm{i}\hat{L}_y\\ \hat{L}_x + \mathrm{i}\hat{L}_y & -\hat{L}_z\end{pmatrix} \equiv \frac{\hbar}{2}\begin{pmatrix}\hat{L}_z & \hat{L}_-\\ \hat{L}_+ & -\hat{L}_z\end{pmatrix}. \tag{16.1}$$

Acting with it on the spinor $\psi_\sigma(\boldsymbol{r})$ gives

$$\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{S}}\begin{pmatrix}\psi_\uparrow(\boldsymbol{r})\\ \psi_\downarrow(\boldsymbol{r})\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}\hat{L}_z & \hat{L}_-\\ \hat{L}_+ & -\hat{L}_z\end{pmatrix}\begin{pmatrix}\psi_\uparrow(\boldsymbol{r})\\ \psi_\downarrow(\boldsymbol{r})\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}\hat{L}_z\psi_\uparrow(\boldsymbol{r}) + \hat{L}_-\psi_\downarrow(\boldsymbol{r})\\ \hat{L}_+\psi_\uparrow(\boldsymbol{r}) - \hat{L}_z\psi_\downarrow(\boldsymbol{r})\end{pmatrix}\,. \tag{16.2}$$

In this expression operators of angular momentum $\hat{L}_z$, $\hat{L}_\pm$ change the angular dependence of the components (by differentiation for example). As the spin-orbit coupling operator is not diagonal, the components $\psi_{\uparrow\downarrow}$ of the spinor become *mixed*. Hence the factorisation in Eqs. (15.15) and (15.19) is no longer applicable. Working with the angular momentum in the abstract Hilbert space would allow us to avoid dealing with matrices of differential operators and we will find appropriate linear combination of the spinors in which the spin-orbit coupling is diagonal. The diagonalisation procedure is known as **addition of angular momenta**.

16.2 Adding angular momenta

Neither orbital angular momentum $\hat{\boldsymbol{L}}$ nor spin $\hat{\boldsymbol{S}}$ commute with the spin-orbit Hamiltonian, Eq. (15.24). This follows from the fact that neither different components of $\hat{\boldsymbol{L}}$ nor different components of $\hat{\boldsymbol{S}}$ commute between themselves.[2] This means that in the presence of the spin-orbit couplint l, m and σ are no longer "good" quantum numbers – they are not conserved.

Instead, one can show (see Problem Sheet 3) that the **total angular momentum**

$$\hat{\boldsymbol{J}} = \hat{\boldsymbol{L}} + \hat{\boldsymbol{S}} \tag{16.3}$$

[1]Such a term would be added to any Hamiltonian of a spin-particle in an orbital motion.

[2]Obviously, $\big[\hat{\boldsymbol{L}}, \hat{\boldsymbol{S}}\big] = 0$ as $\hat{\boldsymbol{L}}$ and $\hat{\boldsymbol{S}}$ act on different variables. Note also that $\hat{H}_{\rm so}$ commutes with both $\hat{L}^2$ and $\hat{S}^2$.

does commute with $H_{\rm so}$. It also commutes with the rotationally-invariant Hamiltonian $\hat{H}_0$, since we know that any component of $\hat{\boldsymbol{L}}$ does, and so obviously does the spin operator since $\hat{H}_0$ is spin-independent.

As the sum of two angular momenta operators, components of $\hat{\boldsymbol{J}}$ have the same algebra so that there exist the joint eigenstates of $\hat{J}^2$ and $\hat{J}_z$,

$$\begin{aligned} \hat{J}^2\,|j,\,m_j\rangle &= \hbar^2 j(j+1)\,|j,\,m_j\rangle \\ \hat{J}_z\,|j,\,m_j\rangle &= \hbar m_j\,|j,\,m_j\rangle\ , \end{aligned} \tag{16.4}$$

and the appropriate ladder operators,[3]

$$\begin{aligned} &\hat{J}_\pm \equiv \hat{J}_x \pm i\hat{J}_y \ \text{ with} \\ &\hat{J}_\pm\,|j,\,m_j\rangle = \hbar c^\pm_{j,m_j}\,|j, m_j \pm 1\rangle\ , \quad c^\pm_{j,m_j} = \sqrt{(j\mp m_j)(j\pm m_j+1)}. \end{aligned} \tag{16.5}$$

The states $|j,\,m_j\rangle$ are also eigenstates of $\hat{L}^2$ and $\hat{S}^2$ with the corresponding eigenvalues,[4] since both commute with $\hat{\boldsymbol{J}}$.

We can write the eigenvalues of $\hat{H}_{\rm so}$ in terms of j, l, s, noticing that

$$2\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{S}} = (\hat{\boldsymbol{L}}+\hat{\boldsymbol{S}})^2 - \hat{L}^2 - \hat{S}^2 = \hat{J}^2 - \hat{L}^2 - \hat{S}^2, \tag{16.6}$$

from which follows

$$\hat{H}_{\rm so}\,|j,\,m_j\rangle = -\tfrac{1}{2}A(r)\hbar^2\Big(j(j+1) - l(l+1) - s(s+1)\Big)\,|j,\,m_j\rangle \tag{16.7}$$

So what is left to find eigenvalues $j,\,m_j$ and the corresponding eigenstates $|j,\,m_j\rangle$.

As illustrated in the figure, classically $\big||\boldsymbol{L}| - |\boldsymbol{S}|\big| \leqslant |\boldsymbol{J}| \leqslant |\boldsymbol{L}| + |\boldsymbol{S}|$. As all the angular momenta quantized, we anticipate that

$$\begin{aligned} |l-s| \leqslant j \leqslant l+s \\ m_j = m + \sigma \end{aligned} \tag{16.8}$$

Any angular momentum state can be represented either in the "old" basis, $\{|l,\,m;\ s,\,\sigma\rangle\}$, or in the "new" one, $\{|j,\,m_j\rangle\}$. The number of eigenkets should remain the same, so that we expect

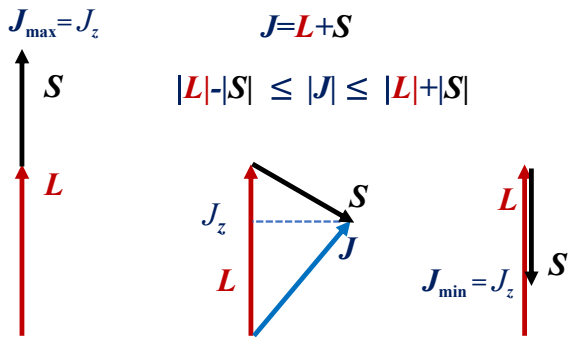


Figure 4: Classical momenta addition.

$$(2l+1)(2s+1) = \sum_{j=|l-s|}^{l+s}(2j+1)\,, \tag{16.9}$$

[3]The coefficients $c^\pm_{j,m_j}$ are the same functions of the total momentum and its projection as $c^\pm_{l,m}$ for the orbital momentum and $c^\pm_{s,\sigma}$ for spin.

[4]It would be more informative to write these eigenstates as $|j, m_j;\, l, s\rangle$ as it is done in some books but we shall use the shortened notation.

which is fairly easy to verify. As each of these basis sets is complete in the appropriate Hilbert subspace, the new eigenket can be expressed in terms of the old one.

The eigenstates $|j,\, m_j\rangle$ can be constructed as linear combinations of eigenstates

$$|l, m; s, \sigma\rangle = |l, m\rangle \otimes |s, \sigma\rangle \tag{16.10}$$

of operators $\hat{L}^2$, $\hat{L}_z$, $\hat{S}^2$, $\hat{S}_z$. This procedure is called addition of angular momenta.

Adding angular momenta is not restricted to an orbital angular momentum and spin. This can be angular momenta of any origin: for example two spins, $\hat{S}_1$ and $\hat{S}_2$ are added in the same way.

16.3 Example: $l = 1$ and $s = \frac{1}{2}$.

Let us go through full details in this case. There are $3 \times 2 = 6$ states in the appropriate Hilbert (sub)space corresponding to all possible combinations of $m = 1, 0, -1$ and $\sigma = \pm\frac{1}{2}$. We expect that j, according to inequality (16.8), could only take values of $\frac{3}{2}$ and $\frac{1}{2}$, with the number of projections m_j equal 4 and 2, respectively. Let us see how it works. It follows from Eq. (16.7) that knowing j and m_j we have already found eigenvalues of $\hat{H}_{\rm so}$ for given l and s whatever is a rotationally-invariant $\hat{H}_0$:

$$\left\langle \tfrac{1}{2}, m_j \,\middle|\, \hat{H}_{\rm so} \,\middle|\, \tfrac{1}{2}, m_j \right\rangle = -\tfrac{1}{2}\hbar^2\left(\tfrac{3}{4} - 2 - \tfrac{3}{4}\right) = \hbar^2 A(r) \tag{16.11}$$

$$\left\langle \tfrac{3}{2}, m_j \,\middle|\, \hat{H}_{\rm so} \,\middle|\, \tfrac{3}{2}, m_j \right\rangle = -\tfrac{1}{2}\hbar^2\left(\tfrac{15}{4} - 2 - \tfrac{3}{4}\right) = -\tfrac{1}{2}\hbar^2 A(r) \tag{16.12}$$

But we need more detailed information such as the probability of measuring a particular value of m_j, *etc.*

As both l and s are fixed and σ takes only two values, we simplify notations for states (16.10) as

$$|1,\, m\rangle \otimes \left|\tfrac{1}{2},\, \sigma\right\rangle \equiv |m\rangle \otimes |\sigma\rangle \to \begin{cases} |m\rangle \otimes |\uparrow\rangle \\ |m\rangle \otimes |\downarrow\rangle \end{cases} \tag{16.13}$$

Two of states (16.13) are already eigenstates of $\hat{J}^2$ and $\hat{J}_z$. These are the states with maximum and minimum values of $m + \sigma$, namely $\left|m = 1;\, \sigma{=}\tfrac{1}{2}\right\rangle$ and $\left|m = -1;\, \sigma{=}-\tfrac{1}{2}\right\rangle$. Indeed,

$$\begin{aligned} \hat{J}_z\,|1\rangle \otimes |\uparrow\rangle &= (\hat{L}_z + \hat{S}_z)\,|1\rangle \otimes |\uparrow\rangle = \hat{L}_z\,|1\rangle \otimes |\uparrow\rangle + |1\rangle \otimes \hat{S}_z\,|\uparrow\rangle \\ &= \hbar\,|1\rangle \otimes |\uparrow\rangle + \tfrac{1}{2}\hbar\,|1\rangle \otimes |\uparrow\rangle = \tfrac{3}{2}\hbar\,|1\rangle \otimes |\uparrow\rangle\ . \end{aligned} \tag{16.14}$$

The crucial point is that *no higher value of $\hat{J}_z$ is possible* so that these states must belong to the subspace of maximal $j = \frac{3}{2}$. Clearly, it works in exactly the same way for $|-1\rangle \otimes |\downarrow\rangle$ so that we have the the most "stretched" states with $m_j = \pm\frac{3}{2}$:

$$\begin{aligned} \left|\tfrac{3}{2},\, \tfrac{3}{2}\right\rangle &= |1\rangle \otimes |\uparrow\rangle \\ \left|\tfrac{3}{2},\, -\tfrac{3}{2}\right\rangle &= |-1\rangle \otimes |\downarrow\rangle\ . \end{aligned} \tag{16.15a}$$

Of course, applying $\hat{J}_z$ to other states $|m\rangle \otimes |\sigma\rangle$ will also give a definitive answer, $m_j = m{+}\sigma{=}\pm\,\frac{1}{2}$, but it will not be clear what is the value of j as such a state could belong to either $j = \frac{1}{2}$ or $j = \frac{3}{2}$. To change the value of m_j while staying in the subspace $j = \frac{3}{2}$ we apply the lowering operator $\hat{J}_- = \hat{L}_- + \hat{S}_-$ to both sides of the 1$^{\rm st}$ of Eq. (16.15a)

$$\hat{J}_-\left|\tfrac{3}{2}, \tfrac{3}{2}\right\rangle = \hbar c^-_{\frac{3}{2},\frac{3}{2}}\left|\tfrac{3}{2}, \tfrac{1}{2}\right\rangle = \hbar\sqrt{\left(\tfrac{3}{2} + \tfrac{3}{2}\right)\left(\tfrac{3}{2} - \tfrac{3}{2} + 1\right)}\left|\tfrac{3}{2}, \tfrac{1}{2}\right\rangle = \hbar\sqrt{3}\left|\tfrac{3}{2}, \tfrac{1}{2}\right\rangle$$

$$\left(\hat{L}_- + \hat{S}_-\right)|1\rangle \otimes |\uparrow\rangle = |\uparrow\rangle \otimes \hat{L}_-|1\rangle + |1\rangle \otimes \hat{S}_-|\uparrow\rangle = \hbar c^-_{1,1}|0\rangle \otimes |\uparrow\rangle + \hbar c^-_{\frac{1}{2},\frac{1}{2}}|1\rangle \otimes |\downarrow\rangle$$
$$= \hbar\sqrt{2}\,|0\rangle \otimes |\uparrow\rangle + \hbar\,|1\rangle \otimes |\downarrow\rangle\ .$$

As the l.h.s. of both equations above express the same quantity in different ways, comparing the r.h.s. we conclude that the eigenstate for $j = \frac{3}{2}$, $m_j = \frac{1}{2}$ is given by the linear combination

$$\left|\tfrac{3}{2},\,\tfrac{1}{2}\right\rangle \equiv \left|\tfrac{3}{2},\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}\,|0\rangle \otimes |\uparrow\rangle + \sqrt{\tfrac{1}{3}}\,|1\rangle \otimes |\downarrow\rangle\ . \tag{16.15b}$$

Similarly, calculating $\hat{J}_+\left|\frac{3}{2},-\frac{3}{2}\right\rangle$ in two different ways and comparing the results we get

$$\left|\tfrac{3}{2},\,-\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}\,|0\rangle \otimes |\downarrow\rangle + \sqrt{\tfrac{1}{3}}\,|-1\rangle \otimes |\uparrow\rangle\ . \tag{16.15c}$$

We have therefore obtained all four states belonging to the subspace $j = \frac{3}{2}$. The remaining two states, which belong to $j = \frac{1}{2}$, are $\left|\frac{1}{2},\pm\frac{1}{2}\right\rangle$. According in Eq. (16.8), they must be linear combinations of states with $m + \sigma = \pm\frac{1}{2}$. For $m_j = +\frac{1}{2}$, we have

$$\left|\tfrac{1}{2},\,\tfrac{1}{2}\right\rangle = a\,|0\rangle \otimes |\uparrow\rangle + b\,|1\rangle \otimes |\downarrow\rangle\,.$$

On the other hand, this state must be orthogonal to all states with $j = \frac{3}{2}$. This is automatic for three of them where $m_j \neq \frac{1}{2}$ so that $a,\,b$ can be found from the orthogonality to state $\left|\frac{3}{2},\frac{1}{2}\right\rangle$, Eq. (16.15b):

$$\left\langle \tfrac{3}{2},\tfrac{1}{2}\,\middle|\,\tfrac{1}{2},\tfrac{1}{2}\right\rangle = \left(\sqrt{\tfrac{2}{3}}\,\langle 0| \otimes \langle\uparrow| + \sqrt{\tfrac{1}{3}}\,\langle 1| \otimes \langle\downarrow|\right)\left(a\,|0\rangle \otimes |\uparrow\rangle + b\,|1\rangle \otimes |\downarrow\rangle\right) = \sqrt{\tfrac{2}{3}}a + \tfrac{1}{\sqrt{3}}b = 0\,.$$

so that $b = -\sqrt{2}a$. We also have $|a|^2 + |b|^2 = 1$ from normalisation. Choosing a, b real we can satisfy these two conditions with $a = \frac{1}{\sqrt{3}}$, $b = -\sqrt{\frac{2}{3}}$. Thus

$$\left|\tfrac{1}{2},\,\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{1}{3}}\,|0\rangle \otimes |\uparrow\rangle - \sqrt{\tfrac{2}{3}}\,|1\rangle \otimes |\downarrow\rangle\,, \quad \left|\tfrac{1}{2},\,-\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{1}{3}}\,|0\rangle \otimes |\downarrow\rangle - \sqrt{\tfrac{2}{3}}\,|-1\rangle \otimes |\uparrow\rangle\,. \tag{16.16}$$

where a similar procedure has been applied to find $\left|\frac{1}{2},\,-\frac{1}{2}\right\rangle$. Hence we have determined all the eigenkets, Eqs. (16.15).

One can generalise this procedure to any values of l and s: first, starting from the most stretched states one obtains, with the ladder operators, all $2(l+s)+1$ states with the highest $j = l+s$. Next one enters the subspace $j = l+s-1$ using orthogonality of eigenstates with the highest m_j and applying the ladder operators to find the rest. The procedure of going down through all subspaces with $|l-s| \leq j \leq l+s$ are exhausted. The coefficients of the linear combinations of eigenstates $|l, m; s, \sigma\rangle$ forming the eigenstates $|j,\,m_j\rangle$ are called *Clebsch - Gordan* coefficients. They are tabulated in the literature for a wide range of l and s (or l_1 and l_2), and the procedure, although being rather cumbersome, is algorithmic and easily programmable.

Lecture 17. Spin and Statistics. Entangled states.

Here, using the addition of two spin-½ systems, we first recall what are fermions and bosons. Then we try to deepen the understanding of entanglement by introducing entanglement entropy with the help of a new, very useful concept of density matrix.

17.1 Singlet and triplet states

In lecture 16.3, we have learned that an eigenbasis for a particle with nonzero orbital angular momentum and spin can be represented either as a set of direct products $\{|l,m\rangle \otimes |s,\sigma\rangle\}$ of eigenstates of $\hat{\boldsymbol{L}}$ and $\hat{\boldsymbol{S}}$, or a set of direct sums $\sum_{j=|l-s|}^{l+s} \{\oplus|j,m_j\rangle\}$ of eigenstates of $\hat{\boldsymbol{J}} = \hat{\boldsymbol{L}} + \hat{\boldsymbol{S}}$. As an example, we have considered,

$$\{|1,m\rangle \otimes |\tfrac{1}{2},\sigma\rangle\} \leftrightarrow \{|\tfrac{1}{2},m_j\rangle \oplus |\tfrac{3}{2},m_j\rangle\}. \tag{17.1}$$

Here factors in the direct product and terms in the direct sum are different non-overlapping subspaces of the same Hilbert space – in this case, the space of states with $l = 1$ and $s = \frac{1}{2}$. Both sets in Eq. (17.1) span the entire space, $|1\rangle_l \otimes |\frac{1}{2}\rangle_s = |\frac{1}{2}\rangle_j + |\frac{3}{2}\rangle_j$, and expressing eigenkets in the direct sum via eigenkets in the direct product was done in Lecture 16.3 as the addition of angular momenta.

What happens if we consider $\hat{\boldsymbol{S}}_{\rm tot} = \hat{\boldsymbol{S}}_1 + \hat{\boldsymbol{S}}_2$? Having two spins means that we consider two particles, and their joint Hilbert space is the direct product of Hilbert spaces for $\hat{\boldsymbol{S}}_1$ and $\hat{\boldsymbol{S}}_2$. An obvious example is the total spin of the Hydrogen atom which consists of proton and electron, each with spin $\frac{1}{2}$. According to Eq. (16.8), we expect that the value of total spin, s, obeys $|s_1 - s_2| \leqslant s \leqslant s_1 + s_2$, i.e. either $s = 0$ or $s = 1$. Hence we have

$$\{|\tfrac{1}{2},\,\sigma\rangle_1 \otimes |\tfrac{1}{2},\,\sigma\rangle_2\} = \{|0,0\rangle \oplus |1,\sigma_{\rm tot}\rangle\}\,, \tag{17.2}$$

where $\sigma = \pm\frac{1}{2}$ and $\sigma_{\rm tot} = -1,\,0,\,1$ with indices $1,2$ indicating the first or second spin. The number of states in each basis is the same: 2×2 in the product bases and $1+3$ in the total spin basis.

Now we express individual states in the total-spin basis in terms of the product states. The most stretched states have the total spin projection equal to 1 or -1. Using, as in Lecture 16.3 the shortened notations, $|\frac{1}{2},\sigma\rangle_i \equiv |\sigma\rangle_i = |\uparrow\rangle$ or $|\downarrow\rangle$, we have

$$|1,1\rangle_{\rm tot} = |\uparrow\rangle_1 \otimes |\uparrow\rangle_2 \quad \text{and} \quad |1,-1\rangle_{\rm tot} = |\downarrow\rangle_1 \otimes |\downarrow\rangle_2\,. \tag{17.3a}$$

Acting by $\hat{S}^- = \hat{S}_1^- + \hat{S}_2^-$ (temporarily renaming $\hat{S}_\pm \to \hat{S}^\pm$) on the first of this state, with $c_{1,1}^- = \sqrt{2}$ and $c_{\frac{1}{2},\frac{1}{2}}^- = 1$, Eq. (16.5), we find

$$\begin{aligned}
\hat{S}^-\,|1,1\rangle_{\rm tot} &= \hbar\sqrt{2}\,|1,0\rangle_{\rm tot} \\
\left(\hat{S}_1^- + \hat{S}_2^-\right)|\uparrow\rangle_1 \otimes |\uparrow\rangle_2 &= \left(\hat{S}_1^-\,|\uparrow\rangle_1\right) \otimes |\uparrow\rangle_2 + |\uparrow\rangle_1 \left(\hat{S}_2^-\,|\uparrow\rangle_2\right) = \hbar\Big(|\downarrow\rangle_1 \otimes |\uparrow\rangle_2 + |\uparrow\rangle_1 \otimes |\downarrow\rangle_2\Big) \\
\Rightarrow\ |1,0\rangle_{\rm tot} &= \frac{1}{\sqrt{2}}\Big(|\downarrow\rangle_1 \otimes |\uparrow\rangle_2 + |\uparrow\rangle_1 \otimes |\downarrow\rangle_2\Big).
\end{aligned} \tag{17.3b}$$

Hence we have found all the three states for the Hilbert subspace with $s_{\rm tot} = 1$, Eqs. (17.3). The one remaining state in the total-spin representation of Eq. (17.1), $|0,\,0\rangle_{\rm tot}$, must be orthogonal to state (17.3b) so that we find it as

$$|0,\,0\rangle_{\rm tot} = \frac{1}{\sqrt{2}}\Big(|\!\downarrow\rangle_1 \otimes |\!\uparrow\rangle_2 - |\!\uparrow\rangle_1 \otimes |\!\downarrow\rangle_2\Big) \tag{17.4}$$

It is said that the three states with total spin $s_{\rm tot} = 1$, Eqs. (17.3), make a triplet, while the single state with $s_{\rm tot} = 0$, Eq. (17.4), is called singlet. One can see that the triplet states are symmetric with respect to swapping the particles, $1 \rightleftarrows 2$, while the singlet state is antisymmetric.

17.2 Spin and statistics. Identical particles

You should remember from QM2 that identical particles are indistinguishable. This follows from the HUP: as we cannot follow a trajectory of a single particle, we cannot tell apart, say, two identical particles. Let us formulate this by denoting a two-particle wavefunction $\Psi(\xi_1, \xi_2)$, where $\xi \equiv (\boldsymbol{r},\, \sigma)$ with $\boldsymbol{r}$ and σ representing the position and spin of a particle. The indistinguishability means that the permutation of two particles result in

$$\Psi(\xi_1, \xi_2) = {\rm e}^{i\alpha}\Psi(\xi_2, \xi_1),$$

since the wavefunction that differ only by a phase factor describe the same physical state. If we repeat the permutation operation, we should get exactly the same phase factor, so that

$$\Psi(\xi_1, \xi_2) = {\rm e}^{2i\alpha}\Psi(\xi_1, \xi_2) \quad \Rightarrow \quad {\rm e}^{2i\alpha} = 1 \quad \Rightarrow \quad {\rm e}^{i\alpha} = \pm 1. \tag{17.5}$$

Hence the two-particle wavefunctions must be either symmetric or antisymmetric with respect to swapping the particles, $1 \rightleftarrows 2$. This property is fundamental for particles of the same kind and is true for many-particle systems as well. In this case we are talking about swapping any two particles from the system.

The symmetry/antisymmetry results in the defining properties for fermions and bosons: no identical fermions could be in the same state (***Pauli's exclusion principle***) as the antisymmetry would make the wavefunction with two fermions having the same quantum numbers to vanish; ***Bose-Einstein condensation*** – at low enough temperatures all bosons prefer to go in the lowest possible energy state thus creating a quantum liquid.

Hence, the **spin-statistics theorem** (which has a relativistic origin) states that identical particles with integer spin – bosons – obey *Bose–Einstein* statistics and are described by wavefunctions which are symmetric w.r.t. particle exchange, while identical particles with half-integer spin – fermions – obey *Fermi–Dirac* statistics and described by anti-symmetric wavefunctions.

Then you should also remember that massive elementary particles are usually spin ½ fermions (electrons, protons, neutrons, etc), while massless particles are typically spin 1 bosons (photons, gluons, etc), with (yet to be experimentally discovered) gravitons being bosons with spin 2. But spin statistics also work for composite particle – hence hydrogen atom is a boson with spin 0 or 1 – that's how our angular momenta addition works in this case.

If we have two spin ½ particles ***without any additional degree of freedom***, all the states of Eqs. (17.3) and (17.4) would be available if they are distinguishable. Hence the total spin of the hydrogen atom could be zero (a singlet) or one (a triplet). Another example are two connected two-level systems which can be effectively described as two spin ½ systems. For indistinguishable fermions, e.g. electrons, only the antisymmetric state (17.4) is possible.

However, additional degrees of freedom are usually available when talking about particles. They have corresponding quantum numbers, e.g. n, l, m for a particle in a rotationally symmetric potential. Let use notations k for a set of all relevant quantum numbers but spin. For two (identical) particles, we can have state $|k_1\rangle_1 \otimes |k_2\rangle_2$, where the index inside the ket corresponds to a particular set of quantum numbers, and index outside – to particle 1 or 2. This set could be either symmetrised or anti-symmetrise with respect to particle exchange:

$$|k_1, k_2\rangle_{\rm S} \equiv \frac{1}{\sqrt{2}} \left(|k_1\rangle_1 \otimes |k_2\rangle_2 + |k_2\rangle_1 \otimes |k_1\rangle_2\right) = +\,|k_2, k_1\rangle_{\rm S} \tag{17.6}$$

$$|k_1, k_2\rangle_{\rm A} \equiv \frac{1}{\sqrt{2}} \left(|k_1\rangle_1 \otimes |k_2\rangle_2 - |k_2\rangle_1 \otimes |k_1\rangle_2\right) = -\,|k_2, k_1\rangle_{\rm A} \tag{17.7}$$

Now the two spin ½ electrons can utilize the entire space spanned by eigenkets of $\hat{\boldsymbol{S}}_{\rm tot}$, as it is the total state (or wavefunction) that must be antisymmetric. The possible four states are

$$|1, \sigma\rangle_{\rm tot} \otimes |k_1, k_2\rangle_{\rm A}\,, \text{ with } \sigma = 1,\, 0,\, -1, \quad \text{and} \quad |0, 0\rangle_{\rm tot} \otimes |k_1, k_2\rangle_{\rm S}\,.$$

It is naturally generalised for an arbitrary number of identical particles: the total state for fermions (or bosons) should be antisymmetric (or symmetric) with respect to exchange of any two particles.

Lecture 18. Density matrix. Entanglement Entropy

Ideally, a particle can be put in a pure state characterised by a ket in the Hilbert space. In a real world, when we are dealing with many particles, their states are mixed, and characterise by the ***density matrix***. It is also a natural tool to quantitatively characterise entangled states via the ***entanglement entropy.***

18.1 Bell states and direct-product states

Let us come back to the system consisted of two subsystems fully described as spin ½ (i.e. two qubits – in modern terminology), with no other degrees of freedom. One-qubit states can be represented similar to $|\pm\rangle$ states, Eq. (15.11), as

$$|\psi\rangle_1 = a_\uparrow |\uparrow\rangle_1 + a_\downarrow |\downarrow\rangle_1 \ , \qquad\qquad |\psi\rangle_2 = b_\uparrow |\uparrow\rangle_2 + b_\downarrow |\downarrow\rangle_2 \ . \tag{18.1}$$

The two bases of Eq. (17.2) are not the only possible in this Hilbert space. If we take direct product of states (18.1),

$$|\Psi\rangle = |\psi\rangle_1 \otimes |\psi\rangle_2 \, , \tag{18.2}$$

four such states will make a basis in the Hilbert space of our two-qubit system, provided that they are orthogonal. The states like that above are called direct-product states. The most general qubit state can be written, simplifying notations like $|\uparrow\rangle_1 \otimes |\uparrow\rangle_2 \to |\uparrow\uparrow\rangle$, as a quantum superposition of four orthonormal states

$$|\Psi\rangle = a_1 |\uparrow\uparrow\rangle + a_2 |\uparrow\downarrow\rangle + a_3 |\downarrow\uparrow\rangle + a_4 |\downarrow\downarrow\rangle \, , \qquad \sum_{\ell=1}^{4} |a_\ell|^2 = 1 \, . \tag{18.3}$$

Such a state is called ***entangled*** if it cannot be reduced to the direct-product form of Eq. (18.2).

Of the eigenstates of $\hat{\boldsymbol{S}}_{\rm tot}$, the two most stretched states, Eq. (17.3a), are clearly direct-product states, while the two states with $\sigma_{\rm tot} = 0$, Eqs. (17.3b) and (17.4), are clearly entangled. A complete set of the two-qubit system in which all basis vectors are entangled can be written as

$$\left|\Psi^\pm\right\rangle = \frac{1}{\sqrt{2}}\Big(\left|\uparrow\downarrow\right\rangle \pm \left|\downarrow\uparrow\right\rangle\Big) \, , \qquad \left|\Phi^\pm\right\rangle = \frac{1}{\sqrt{2}}\Big(\left|\uparrow\uparrow\right\rangle \pm \left|\downarrow\downarrow\right\rangle\Big) \, . \tag{18.4}$$

The states above are called the Bell states (or, sometimes, the cat states in reference to the Schrödinger cat), ahd we have implicitly used $|\Psi_\pm\rangle$ when describing Bell's inequalities and the EPR paradox in Lecture 4.3. Note that $|\Psi_\pm\rangle$ are triplet and singlet states with $\sigma = 0$ which are eigenkets of $\hat{\boldsymbol{S}}_{\rm tot}$, but states $|\Phi_\pm\rangle$ are not.

While it is easy to see that the Bell states are entangled and direct product states of Eq. (18.2) are not, it is not that easy to see, given coefficients a_ℓ, whether the general state of Eq. (18.3) is entangled or not. So some criterion and a measure of entanglement is required. To this end we shall start with noting that a state of a QM system is not always representable by a state vector in a Hilbert space, $|Psi\rangle$. Up to now, we have dealt only with such states, which are called **pure states**. But a QM system can be in a **mixed state** which is not a quantum superposition of basis states.

18.2 Pure and mixed states

The reason of not being in a pure state is insufficient knowledge or insufficient accuracy of state preparation. For example, sending a beam of particles through z-oriented Stern–Gerlach device we will get all of them in state $|\uparrow\rangle$ only in ideal world. In the real world, the device is not ideal an only puts each particle into this state with some – ***classical*** – probability $p_\uparrow$. As in this case $|\uparrow\rangle$ and $|\downarrow\rangle$ make a complete basis, we use the **density matrix**, $\hat\rho \equiv p_\uparrow|\uparrow\rangle\langle\uparrow| + p_\downarrow|\downarrow\rangle\langle\downarrow|$, to characterise such a mixed state. More generally, for any complete basis set $\{|k\rangle\}$, we define the density matrix

$$\hat\rho = \sum_k p_k|k\rangle\langle k| \text{ with } \sum_k p_k = 1\,. \tag{18.5}$$

Obviously, $\hat\rho$ is Hermitian, and for any state $|\psi\rangle$ we have

$$\left\langle \psi \,\middle|\, \hat\rho \,\middle|\, \psi \right\rangle = \sum_k p_k |\langle \psi \mid k\rangle|^2 \geqslant 0. \tag{18.6}$$

The trace of this matrix, $\mathrm{Tr}\,\hat\rho = 1$. Indeed, using the same basis $\{|k\rangle\}$:

$$\mathrm{Tr}\,\hat\rho \equiv \sum_k \left\langle k \,\middle|\, \hat\rho \,\middle|\, k \right\rangle = \sum_k \left\langle k \,\middle|\, \sum_{k'} p_{k'}|k'\rangle\langle k'| \,\middle|\, k \right\rangle = \sum_{kk'} p_{k'} \langle k \mid k'\rangle\langle k' \mid k\rangle = \sum_k p_k = 1. \tag{18.7}$$

Similarly, one finds that

$$\mathrm{Tr}\,\hat\rho^2 = \sum_k p_k^2 \leqslant 1, \tag{18.8}$$

as each $p_k^2 < p_k < 1$, unless we are in a pure state where all p_k but one, say p_{k_0}, vanish so that $\sum_k p_k^2 = p_{k_0}^2 = 1$. So using the density matrix is generic (and totally widespread in modern QM), and the equality in Eq. (18.8) is the criterion for a system being in a pure state.

Finally, the expectation value of an operator $\hat A$ in the mixed state is

$$\begin{aligned}\langle\hat A\rangle &= \sum_k p_k \langle k \,|\, \hat A \,|\, k\rangle = \sum_{k,\ell} \underbrace{\langle k \mid \ell\rangle}_{=\delta_{\ell k}} \langle \ell \,|\, p_k \hat A \,|\, k\rangle = \sum_k \langle k \,|\, \sum_\ell p_\ell |\ell\rangle\langle \ell|\hat A \,|\, k\rangle \\ &\equiv \mathrm{Tr}\,\hat\rho\hat A \to \langle k_0 \,|\, \hat A \,|\, k_0\rangle \text{ for a pure state with } p_{k_0} = 1\,.\end{aligned} \tag{18.9}$$

Aside. *The **von Neumann entropy** of the system is defined as*

$$S = -\,\mathrm{Tr}\,\hat\rho \ln \hat\rho. \tag{18.10}$$

The system can be in a mixed state when it interacts with the environment at a temperature T. In this case the probabilities p_k in Eq. (18.5) *are simply Boltzmann–Gibbs probabilities, $p_k = \exp[-\beta\varepsilon_k]$, where ε_k is the energy of state $|k\rangle$ and $\beta = 1/k_{\mathrm{B}}T$. If the spectrum $\{\varepsilon_k\}$ is known, the von Neumann entropy is reduced to the standard Gibbs entropy of statistical mechanics. Below, we shall use it in a different way for characterising entanglement between two systems.*

18.3 Entanglement Entropy

Now we are able to formulate a criterion of entanglement, introducing the entanglement entropy S. Suppose we have a composite system comprising subsystems A and B. Its Hilbert space is a direct product of two Hilbert subspaces, $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. It obviously includes the above example of two spin-½ systems. Assume that the entire system is in pure state $|\Psi\rangle$. Its density matrix is $\hat{\rho} = \big|\Psi\big\rangle\big\langle\Psi\big|$ with $\mathrm{Tr}\,\hat{\rho}^2 = \mathrm{Tr}\,\hat{\rho} = 1$. The question to be asked is whether the two subsystems are mutually entangled. If they are, then the state $|\Psi\rangle$ cannot be reduced to the direct product form $|\psi\rangle_{\mathrm{A}} \otimes |\psi\rangle_{\mathrm{B}}$.

Now we define the **reduced density matrix** of subsystem A as $\hat{\rho}_{\mathrm{A}} = \mathrm{Tr}_{\mathrm{B}}\,\hat{\rho} \equiv \sum_{n_{\mathrm{B}}} \big\langle n_{\mathrm{B}} \,\big|\, \hat{\rho} \,\big|\, n_{\mathrm{B}}\big\rangle$, where $\{|n_{\mathrm{B}}\rangle\}$ is any compete set in subspace $\mathcal{H}_{\mathrm{B}}$. Then the ***Entanglement Entropy*** of subsystem A is given by[1]

$$S_{\mathrm{A}} = -\,\mathrm{Tr}_{\mathrm{A}}\,(\hat{\rho}_{\mathrm{A}} \ln \hat{\rho}_{\mathrm{A}}) = -\sum_{n_{\mathrm{A}}} \Big\langle n_{\mathrm{A}} \,\Big|\, \hat{\rho}_{\mathrm{A}} \ln \hat{\rho}_{\mathrm{A}} \,\Big|\, n_{\mathrm{A}} \Big\rangle. \tag{18.11}$$

Naturally, S_{B} can be defined in a similar way. The meaning of this is to check whether a full measurement on B affects in any way subsystem A.

It is easy to see that $S_{\mathrm{A}} = 0$ if the subsystems are not entangled, *i.e.* $|\psi\rangle_{\mathrm{A}} \otimes |\psi\rangle_{\mathrm{B}}$. Indeed, in this case $\hat{\rho} = \big|\psi_{\mathrm{A}}\big\rangle\big\langle\psi_{\mathrm{A}}\big| \otimes \big|\psi_{\mathrm{B}}\big\rangle\big\langle\psi_{\mathrm{B}}\big|$ so that

$$\hat{\rho}_{\mathrm{A}} = \big|\psi_{\mathrm{A}}\big\rangle\big\langle\psi_{\mathrm{A}}\big|\ \mathrm{Tr}_{\mathrm{B}}\,\big|\psi_{\mathrm{B}}\big\rangle\big\langle\psi_{\mathrm{B}}\big| = \big|\psi_{\mathrm{A}}\big\rangle\big\langle\psi_{\mathrm{A}}\big|\underbrace{\mathrm{Tr}_{\mathrm{B}}\,\hat{\rho}_{\mathrm{B}}}_{=1} = \big|\psi_{\mathrm{A}}\big\rangle\big\langle\psi_{\mathrm{A}}\big|,$$

Hence, subsystem A is in a pure state $|\psi\rangle_{\mathrm{A}}$. We write matrix $\hat{\rho}_{\mathrm{A}}$ in an orthonormal basis where $|\psi\rangle_{\mathrm{A}}$ is one of the eigenvectors. By definition of the pure state, the statistical probability of being in any other state of this basis $p_{n_{\mathrm{A}}} = 0$ so that matrix $\hat{\rho}_{\mathrm{A}} = \mathrm{diag}\,\{1, 0, \cdots\}$ with only a single nonzero matrix element, $\Big\langle \psi_{\mathrm{A}} \,\Big|\, \big|\psi_{\mathrm{A}}\big\rangle\big\langle\psi_{\mathrm{A}}\big| \,\Big|\, \psi_{\mathrm{A}} \Big\rangle = 1$. Then[2] $S_{\mathrm{A}} = \mathrm{Tr}_{\mathrm{A}}\,\hat{\rho}_{\mathrm{A}} \ln \hat{\rho}_{\mathrm{A}} = 1 \ln 1 = 0$.

Now, coming back to our example of the two spin ½ system, let us consider one of the Bell states, $\big|\Psi^+\big\rangle$, Eq. (18.4). The entire system is in a pure state $|\Psi\rangle^+$ so that

$$\hat{\rho} = \big|\Psi_+\big\rangle\big\langle\Psi_+\big| = \tfrac{1}{2}\Big(\big|\uparrow\downarrow\big\rangle\big\langle\uparrow\downarrow\big| + \big|\downarrow\uparrow\big\rangle\big\langle\downarrow\uparrow\big| + \big|\uparrow\downarrow\big\rangle\big\langle\uparrow\downarrow\big|\downarrow\uparrow + \big|\downarrow\uparrow\big\rangle\big\langle\downarrow\uparrow\big|\uparrow\downarrow\Big) \tag{18.12}$$

Taking trace over all states of B means that only terms diagonal in the second spin survive, e.g. $\langle\uparrow_B|\downarrow\uparrow\rangle\langle\downarrow\uparrow|\uparrow_B\rangle = \langle\uparrow_B|\uparrow_B\rangle\,|\downarrow_A\rangle\,\langle\downarrow_A|\,\langle\uparrow_B|\uparrow_B\rangle = \big|\downarrow_A\big\rangle\big\langle\downarrow_A\big|\downarrow_A$, so that

$$\hat{\rho}_{\mathrm{A}} = \mathrm{Tr}_{\mathrm{B}}\,\hat{\rho} = \big\langle\uparrow_B \,\big|\, \hat{\rho} \,\big|\, \uparrow_B\big\rangle + \big\langle\downarrow_B \,\big|\, \hat{\rho} \,\big|\, \downarrow_B\big\rangle = \tfrac{1}{2}\Big(\big|\downarrow_A\big\rangle\big\langle\downarrow_A\big| + \big|\uparrow_A\big\rangle\big\langle\uparrow_A\big|\Big). \tag{18.13}$$

In matrix form this becomes $\big\langle\sigma_A \,\big|\, \hat{\rho}_{\mathrm{A}} \,\big|\, \sigma_A\big\rangle = \mathrm{diag}\left(\frac{1}{2}, \frac{1}{2}\right)$. From Maths2B, you might remember that if A $= \mathrm{diag}\,(a_1 \cdots a_n)$, then any function $f(\mathrm{A}) = \mathrm{diag}\,(f(a_1) \cdots f(a_n))$. Hence $\ln \hat{\rho}_{\mathrm{A}} \to \mathrm{diag}\left(\ln \frac{1}{2}, \ln \frac{1}{2}\right) = -\mathrm{diag}\,(\ln 2,\, \ln 2)$ and $\hat{\rho}_{\mathrm{A}} \ln \hat{\rho}_{\mathrm{A}}$ is the product of two diagonal matrices equal to $-\frac{1}{2}\ln 2\mathrm{diag}\,(1,\, 1)$ so that the entanglement entropy $S_{\mathrm{A}} = -\,\mathrm{Tr}\,\rho_{\mathrm{A}} \ln \rho_{\mathrm{A}} = \ln 2$. This happens to be a maximal possible value of S_A for a two-qubit system.

[1] In quantum information theory, both the von Neumann entropy and the entanglement entropy are defined with $\log_2$ instead of $\ln$, the latter is natural for statistical physics where it is also multiplied by Boltzmann's k_{B}.

[2] Reminder: although we have chosen a particular basis to calculate the matrix trace, the result is invariant, *i.e.* basis independent – see Maths2B lectures.

Lecture 19. Time-independent perturbation theory

QM Hamiltonians can be exactly diagonalised only in a few exceptional cases. For a generic system, even the simplest possible – one particle in an external potential – approximate methods are required. One of the most powerful is the perturbation theory, directly applicable to the situations when a small additional term is included in the Hamiltonian.

19.1 Secular equation

It is reasonable to believe that the eigenstates and eigenenergies of a system "***perturbed***" by a small[1] addition, $\hat{V}$, to the Hamiltonian $\hat{H}_0$ are somewhat "close" to the eigenenergies and eigenstates of the unperturbed $\hat{H}_0$. We assume that $\hat{H}_0$ is solved, i.e. the eigenenergies ε_n and the corresponding ***normalised*** eigenstates $|n\rangle$ of the time-independent Schrödinger equation,

$$\hat{H}_0|n\rangle = \varepsilon_n|n\rangle\,. \tag{19.1}$$

are known. Here we use n for brevity but it could a few quantum numbers, *e.g.*, $\{n, l, m; s, \sigma\}$, *etc.*

Our task is to find ***approximate*** solutions of the EVE for the full Hamiltonian,[2]

$$\hat{H}|\psi_n\rangle = \left(\hat{H}_0 + \hat{V}\right)|\psi_n\rangle = E_n|\psi_n\rangle\,. \tag{19.2}$$

Here $|\psi_n\rangle$ is an unknown eigenfunction in the Hilbert space $\mathcal{H}$ spanned by the complete set $\{|n\rangle\}$ of eigenvectors of Eq. (19.1). Let us apply the resolution of unity to $|\psi_n\rangle$:

$$|\psi_n\rangle = \left(\sum_m |m\rangle\langle m|\right)|\psi_n\rangle = \sum_m |m\rangle \underbrace{\langle m\,|\,\psi_n\rangle}_{\equiv c_{mn}} = \sum_m c_{mn}\,|m\rangle\ , \tag{19.3}$$

and substitute Eq. (19.3) into the Schrödinger equation (19.2):

$$\left(\hat{H}_0 + \hat{V}\right)\sum_m c_{mn}\,|m\rangle = \sum_m \varepsilon_m c_{mn}\,|m\rangle + \sum_m c_{mn}\hat{V}\,|m\rangle = E_n\,|\psi_n\rangle\ . \tag{19.4}$$

Now we bra-ket Eq. (19.4) with $\langle\ell|$ from the conjugate Hilbert space corresponding to $\mathcal{H}$:

$$\sum_m \varepsilon_m c_{mn}\underbrace{\langle\ell\,|\,m\rangle}_{=\delta_{\ell m}} + \sum_m c_{mn}\underbrace{\langle\ell|\,\hat{V}\,|m\rangle}_{\equiv V_{\ell m}} = E_n\underbrace{\langle\ell\,|\,\psi_n\rangle}_{\equiv c_{\ell n}}$$

$$\| \qquad\qquad\qquad\

This equation is exact, with known unperturbed eigenenergies ε_ℓ and unknown perturbed energies E_n and coefficients $c_{mn} \equiv \langle m \,|\, \psi_n \rangle$ – the bra-ket of known $\langle m|$ with yet unknown $|\psi_n\rangle$. Further steps require approximations which depend on whether the eigenset $\{|n\rangle\}$ is degenerate or not. I remind you that the set is degenerate when there is more than one eigenket corresponding to an eigenvalue,[3] or, equivalently, when $\varepsilon_m \neq \varepsilon_n$ for any n and m.

19.2 Perturbation expansion for non-degenerate systems

We will look for the solution in the form of perturbation series,

$$\begin{aligned} E_n &= \varepsilon_n + \varepsilon_n^{(1)} + \varepsilon_n^{(2)} + \dots\,, \\ |\psi_n\rangle &= |n\rangle + |\psi_n^{(1)}\rangle + |\psi_n^{(2)}\rangle + \dots \\ \Rightarrow\quad c_{mn} &\equiv \langle m \,|\, \psi_n \rangle = \langle m \,|\, n \rangle + \langle m \,|\, \psi_n^{(1)} \rangle + \langle m \,|\, \psi_n^{(2)} \rangle + \dots \equiv \delta_{mn} + c_{mn}^{(1)} + c_{mn}^{(2)} + \cdots \end{aligned} \tag{19.6}$$

which are power series in the perturbation $\hat{V}$, where indices (1), (2), $\cdots$ means the order in (matrix elements of) $\hat{V}$. In these notations, the quantum superposition of Eq. (19.3) can be rewritten order by order as

$$|\psi_n^{(j)}\rangle = \sum_m c_{mn}^{(j)} |m\rangle\ . \tag{19.7}$$

Now we substitute expansion (19.6) into the secular equation (19.5):

$$\left(\varepsilon_n - \varepsilon_\ell + \varepsilon_n^{(1)} + \varepsilon_n^{(2)} + \dots\right)\left(\delta_{\ell n} + c_{\ell n}^{(1)} + c_{\ell n}^{(2)} + \cdots\right) = \sum_m V_{\ell m}\left(\delta_{mn} + c_{mn}^{(1)} + c_{mn}^{(2)} + \cdots\right) \tag{19.8}$$

We need to equate the l.h.s. and r.h.s. of this equation order by order in V, intending to go up to the 2$^{\rm nd}$ order corrections to the unperturbed energy ε_n and state vector $|n\rangle$, noting that the terms $\varepsilon_n^{(i)} c_{nn}^{(j)}$ are of the $(i+j)^{\rm th}$ order, and $V_{nm} c_{mn}^{(j)}$ are of the $(1+j)^{\rm th}$ order.

Important: the unperturbed basis, $\{|n\rangle\}$ is assumed orthonormal, $\langle n \,|\, m \rangle = \delta_{nm}$, and we need to preserve this in any order. This means that if we have calculated $|\psi_n\rangle$ up to the $j^{\rm th}$ order, then $\langle \psi_n \,|\, \psi_m \rangle = \delta_{mn}$ up to possible corrections of the $(j{+}1)^{\rm th}$ order or higher.

19.3 First order perturbation

The 1$^{\rm st}$ order correction to ε_n follows from keeping only the 1$^{\rm st}$ order terms in Eq. (19.8) for $n = \ell$:

$$\varepsilon_n^{(1)} = \sum_m V_{nm}\delta_{mn} = V_{nn} \quad \Leftrightarrow \quad \boxed{\varepsilon_n^{(1)} = \left\langle n \middle| \hat{V} \middle| n \right\rangle}\,. \tag{19.9}$$

Similarly, keeping only the 1$^{\rm st}$ order terms in the secular equation (19.5) for $n \neq \ell$ we find

$$(\varepsilon_n - \varepsilon_\ell) c_{\ell n}^{(1)} = \sum_m V_{\ell m}\delta_{mn} = V_{\ell n} \quad \Rightarrow \quad c_{mn}^{(1)} = \frac{V_{mn}}{\varepsilon_n - \varepsilon_m}\,, \tag{19.10}$$

[3] We do not include here a trivial degeneracy, like that with respect to spin when $\hat{H}$ is spin-independent. We have seen that such a degeneracy resulted in the appearance of the spin-degeneracy factor ($= 2$ for electrons) in certain results.

where we have relabelled $\ell \to m$ for convenience, so that up to the 1st order

$$|\psi_n\rangle = |n\rangle + \left|\psi_n^{(1)}\right\rangle = |n\rangle \left(1 + c_{nn}^{(1)}\right) + \sum_m{}' c_{mn}^{(1)} |m\rangle \ . \tag{19.11}$$

The symbol $\sum_m{}'$ means that summation is performed over all m apart from $m = n$ (or, in general, apart from those m that makes a summation term diverging). Equation (19.10) does not define $c_{nn}^{(1)}$, the 1st order diagonal correction to $\left|n\right\rangle$. The normalisation requirement to this order gives, neglecting higher order corrections, the following:

$$1 = \langle\psi_n \,|\, \psi_n\rangle = \underbrace{\langle n \,|\, n\rangle}_{=1} + \underbrace{\langle n \,|\, \psi_n^{(1)}\rangle}_{\equiv c_{nn}^{(1)}} + \underbrace{\langle \psi_n^{(1)} \,|\, n\rangle}_{\equiv c_{nn}^{(1)*}} + \sum_m{}' c_{mn}^{(1)} \underbrace{\langle n \,|\, m\rangle}_{=0} \ \Rightarrow \ c_{nn}^{(1)} + \left(c_{nn}^{(1)}\right)^* = 0$$

Hence $c_{nn}^{(1)} = i\alpha$ is a purely imaginary 1st order correction, and to this order $|\psi_n\rangle = |n\rangle \, \mathrm{e}^{i\alpha} + \sum_{m\neq n} c_{mn}^{(1)} |m\rangle$. Since the states are defined up to an arbitrary phase, multiplying this by $\mathrm{e}^{-i\alpha}$ removes diagonal correction and does not change, to the 1st order, the off-diagonal one.[4] Therefore, the 1st order correction to the nth state can be written as

$$\boxed{\left|\psi_n^{(1)}\right\rangle = \sum_m{}' c_{mn}^{(1)} |m\rangle = \sum_m{}' \frac{|m\rangle \langle m \,|\, \hat{V} \,|\, n\rangle}{\varepsilon_n - \varepsilon_m}} \ . \tag{19.12}$$

You can see now why this is only suitable for non-degenerate systems, $\varepsilon_n \neq \varepsilon_m$ for all levels.

We are at the position to formulate the criterion for the *smallness of the perturbation* $\hat{V}$. Small perturbation causes small changes to the eigenstates (and eigenenergies) and this means that the coefficients $c_{mn}^{(1)}$ must be small:

$$\left|c_{mn}^{(1)}\right| = \left|\frac{\langle m \,|\, \hat{V} \,|\, n\rangle}{\varepsilon_n - \varepsilon_m}\right| \ll 1 \, . \tag{19.13}$$

or, in words, the criterion of validity of perturbation theory for non-degenerate systems is the following:

Matrix elements of the perturbation must be smaller than a typical **level spacing**, which is the energy difference between two consecutive unperturbed levels.

19.4 Second order perturbation

Here we only interested in $\varepsilon_n^{(2)}$, albeit finding the 2nd order correction to $|n\rangle$ is also straightforward. Keeping only the 2nd order terms in Eq. (19.8) for $n{=}\ell$ (as the 1st order terms have already been equalised) and remembering that $c_{nn} = 1$ to all orders, we have

$$\boxed{\varepsilon_n^{(2)} = \sum_m{}' V_{nm} c_{mn}^{(1)} = \sum_m{}' \frac{|V_{mn}|^2}{\varepsilon_n - \varepsilon_m}} \ . \tag{19.14}$$

It happens rather often that the 1st order correction $\varepsilon_0^{(1)}$ to the groundstate energy, $\varepsilon_0 = 0$, vanishes. In this case the leading contribution given by Eq. (19.14) is always negative.

[4] This could be repeated order by order effectively resulting in $c_{nn} = 1$ exactly.

Lecture 20. Examples of perturbations. Degenerate systems.

Here, after considering a few examples of applying perturbation theory to non-degenerate systems, we shall learn how to extend it to degenerate systems, and then to time-dependent systems.

We start with considering a few examples of applying the perturbation theory to non-degenerate systems. In many cases, symmetry makes the first-order corrections to energy vanishing, but I will also include an easier example where this is not the case.

20.1 Harmonic oscillator

Consider the QHO Hamiltonian

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\right) \tag{20.1}$$

and the linear perturbation

$$\hat{V} = -f\hat{x} = -\frac{f\ell}{\sqrt{2}}\left(\hat{a} + \hat{a}^\dagger\right), \qquad \ell = \sqrt{\frac{\hbar}{m\omega}}. \tag{20.2}$$

We start by calculating the first order corrections to the energy

$$\varepsilon_n^{(1)} = \langle n|\hat{V}|n\rangle = -\frac{f\ell}{\sqrt{2}}\langle n\,|\,\hat{a} + \hat{a}^\dagger\,|\,n\rangle \tag{20.3}$$

Since diagonal matrix elements of creation and annihilation operators are zero, $\langle n|\hat{a}|n\rangle = \langle n|\hat{a}^\dagger|n\rangle = 0$, the result is $\varepsilon_n^{(1)} = 0$.

The only nonzero matrix elements of $\hat{V}$ are between neighbouring states n and $m = n \pm 1$,

$$\begin{aligned}\langle m|\hat{V}|n\rangle &= -\frac{f\ell}{\sqrt{2}}\langle m|\left(\hat{a} + \hat{a}^\dagger\right)|n\rangle = -\frac{f\ell}{\sqrt{2}}\left(\langle m|\hat{a}|n\rangle + \langle m|\hat{a}|n\rangle\right)\\ &= -\frac{f\ell}{\sqrt{2}}\left(\sqrt{n}\delta_{n,m+1} + \sqrt{n+1}\delta_{n,m-1}\right).\end{aligned} \tag{20.4}$$

Hence there are only two nonvanishing matrix elements of $\hat{V}$,

$$\langle n-1\,|\,\hat{V}\,|\,n\rangle = -f\ell\sqrt{\frac{n}{2}} \text{ and } \langle n+1\,|\,\hat{V}\,|\,n\rangle = -f\ell\sqrt{\frac{n+1}{2}},$$

and we have the correction

$$|\psi_n^{(1)}\rangle = \sum_m{}' \frac{\langle m|\hat{V}|n\rangle}{\varepsilon_n - \varepsilon_m}|m\rangle = -\frac{f\ell}{\sqrt{2}}\left(\frac{\sqrt{n}}{\varepsilon_n - \varepsilon_{n-1}}|n-1\rangle + \frac{\sqrt{n+1}}{\varepsilon_n - \varepsilon_{n+1}}|n+1\rangle\right). \tag{20.5}$$

Using $\varepsilon_n - \varepsilon_{n-1} = \hbar\omega$, and $\varepsilon_n - \varepsilon_{n+1} = -\hbar\omega$ we obtain

$$|\psi_n^{(1)}\rangle = -\frac{f\ell}{\sqrt{2}\hbar\omega}\left(\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle\right). \tag{20.6}$$

orthogonal to the state $|n\rangle$.

We now calculate the second order correction to energy levels:

$$\varepsilon_n^{(2)} = \sum_m{}' \frac{\left|\langle m|\hat{V}|n\rangle\right|^2}{\varepsilon_n - \varepsilon_m} = \frac{f^2\ell^2}{2}\left(\frac{n}{\varepsilon_n - \varepsilon_{n-1}} + \frac{n+1}{\varepsilon_n - \varepsilon_{n+1}}\right) \tag{20.7}$$

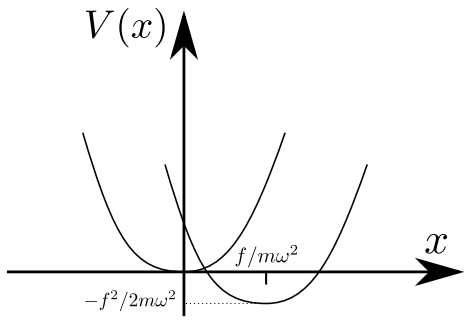
$$= \frac{f^2\ell^2}{2\hbar\omega}(n - n - 1) = -\frac{f^2\ell^2}{2\hbar\omega} = -\frac{f^2}{2m\omega}\,. \tag{20.8}$$

It is interesting to see that this result does not contain $\hbar$ and is therefore can be obtained from classical considerations. Indeed, let us absorb the linear term into potential energy,

$$\frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2 - f\hat{x} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\left(x - \frac{f}{m\omega^2}\right)^2 - \frac{m\omega^2}{2}\frac{f^2}{m^2\omega^4} \tag{20.9}$$

$$= \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2_{\text{shifted}} - \frac{f^2}{2m\omega^2}\,. \tag{20.10}$$

We see that as a result of linear term $-f\hat{x}$ corresponding to the force f acting on the harmonic oscillator its equilibrium position has been shifted $x \to x_{\text{shifted}} = x - f/m\omega^2$ and the equilibrium energy has been lowered by the amount $\Delta E = -f^2/2m\omega^2$. This is captured by the second perturbation theory and is, in fact, exact.



20.2 A tilted potential well.

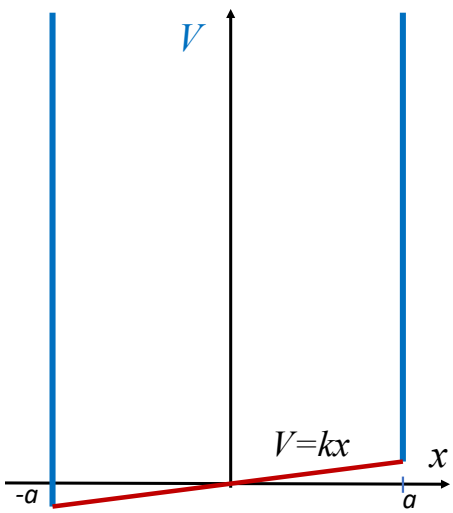
Consider an infinitely deep potential well with a tilted bottom,

$$V = \begin{cases} kx, & |x| \leqslant a \\ \infty, & |x| > a \end{cases} \qquad \text{with } ka \ll \frac{\hbar^2}{\mu a^2}$$

with a particle of mass μ inside. This potential can be considered as a perturbation to the standard $\hat{V}(x) = 0$ for $|x| \leqslant a$. As you know, a particle in the unperturbed infinitely deep potential well has the following energy and wavefunctions:

$$\varepsilon_n = \tfrac{\hbar^2 n^2}{8\mu a^2},\ \langle x\,|\,2n{+}1\rangle = \tfrac{1}{\sqrt{a}}\cos\tfrac{\pi x(2n+1)}{2a},\ \langle x\,|\,2n\rangle = \tfrac{1}{\sqrt{a}}\sin\tfrac{\pi x n}{a}.$$

We want to find the leading perturbative correction to the ground-state energy $\varepsilon_1 = \hbar^2/(8\mu a^2)$.



The first order correction, Eq. (19.9), to any energy level vanishes, since in

$$\langle n\,|\,\hat{V}\,|\,n\rangle = \int_{-\infty}^{\infty}\langle n\,|\,x\rangle V(x)\langle x\,|\,n\rangle\,\mathrm{d}x$$

we have the product of potential $V(x)$, odd in x , and the square of wavefunction, $|\langle n \,|\, x\rangle|^2$, even in x for any n. Hence we should use the 2$^{\rm nd}$ order correction, Eq. (19.14).

As the product of wavefunctions in $\left\langle n \,\middle|\, \hat{V} \,\middle|\, m \right\rangle$ must be odd, the corrections to the groundstate is contributed only by transitions to the even levels so that

$$\varepsilon_1^{(2)} = \sum_n{}' \frac{\left|\left\langle 1 \,\middle|\, \hat{V} \,\middle|\, 2n \right\rangle\right|^2}{\varepsilon_1 - \varepsilon_{2n}} = \frac{k^2}{\varepsilon_1} \sum_{n=1}^{\infty} \frac{\left|\left\langle 1 \,\middle|\, \hat{x} \,\middle|\, 2n \right\rangle\right|^2}{1-4n^2} \,.$$

The transition matrix element (with the substitution $t \equiv \pi x/2a$)

$$\left\langle 1 \,\middle|\, \hat{x} \,\middle|\, 2n \right\rangle = \int_{-\infty}^{\infty} x \langle 1 \,|\, x\rangle \langle x \,|\, 2n\rangle \,\mathrm{d}x = \frac{1}{a}\int_{-a}^{a} x \cos\frac{\pi x}{2a} \sin\frac{2\pi x n}{a} \,\mathrm{d}x = \frac{8a}{\pi^2}\int_0^{\pi/2} t\cos t \sin(2nt)\,\mathrm{d}t \equiv \frac{8a\mathcal{I}_{2n}}{\pi^2} \,.$$

Even without calculating the integral, one can see that the relative correction

$$\frac{\varepsilon_1^{(2)}}{\varepsilon_1} = -\left(\frac{ka}{\varepsilon_1}\right)^2 C, \text{ where } C = \left(\frac{8}{\pi^2}\right)^2 \sum_{n=1}^{\infty} \frac{\mathcal{I}_{2n}^2}{4n^2-1}$$

is proportional to the square of the small parameter ka/ε_1, and the numerical constant C is given by a rapidly convergent series that is mostly contributed by the first term.

Calculating the integral reveals that we do not need the series:

$$\mathcal{I}_{2n} = \int_0^{\pi/2} t\cos t\sin(2nt)\,\mathrm{d}t = \tfrac{1}{2}\int_0^{\pi/2} t\left[\sin\big((2n+1)t\big) + \sin\big((2n-1)t\big)\right]\mathrm{d}t = \frac{(-1)^n}{2(2n+1)^2} - \frac{(-1)^n}{2(2n-1)^2}$$
$$= \frac{(-1)^{n+1}4n}{(4n^2-1)^2} \text{ so that } C = \left(\frac{32}{\pi^2}\right)^2 \sum_{n=1}^{\infty} \frac{n^2}{(4n^2-1)^5}$$

Hence the 2$^{\rm nd}$ term in this sum is small compared to the 1$^{\rm st}$ by the factor $4/5^5 \sim 10^{-3}$, so that $C \approx (4/3)^5/\pi^4 \approx 0.04$.

20.3 Infinitely deep well with a step inside

The previous two examples might create a wrong impression that we always need to start with the 2$^{\rm nd}$ order perturbation. Naturally, this is not the case – the 1$^{\rm st}$ order is sometimes but not always suppressed by symmetry. Consider, as an example, a particle of mass μ in the potential

$$\hat{V} = \begin{cases} V_0 & |x| \leqslant b \\ 0 & b \leqslant |x| \leqslant a \\ \infty & |x| \geqslant a \end{cases}, \text{ with } 0 < b \ll a \text{ and } V_0 \lesssim \frac{\hbar^2}{\mu a^2}$$

Let us treat V_0 as a perturbation in spite of the fact that it can be of order of ε_1. The 1$^{\rm st}$ order correction to the ground state is given by

$$\varepsilon_1^{(1)} = \left\langle 1 \,\middle|\, V_0 \,\middle|\, 1 \right\rangle = 2V_0 \int_0^b |\langle x \,|\, 1\rangle|^2 \,\mathrm{d}x = \frac{2V_0}{a}\int_0^b \cos^2\frac{\pi x}{2a}\,\mathrm{d}x \approx \frac{2V_0 b}{a} \ll \varepsilon_1 \,,$$

since cosine is ≈ 1 for $b \ll a$. If you want the next order correction, you need to expand cosine in Taylor's series, but they be of the same order in small b/a as the 2$^{\rm nd}$ order perturbative corrections, so that they cannot be legitimately kept. Note that the 1$^{\rm st}$ order corrections are of a similar magnitude for all levels with odd n but would be proportional to $(b/a)^3$ for levels with even n.

20.4 Degenerate systems

For systems with degeneracy, where two (or more) states, say $|n\rangle$ and $|m\rangle$ have the same energy, Eqs. (19.12) and (19.14) obviously do not work – you cannot have zero in the denominator. The reason is that any linear combination of eigenstates with the same energy is also an eigenstate, and we need to find the "best" linear combinations, $\psi_n^{(0)} = \sum_{m=1}^{g} c_m |m\rangle$. Here we assume that the sum goes over all the degenerate levels, from 1 to g – the degree of degeneracy. The "best" means such that is least affected by the perturbation.

The starting point is again the secular equation (19.5).When the states $|n\rangle$ and $|\ell\rangle$ are degenerate with energy $\varepsilon_n = \varepsilon_\ell$, the zero-order terms on the l.h.s. of Eq. (19.5) cancel, so that we are left with $\varepsilon_n^{(1)} c_{\ell n}$ there. In order this to be of the 1$^{\rm st}$ perturbative order, we must take $c_{\ell n} \to c_{nn}^{(0)} \equiv c_n$, as there are no mixed coefficients, $c_{\ell n}$, in the zeroth order. In the r.h.s. we have $c_{mn} \to c_n \delta_{mn}$ so that we have

$$\varepsilon_n^{(1)} c_n = \sum_{m=1}^{g} V_{nm} c_n \quad \Leftrightarrow \quad \sum_{m=1}^{g} \left[V_{nm} - \varepsilon_n^{(1)} \delta_{mn} \right] c_n = 0\,. \tag{20.11}$$

So this is the EVE for the zeroth order coefficients c_n which would give the best linear superposition. The energy corrections $\varepsilon_n^{(1)}$ play the role of eigenvalues λ. In general, there should be g solutions that fully lift the degeneracy. First one solves the appropriate characteristic equation for the known $g \times g$ matrix V_{nm} to find the 1$^{\rm st}$ order corrections to energy that lift degeneracy. With known eigenvalues, one solve the EVE to find correct to the 1$^{\rm st}$ order linear combinations of unperturbed states.

20.5 Level repulsion

As an example of a system with energy degeneracy, consider a two-level system with the energies of each level being a function of the same single parameter. For example, this could be a weakly bound diatomic molecule where energies of each atom are certain functions of their separation with minima at equilibrium positions corresponding to zero point oscillations. At some point, the two energy levels cross so that energies are degenerate. We can model this with

$$\hat{H}_0 = \varepsilon_1(\lambda)\big|1\big\rangle\big\langle 1\big| + \varepsilon_2(\lambda)\big|2\big\rangle\big\langle 2\big|,$$

where we have already solved the problem in some good approximation so that we do not need to consider separately kinetic and potential energy of each atom. However, atoms are interacting so that H_0 is inevitably perturbed by

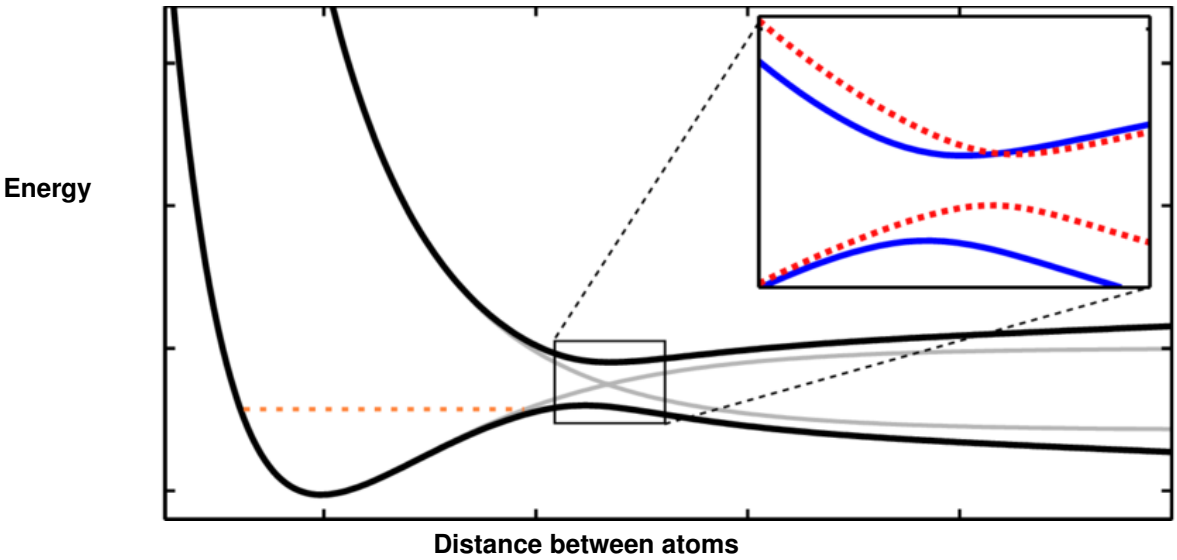
$$\hat{V} = w\Big(\big|1\big\rangle\big\langle 2\big| + \big|2\big\rangle\big\langle 1\big|\Big) \quad \text{with} \quad w \ll \varepsilon.$$

We keep in the perturbation only off-diagonal terms as the diagonal ones are well described by $\varepsilon_{1,2}$ as w is small. The point here is that however small w is, the spectrum changes dramatically at the crossing point where the two levels have the same energy, $\varepsilon_1(\lambda_0) = \varepsilon_2(\lambda_0)$.

To see this, we find the first order energy corrections, $\varepsilon^{(1)}$, governed by the secular (characteristic) equation:

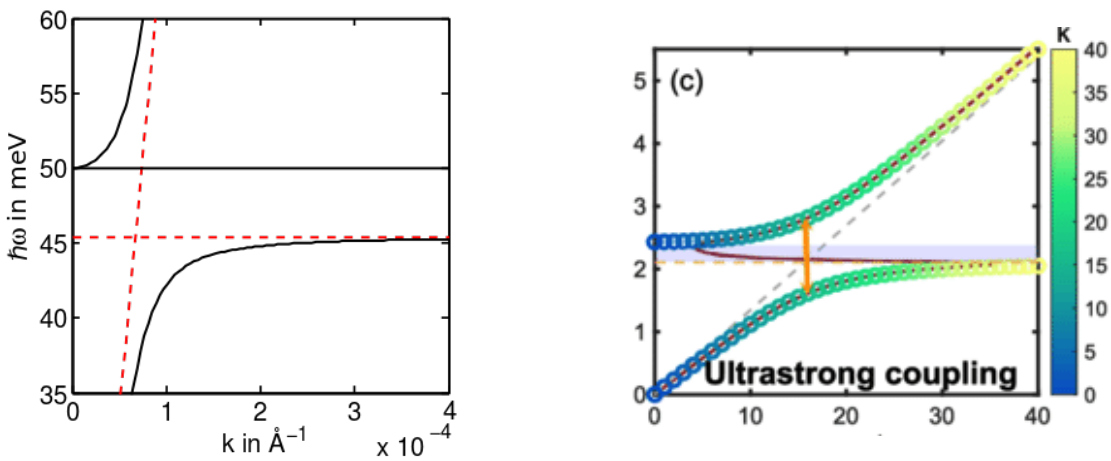
$$\begin{vmatrix} \varepsilon_1(\lambda) - \varepsilon^{(1)} & w \\ w & \varepsilon_2(\lambda) - \varepsilon^{(1)} \end{vmatrix} = 0 \;\Rightarrow \varepsilon_\pm^{(1)} = \frac{1}{2}\left(\varepsilon_1 + \varepsilon_2 \pm \sqrt{(\varepsilon_1 - \varepsilon_2)^2 + 4w^2} \right).$$

At a point λ_0, the unperturbed energies coincide, $\varepsilon_1 = \varepsilon_2$, and the perturbation gives corrections $\varepsilon_\pm^{(1)} = \pm w$. It is illustrated in the picture below where unperturbed levels are grey – at the point of crossing, the levels "repel" each other – hence level repulsion (also known as *avoided crossing).*



Now both the lower and the upper branch belong to the two atoms, not to one of them separately. The branches are separated by $\Delta = 2w$ which is called the energy gap.

The level repulsion is unavoidable unless there is symmetry (or topological) "protection" that does not allow energy levels to mix. The latter is a rare situation while ***avoided crossing*** is common, explaining energy gaps between energy bands, or various quasiparticle, such as ***polaritons*** – a mixed state of photons and phonons in certain solid state materials.



Schematic (left) and experimental (right) pictures of the polariton – the quasiparticle made of optical phonon (horizontal dashed line) and of light (tilted dashed line). Any small interaction leads to avoided crossing and creating a gap between the lower and upper branches, each is a mixture of the lattice vibrations (phonon) and electromagnetic wave (light).

Lecture 21. Time-dependent perturbation theory

Here we learn how to treat "small" time-dependent perturbations. The inverted commas mean that the definition of what perturbation can be considered as small is not that straightforward.

21.1 Perturbed time-evolution

We have learned in Lecture 7 that the time-dependent state corresponding to $\hat{H}_0 |\varepsilon_n\rangle = \varepsilon_n |\varepsilon_n\rangle$ are $\varepsilon_n(t) = \mathrm{e}^{-i\varepsilon_n t/\hbar}$, Eq. (7.13). This states are called stationary since the probability $P_n(t)$ to find the system in $|\varepsilon_n\rangle$ does not change with time. Now we shall find what happens if a small *time-dependent* perturbation, $\hat{V}(t)$, is added to $\hat{H}_0$ (e.g., electromagnetic wave interacting with an atom at a given position in space). The main problem is to find the probability $P_{\mathrm{fi}}(t)$ of transition between eigenstates,

$$\text{initial state} \quad |\varepsilon_i\rangle \to |\varepsilon_f\rangle \quad \text{final state.} \tag{21.1}$$

of unperturbed Hamiltonian $\hat{H}_0$ in a given interval of time t, where each $|\varepsilon_{i,f}\rangle$ is any one of $\{|\varepsilon_n\rangle\}$.

Since at any time t the set $\varepsilon_n(t)$ is a complete basis, we write the time-dependent state of the *combined Hamiltonian* $\hat{H} = \hat{H}_0 + \hat{V}(t)$ as a quantum superposition

$$|\Psi(t)\rangle = \sum_n d_n(t)\mathrm{e}^{-\mathrm{i}\varepsilon_n t/\hbar} |\varepsilon_n\rangle \ , \tag{21.2}$$

We have explicitly extracted the time dependence due to “unperturbed” time evolution, so in the absence of perturbation, $\hat{V}(t) = 0$, the coefficients d_n are constant. The only effect of the perturbation is to make them time-dependant. Once we find this time dependence, the transition probabilities can be expressed via these coefficients.

Substituting the quantum superposition $\Psi(t)$, Eq. (21.2), into the time-dependent Schrödinger equation we find (with $\partial_t \equiv \frac{\mathrm{d}}{\mathrm{d}t}$):

$$0 = \left(\mathrm{i}\hbar\,\partial_t - \hat{H}\right)|\Psi(t)\rangle = \left(\mathrm{i}\hbar\,\partial_t - \hat{H}_0 - \hat{V}(t)\right)|\Psi(t)\rangle \tag{21.3}$$

$$= \sum_n \left[\left(\mathrm{i}\hbar\dot{d}_n(t) + \varepsilon_n d_n(t)\right)\mathrm{e}^{-\mathrm{i}\varepsilon_n t/\hbar}|\varepsilon_n\rangle \ - \mathrm{e}^{-\mathrm{i}\varepsilon_n t/\hbar} d_n(t)\left(\hat{H}_0 + \hat{V}\right)|\varepsilon_n\rangle\right] \tag{21.4}$$

Using $\hat{H}_0 |\varepsilon_n\rangle = \varepsilon_n |\varepsilon_n\rangle$, we cancel the 2nd and 3rd terms above to see that the time-dependence in d_n comes from the perturbation,

$$\sum_n \mathrm{i}\hbar\dot{d}_n(t)\mathrm{e}^{-\mathrm{i}\varepsilon_n t/\hbar}|\varepsilon_n\rangle = \sum_n \mathrm{e}^{-\mathrm{i}\varepsilon_n t/\hbar} d_n(t)\hat{V}(t)|\varepsilon_n\rangle \ . \tag{21.5}$$

We take the bra-ket with $\langle\varepsilon_f|$ and obtain

$$\sum_n \mathrm{i}\hbar\,\dot{d}_n(t)\mathrm{e}^{-\mathrm{i}\varepsilon_n t/\hbar}\langle\varepsilon_f \,|\, \varepsilon_n\rangle = \sum_n \mathrm{e}^{-\mathrm{i}\varepsilon_n t/\hbar} d_n(t)\langle\varepsilon_f|\hat{V}(t)|\varepsilon_n\rangle \tag{21.6}$$

and, using orthogonality $\langle\varepsilon_f \,|\, \varepsilon_n\rangle = \delta_{fn}$ we have

$$\mathrm{i}\hbar\,\dot{d}_f(t) = \sum_n \langle\varepsilon_f|\hat{V}(t)|\varepsilon_n\rangle\,\mathrm{e}^{\mathrm{i}\omega_{fn}t} d_n(t)\,, \tag{21.7}$$

where we have defined the *transition frequency*

$$\omega_{fn} = \frac{\varepsilon_f - \varepsilon_n}{\hbar} \,. \tag{21.8}$$

We have not done any approximations when deriving Eq. (21.7). The only thing we did was to rearrange the terms in the time-dependent Schrödinger equation in such a way that the rate of temporal change of coefficients d_n is manifestly proportional to matrix elements of the perturbation. The equation is still difficult (if at all possible) to solve, except for some simple models with few states (like a TLS that we will consider later). To proceed, we write the initial condition

$$d_n(0) = \delta_{ni} \,. \tag{21.9}$$

and solve Eq. (21.7) iterating order by order in $\hat{V}$.

- zeroth order $\dot{d}_n(t)=0 \;\Rightarrow\; d_f(t)=\delta_{fi}$: no transition without $\hat{V}$ – the system remains in state $|\varepsilon_i\rangle$.
- first order: we substitute into the r.h.s. of Eq. (21.7) the zeroth order expression for $d_n(t)$:

$$\mathrm{i}\hbar\dot{d}_f = \sum_n \langle\varepsilon_f|\hat{V}(t)\,|\varepsilon_n\rangle\,\mathrm{e}^{\mathrm{i}\omega_{fn}t}\delta_{ni} \quad\Rightarrow\quad \dot{d}_f(t) = \frac{1}{\mathrm{i}\hbar}\langle\varepsilon_f|\hat{V}(t)\,|\varepsilon_i\rangle\,\mathrm{e}^{\mathrm{i}\omega_{fn}t} \,. \tag{21.10}$$

Integrating the last equation from 0 to t using the initial condition we obtain

$$\boxed{d_f(t) = \delta_{fi} - \frac{\mathrm{i}}{\hbar}\int_0^t \langle\varepsilon_f|\hat{V}(t)\,|\varepsilon_i\rangle\,\mathrm{e}^{\mathrm{i}\omega_{fi}t}\,\mathrm{d}t \,.} \tag{21.11}$$

We will stop at this order and calculate the probability for the transition $|\varepsilon_i\rangle \to |\varepsilon_f\rangle$. Assuming $i \neq f$ we can drop the Krönecker delta and write that to the lowest (second) order in perturbation the probability is given by the absolute value squared of the amplitude,

$$P_{\mathrm{fi}}(t) = |d_f(t)|^2 = \frac{1}{\hbar^2}\left|\int_0^t \langle\varepsilon_f\,|\,\hat{V}(t)\,|\,\varepsilon_i\rangle\mathrm{e}^{\mathrm{i}\omega_{fi}t}\,\mathrm{d}t\right|^2 \,. \tag{21.12}$$

This does not work for the probability to remain in the same state, since for $i = f$ one needs to know *amplitudes* $d_f(t)$ to the 2nd order in $\hat{V}$ since in the expansion

$$P_{ii}(t) = \left|1 + d_i^{(1)} + d_i^{(2)}\right|^2 = 1 + \underbrace{2\mathrm{Re}(d_i^{(1)})}_{=0} + \underbrace{2\mathrm{Re}(d_i^{(2)})}_{\sim\hat{V}^2} + \underbrace{\left|d_i^{(1)}\right|^2}_{\sim\hat{V}^2} + \ldots \tag{21.13}$$

the second order terms come both from the square of the first order amplitude $d_i^{(1)} \sim \hat{V}$ (the last term) and the second order amplitude $d_i^{(2)} \sim \hat{V}^2$ (the third term).

The validity of our perturbation scheme can be formulated by comparing zeroth and second order expressions for probabilities which gives simply

$$|d_f| \ll 1 \quad \text{for} \quad f \neq i \,. \tag{21.14}$$

However, we shall see that this condition does not need to be fulfilled for all times – perturbation amplitudes can be large (or even huge) for a period of time short as compared to the entire duration of the perturbation.

Lecture 22. Sudden, adiabatic and periodic perturbations. Fermi Golden rule

Here we consider three types of time-dependent perturbations and formulate the Fermi Golden Rule for the rate of intra-level transitions for perturbations periodic in time.

22.1 Sudden and adiabatic perturbation

Consider, as an example of using the time-dependent perturbation, Eq. (21.11), the QHO with frequency ω initially in the ground state. Here we will use convention that initial state corresponds to time $t = -\infty$ and consider time dependent perturbation obeying the condition $\lim_{|t|\to\infty} V(t) = 0$. We choose a particular t-dependence of perturbation $\hat{V}(t)$ acting on the QHO which satisfies this condition:

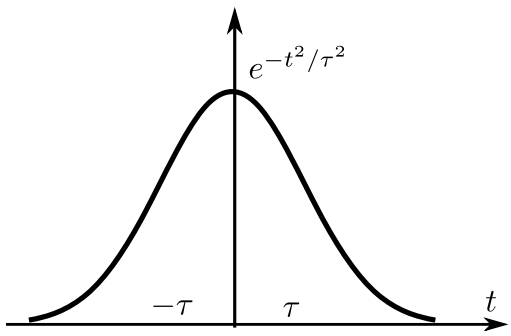
$$\hat{V}(t) = \lambda \hat{x} \mathrm{e}^{-t^2/\tau^2} \,. \tag{22.1}$$

We want to find probability $P_{n0}(+\infty)$ that at time $t = \infty$ the oscillator will be excited to a higher state $n > 0$. Using Eq. (21.11) with $f \neq i$ we have $P_{n0}(+\infty) = |d_{n0}(+\infty)|^2$ where the amplitude

$$d_{n0}(+\infty) = -\frac{\mathrm{i}}{\hbar}\int\limits_{-\infty}^{\infty}\langle n|\hat{V}(t)\,|0\rangle \exp\left(\mathrm{i}\omega_{n0}t\right)\mathrm{d}t = -\frac{\mathrm{i}\lambda}{\hbar}\int\limits_{-\infty}^{\infty}\langle n|\hat{x}\,|0\rangle\,\mathrm{e}^{-t^2/\tau^2+\mathrm{i}\omega t}\,\mathrm{d}t\,.$$

The matrix element of the position operator is non-zero only for $n = 1$,

$$\langle n|\hat{x}\,|0\rangle = \frac{\ell}{\sqrt{2}}\langle n|\left(\hat{a}+\hat{a}^\dagger\right)|0\rangle = \frac{\ell}{\sqrt{2}}\delta_{n,1}\,, \qquad \ell^2 = \frac{\hbar}{m\omega}$$



and the corresponding transition frequency $\omega_{10} = \omega$. Calculating the integral over time,[1]

$$d_{10}(+\infty) = -\frac{\mathrm{i}\lambda\ell}{\sqrt{2}\hbar}\int\limits_{-\infty}^{\infty}\mathrm{e}^{-t^2/\tau^2+\mathrm{i}\omega t}\,\mathrm{d}t = -\frac{\mathrm{i}\lambda\ell}{\sqrt{2}\hbar}\tau\sqrt{\pi}\mathrm{e}^{-\omega^2\tau^2/4}$$

we get the probability to excite harmonic oscillator to the first excited level,[2]

$$P_{10}(+\infty) = |d_{10}(+\infty)|^2 = \frac{\pi}{2}\frac{\lambda^2\ell^2\tau^2}{\hbar^2}\mathrm{e}^{-\omega^2\tau^2/2} \tag{22.2}$$

[1]The Fourier transform of the Gaussian has been considered in Maths2B, Lec. 25. Using the trick which is called *completing the square* and shifting the variable t by a constant (imaginary, but this does not matter), $y \equiv t - \frac{1}{2}\mathrm{i}\omega\tau^2$):

$$\int\limits_{-\infty}^{\infty}\exp\left(-\frac{t^2}{\tau^2}+\mathrm{i}\omega t\right)\mathrm{d}t = \int\limits_{-\infty}^{\infty}\exp\left[-\frac{1}{\tau^2}\left(t^2 - 2t\frac{\mathrm{i}\omega\tau^2}{2} - \frac{\omega^2\tau^4}{4}\right) - \frac{\omega^2\tau^2}{4}\right]\mathrm{d}t$$
$$= \mathrm{e}^{-\frac{1}{4}\omega^2\tau^4}\int\limits_{-\infty}^{\infty}\exp\left[-\left(\frac{t-\frac{1}{2}\mathrm{i}\omega\tau^2}{\tau}\right)^2\right]\mathrm{d}t = \mathrm{e}^{-\frac{1}{4}\omega^2\tau^4}\int\limits_{-\infty}^{\infty}\exp\left[-\left(\frac{y}{\tau}\right)^2\right]\mathrm{d}y = \tau\sqrt{\pi}\mathrm{e}^{-\frac{1}{4}\omega^2\tau^4}$$

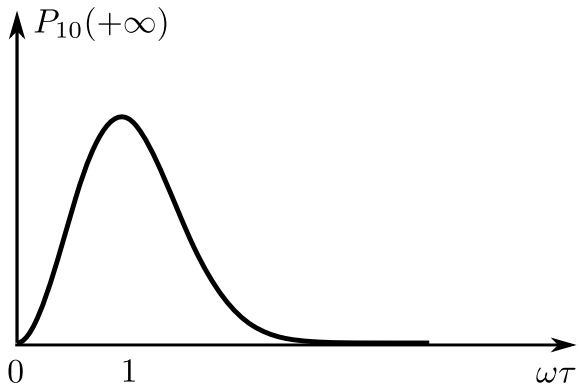
[2]Can you show that it is dimensionless as should be?

The probability to excite any other level is zero (in this order of perturbation theory). It is interesting to consider the two limiting cases:

Sudden perturbation $\tau \to 0$. In this limit $P_{10} \to 0$ as the perturbation does not have time to act and the state does not have time to change.

Adiabatic perturbation $\tau \to \infty$. The transition probability is also zero in this limit as the system manages to "adjust" to the new equilibrium position induced by the linear force: at all times the system is in the instantaneous ground state of the modified Hamiltonian. At the end the harmonic oscillator returns in the initial state and no transition occurs.

One sees that the typical time scale (the time interval with respect to which the perturbation can be considered fast or slow) is given by the natural period of oscillations, $T = 2\pi/\omega$. The probability $P_{10}(+\infty)$ peaks (as function of the parameter τ) around $\tau \sim 1/\omega \sim T$.



An ultimate example of sudden perturbation is a so-called ***quantum quench***, where during a time $\tau \ll \hbar/\omega_{nm}$ the system switches from one potential to another, e.g. the width of a quantum well or the frequency of a QHO is suddenly changed. The change does not need to be small but must be fast – then, the perturbation treatment is still possible (but will not be considered here).

22.2 Periodic perturbation

Consider a system under a constant action of time-periodic potential

$$\hat{V}(t) = \hat{v}\mathrm{e}^{-\mathrm{i}\omega t} + \hat{v}^{\dagger}\mathrm{e}^{\mathrm{i}\omega t} \tag{22.3}$$

characterised by an external frequency ω. Assuming $\hat{v} = \hat{v}^{\dagger}$, we consider the first term only. Substituting this into the expression for the amplitude, Eq. (21.11), we find for $f \neq i$

$$d_{fi}(t) = -\frac{\mathrm{i}}{\hbar}\int_0^t \langle f|\hat{v}|i\rangle \mathrm{e}^{\mathrm{i}(\omega_{fi}-\omega)t}\,\mathrm{d}t = \frac{1}{\hbar}\frac{1-\mathrm{e}^{\mathrm{i}(\omega_{fi}-\omega)t}}{\omega_{fi}-\omega}\langle f|\hat{v}|i\rangle$$

Hence the transition probability from $|i\rangle$ to $|f\rangle$ (in the 1st order in $\hat{v}$) is

$$\begin{aligned} P_{fi}(t) = |d_{fi}(t)|^2 &= \frac{1}{\hbar^2}\left|\frac{1-\mathrm{e}^{\mathrm{i}(\omega_{fi}-\omega)t}}{\omega_{fi}-\omega}\langle f|\hat{v}|i\rangle\right|^2 \\ &= \frac{1}{\hbar^2}\frac{\left(1-\mathrm{e}^{\mathrm{i}(\omega_{fi}-\omega)t}\right)\left(1-\mathrm{e}^{-\mathrm{i}(\omega_{fi}-\omega)t}\right)}{(\omega_{fi}-\omega)^2}\left|\langle f|\hat{v}|i\rangle\right|^2 \\ &= \frac{1}{\hbar^2}\frac{\Big(2-2\cos(\omega-\omega_{fi})t\Big)}{(\omega-\omega_{fi})^2}\left|\langle f|\hat{v}|i\rangle\right|^2 = \frac{t^2}{\hbar^2}\mathrm{sinc}^2\left[\tfrac{1}{2}(\omega-\omega_{fi})t\right]\left|\langle f|\hat{v}|i\rangle\right|^2 . \end{aligned} \tag{22.4}$$

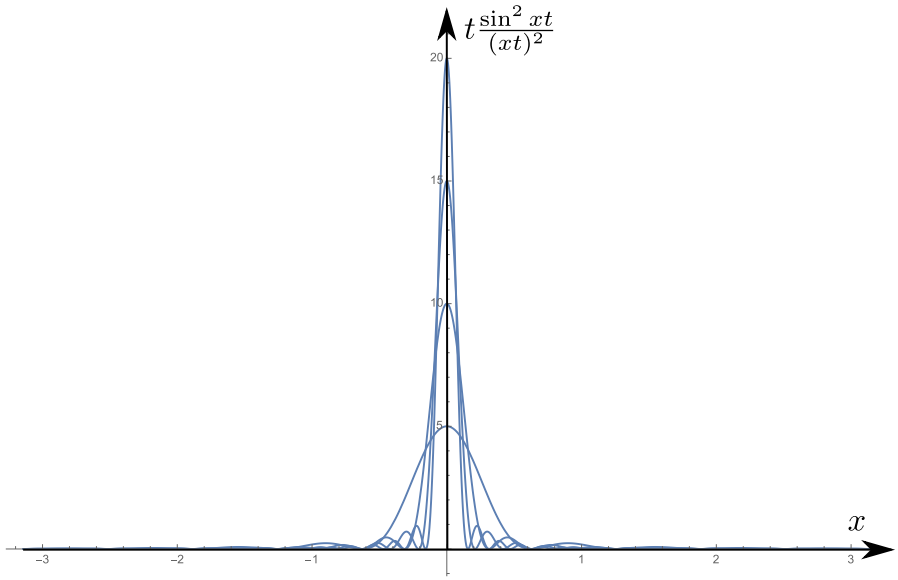
Here the function sinc is defined by $\mathrm{sinc}\, x \equiv \dfrac{\sin x}{x}$. Overleaf is the plot of the function $t\,\mathrm{sinc}^2(xt)$ for $t = 5, 10, 15, 20$. One can see that as t grows, this function becomes thinner and higher while the area below the curve is constant:

$$\int_{-\infty}^{\infty} t\,\mathrm{sinc}^2(xt)\mathrm{d}x = \int_{-\infty}^{\infty} \frac{\sin^2 y}{y^2}\,\mathrm{d}y = \pi\,. \tag{22.5}$$

It is therefore reasonable to approximate $t\,\mathrm{sinc}^2(xt)/\pi$ by Dirac $\delta(x)$, so that the probability of the transition becomes proportional to

$$t\left\{t\,\mathrm{sinc}^2\left[\tfrac{1}{2}(\omega-\omega_{fi})t\right]\right\} \to 2\pi t\delta(\omega-\omega_{fi})\,, \tag{22.6}$$

i.e. it grows linearly with time. if the perturbation acts for sufficiently long time, $t \gg \hbar/\omega_{fi}$.



We have thus calculated the transition rate, *i.e.* the probability of transition $|i\rangle \to |f\rangle$ *per unit time*

$$\boxed{\Gamma_{fi} = \lim_{t\to\infty}\frac{P_{fi}(t)}{t} = \frac{2\pi}{\hbar^2}\left|\langle f|\hat{v}|i\rangle\right|^2\delta(\omega-\omega_{fi})} \tag{22.7}$$

This formula is known as the **"Fermi Golden Rule"**. Here the δ function is responsible for the energy conservation – transition is possible only between the energy levels, while $\hat{v}_{fi}$ is time-independent amplitude of the transition that defines selection rules (e.g., transitions are only possible between the neighbouring levels in the QHO). The Golden Rule is used to estimate the rate of transitions in nuclear reactions, fusion and fission, the life-time of elementary particles and so on.

We see that periodic perturbation with $\omega > 0$ induces transition from state $|i\rangle$ with energy ε_i to the state $|f\rangle$ with energy $\varepsilon_f = \varepsilon_i + \hbar\omega > \varepsilon_i$. The second term in Eq. (22.3) for the amplitude d_{fi} gives a similar result but with $\omega \to -\omega$ so that it contributes to the transition rate for processes with $\varepsilon_f = \varepsilon_i - \hbar\omega$ describing *stimulated emission* of radiation (if the periodic field is electromagnetic) – the processes driving the coherent laser radiation. For $\omega \neq 0$ the two rates are peaked at different frequencies $\pm\omega_{fi}$ and therefore there is no contribution from the mixed terms in the square of the transition amplitude (22.3).

22.3 TLS and Rabi oscillations

For a two-level system (a qubit, a spin ½, etc) the time-dependent Hamiltonian can be solved exactly. Consider

$$\hat{H}_0 = \epsilon_1 \big|1\big\rangle\big\langle 1\big| + \epsilon_2 \big|2\big\rangle\big\langle 2\big| \tag{22.8}$$

and induce transition between the levels with the time-dependent off-diagonal term,

$$\hat{V}(t) = \hbar\gamma\big(\mathrm{e}^{\mathrm{i}\omega t}\big|1\big\rangle\big\langle 2\big| + \mathrm{e}^{-\mathrm{i}\omega t}\big|2\big\rangle\big\langle 1\big|\big)\,. \tag{22.9}$$

As before we represent the time-dependent state as the quantum superposition

$$\big|\Psi(t)\big\rangle = d_1(t)\mathrm{e}^{-\mathrm{i}\epsilon_1 t/\hbar}\big|1\big\rangle + d_2(t)\mathrm{e}^{-\mathrm{i}\epsilon_2 t/\hbar}\big|2\big\rangle \tag{22.10}$$

We have found in Lecture 21 that from the time-dependent Schrödinger equation follows the exact differential equations (21.7) for $d_f(t)$ where in this case $|f\rangle = |1\rangle$ or $|2\rangle$:

$$\mathrm{i}\dot{d}_1 = \frac{1}{\hbar}\sum_{n=1}^{2}\big\langle 1\big|\hat{V}(t)\big|n\big\rangle\mathrm{e}^{\mathrm{i}\omega_{1n}t}d_n = \gamma\mathrm{e}^{\mathrm{i}(\omega-\omega_{21})t}d_2 \tag{22.11}$$

$$\mathrm{i}\dot{d}_2 = \frac{1}{\hbar}\sum_{n=1}^{2}\big\langle 2\big|\hat{V}(t)\big|n\big\rangle\mathrm{e}^{\mathrm{i}\omega_{2n}t}d_n = \gamma\mathrm{e}^{-\mathrm{i}(\omega-\omega_{21})t}d_1 \tag{22.12}$$

since $\omega_{21} = (\epsilon_2 - \epsilon_1)/\hbar = -\omega_{12}$. We differentiate the second equation again and substitute $\dot{d}_1$ from the first equation:

$$\ddot{d}_2 + \mathrm{i}(\omega - \omega_{21})\dot{d}_2 + \gamma^2 d_2 = 0\,. \tag{22.13}$$

We have used the equation for $\dot{d}_2$ in the second term. This linear 2nd order differential equation is solved by the substitution $d_2 = \mathrm{e}^{-\mathrm{i}\lambda t}$ which results in the quadratic equation for λ:

$$\lambda^2 - (\omega - \omega_{21})\lambda - \gamma^2 = 0 \quad \Rightarrow \quad \lambda_{1,2} = \frac{1}{2}\left(\omega - \omega_{21} \pm \Omega\right)\,, \quad \Omega \equiv \sqrt{\frac{(\omega-\omega_{21})^2}{4} + \gamma^2}\,. \tag{22.14}$$

With the initial condition $d_1(0) = 1$ and $d_2(0) = 0$, we find that

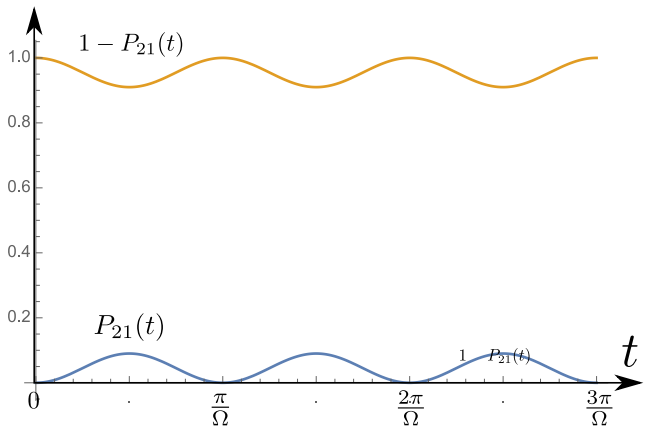
$$d_2(t) = A\mathrm{e}^{-\mathrm{i}\lambda_1 t} + B\mathrm{e}^{-\mathrm{i}\lambda_2 t} \to \frac{\mathrm{i}\gamma}{\Omega}\mathrm{e}^{-\frac{\mathrm{i}}{2}(\omega-\omega_{12})t}\sin\Omega t \tag{22.15}$$

$$d_1(t) = \frac{\mathrm{i}}{\gamma}\mathrm{e}^{\mathrm{i}(\omega-\omega_{21})t}\dot{d}_2(t). \tag{22.16}$$

The probability to find the system excited to the state $\big|2\big\rangle$ is therefore

$$P_{21}(t) = |d_2(t)|^2 = \frac{\gamma^2}{\Omega^2}\sin^2\Omega t\,. \tag{22.17}$$

The probability to remain in the ground state is $|d_1(t)|^2 = 1 - |d_2(t)|^2 = 1 - (\gamma^2\Omega^2)\sin^2\Omega t$. These oscillating probabilities – the Rabi oscillations – are shown below for $\gamma/\Omega = 0.4$.



In the limit $\gamma \to 0$ one has $\Omega \to (\omega - \omega_{21})/2$ and the expression for the transition probability $P_{21}(t)$ converges to the result of time-independent perturbation theory. On the other hand, in the resonance, $\omega = \omega_{21}$, the oscillation frequency $\Omega = \gamma$ is fully determined by the amplitude (not frequency!) of the external field $\hat{V}$.